Tectonophysics, 64 (1980) 249—332

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EARLY PALAEOZOIC PALAEOMAGNETISM IN AUSTRALIA

I. Cambrian results from the Flinders Ranges, South Australia
II. Late Early Cambrian results from Kangaroo Island, South Australia
III. Middle to early-Late Cambrian results from the Amadeus Basin, Northern Territory

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(Received September 27, 1978; accepted February 27, 1979)

ABSTRACT

Klootwijk, C.T., 1980. Early Palaeozoic palaeomagnetism in Australia. Tectonophysics, 64: 249—332.

I. Cambrian results from the Flinders Ranges, South Australia

A total of 460 samples from six sequences spanning the Cambrian succession of the Flinders Ranges (Adelaide "Geosyncline", South Australia) has been analyzed through thermal demagnetization studies. All samples showed a recent field component, generally constituting more than 50% of the initial intensity, which in most cases was removed by 200—400°C. Two characteristic magnetic components have been identified:

(A) A secondary magnetic component of Cambro-Ordovician age (S-pole at 75.3°E 26.0°N, $\alpha_{95} = 7.4°$, $N = 5$ localities) interpreted as having been induced by thermochemical activity during a period of enhanced heat flux prior to the Late Cambrian—Early Ordovician diastrophistic phases of the Delamarian Orogeny.

(B) A primary magnetic component, which indicates rapid polar motion during the Early Cambrian and a much reduced polar motion during the Middle Cambrian.

Representative palaeomagnetic pole positions for the primary component are:

(1) *Basal Hawker Group* (earliest Cambrian): S-pole at 2.3°E 26.7°S, $dp = 8.1°$, $dm = 14.3°$, $N = 10$ (sites).

(2) *Billy Creek Formation—Wirrealpa Limestone—Aroona Creek Limestone* (late Early Cambrian to early Middle Cambrian): S-pole at 20.1°E 37.4°S, $dp = 7.2°$, $dm = 14.4°$, $N = 11$ (sites).

(3) *Basal Lake Frome Group* (Middle Cambrian): S-pole at 26.1°E 29.3°S, $dp = 6.6°$, $dm = 13.1°$, $N = 10$ (sites).

(4) *Pantapinna Formation* (late Middle Cambrian?): S-pole at 29.2°E 36.4°S, $dp = 8.4°$, $dm = 16.7°$, $N = 4$ (sites).

Available data suggest that deposition of the Lake Frome Group beds probably did not continue into the Late Cambrian.

II. Late Early Cambrian results from Kangaroo Island, South Australia

A total of 108 block samples from a late Early Cambrian red-bed sequence on Kangaroo Island (Adelaide "Geosyncline", South Australia) has been analysed through thermal demagnetization studies.

All samples contained a recent field component of considerable intensity. Two characteristic magnetic components have been identified:

(A) A secondary magnetic component of Late Cambrian—Early Ordovician age (S-pole at: 75.8°E 17.4°N, $dp = 4.2°$, $dm = 7.9°$, $N = 54$ specimens), attributed to thermo-chemical activity predating the main folding phases of the Delamarian Orogeny.

(B) A primary magnetic component corresponding to a S-pole position at 15.1°E 33.8°S ($dp = 6.2°$, $dm = 12.3°$, $N = 16$ sites).

Both the primary and the secondary magnetic component are in very good directional agreement with the magnetization pattern from the correlated Billy Creek Formation of the Flinders Ranges (I). Consequently, noticeable rotational movement since late-Early Cambrian times between Kangaroo Island and the northwestern part of the Adelaide "Geosyncline" can be ruled out.

III. Middle to early-Late Cambrian results from the Amadeus Basin (Northern Territory)

A total of 328 samples from a Middle Cambrian red-bed succession and a Middle to early-Late Cambrian carbonate succession in the Amadeus Basin (Central Australia) have been analyzed through thermal demagnetization studies.

All samples contained a recent field component of considerable intensity which became broken down, respectively below 200°C in the carbonate samples and between 300°C and 500°C in the red-bed samples. Another recent field component, broken down between 600°C and 675°C, was noted in some of the red-bed samples.

Three characteristic magnetic components have been identified:

(A) A secondary magnetic component of Late Devonian—Early Carboniferous age (S-pole at 110.5°E 46.9°S, $N = 2$ localities) which predates the main folding phase of the Early Carboniferous Alice Springs Orogeny.

(B) Another secondary magnetic component (S-pole at 60.8°E 33.8°N, $N = 2$ localities) which is very similar to a thermo-chemically induced Cambro-Ordovician magnetic component, noted in rocks from the Adelaide "Geosyncline".

(C) A primary magnetic component which suggests reduced polar motion during the Middle Cambrian (S-pole at 16.6°E 35.0°S, $\alpha_{95} = 7.4°$, $N = 5$ sites) and rapid polar motion during the early-Late Cambrian (S-pole at 34.9°E 14.9°N, $dp = 6.3°$, $dm = 11.6°$, $N = 9$ specimens).

The thermally obtained palaeomagnetic record from the red-bed sequences seems less well suited for magnetostratigraphic purposes than that from the carbonate sequences. The magnetic polarity record in the former has become obscured as result of a prolonged acquisition phase. In contrast, the red-bed sequences generally showed a better concentration of component directions when compared with the carbonate sequences. Combined results from the Adelaide "Geosyncline" and the Amadeus Basin indicate the Early and Middle Cambrian as a predominantly reversed polarity interval during which several shorter periods of mixed polarity occurred. A succeeding normal polarity interval started in the early-Late Cambrian and rapid polar motion took place during the Early Cambrian and the Cambro-Ordovician, whereas the Precambrian—Cambrian boundary period and the Middle Cambrian were quasi-static intervals.

The agreement between Middle Cambrian data from the Amadeus Basin and from the Adelaide "Geosyncline" indicates unity of the Australian plate at least since that time. A common apparent polar wander path for the Gondwanaland continents backwards till at least 750 m.y. ago can be constructed through available palaeomagnetic data. The suggested late Proterozoic part of this APW remains open to alternative interpretations.

I. Cambrian results from the Flinders Ranges, South Australia

INTRODUCTION

Late Proterozoic and early Phanerozoic palaeomagnetic data from the Gondwanaland continents were reinterpreted recently by McElhinny et al. (1974b) and McElhinny and Embleton (1976). Using selected pole positions, they showed that a common apparent polar wanderpath (APW) can be drawn when the main Gondwanaland fragments are reunited into the Smith and Hallam (1970) reconstruction. This common APW stretches from the ultimate Late Jurassic breakup backwards to at least 750 m.y. ago, thus implying unity of Gondwanaland during this period. Their proposed APW (south poles) has the shape of double question mark; one half spanning the late Proterozoic and the other half spanning the early Phanerozoic. The overall structure of this APW is certainly acceptable, but detailed examination indicates several discrepancies which make their proposed APW open to improvement and reinterpretation.

One of the major uncertainties in constructing APW's which span periods of appreciable polar movement is the inherent difficulty in assigning poles to either the normal or the reversed polarity group. McElhinny et al.'s interpretation of polarities presently seems to be the more viable and less complicated approach. However, alternative backwards continuations of the Gondwanaland APW from its well defined late Palaeozoic trajectory, by reversing polarity of early Phanerozoic and latest Proterozoic palaeomagnetic poles (Piper, 1973; Schmidt and Morris, 1977) certainly cannot be ruled out. More palaeomagnetic poles are required in the less densely populated trajectories of appreciable polar wandering to solve this polarity ambiguity and for a better definition of the Gondwanaland APW.

The late Proterozoic segment of McElhinny et al.'s common APW is long and is only sparsely covered with pole positions. Recently, new results from sections in the Amadeus Basin (Central Australia) spanning the Precambrian—Cambrian boundary period were reported by Kirschvink (1978a, b), and McWilliams (1977) finalized comprehensive palaeomagnetic studies on the late Proterozoic from Australia, mainly from the Adelaide "Geosyncline" and the Amadeus Basin, and from southern Africa. McWilliams' interpretation of late Proterozoic palaeomagnetic data from Gondwanaland differs notably from McElhinny et al.'s APW, but the Cambrian data presented in this series of papers (Sections I, II, III) show that McWilliams' late Proterozoic results also can be interpreted in support of McElhinny et al.'s APW. (cf. Fig. III.13).

The early Palaeozoic part of McElhinny et al.'s common APW also appears to be somewhat problematic. Several pole positions, especially those from the Adelaide "Geosyncline", are out of sequence, whilst others — notably some South American and African data — appear to diverge from the proposed path. Therefore, I have initiated palaeomagnetic studies on well

developed early Palaeozoic sequences in Australia, namely in the Amadeus Basin and in the Adelaide "Geosyncline".

The recent availability of a sensitive cryogenic magnetometer facilitates palaeomagnetic studies on weakly magnetized sedimentary rocks such as carbonates. Such a widening spectrum of prospective lithologies and a consequent better documentation of the palaeomagnetic record ultimately may clarify the ambiguous polarity assignment of early Palaeozoic and older rocks.

I report here new results from carbonate and red beds successions spanning the Cambrian sequence of the Flinders Ranges, South Australia.

GEOLOGICAL SETTING

The Adelaide "Geosyncline" in southern Australia (Fig. I.1; Thompson et al., 1976) represents a mainly meridionally aligned downwarped region of complex fractured basement, with a thick intracratonic accumulation of sedimentary rocks. Sedimentation commenced probably between 1300 and 1400 m.y. ago, although a possible much younger age limit of about 900 m.y. has been suggested by Cooper (1975) and McWilliams (1977), and con-

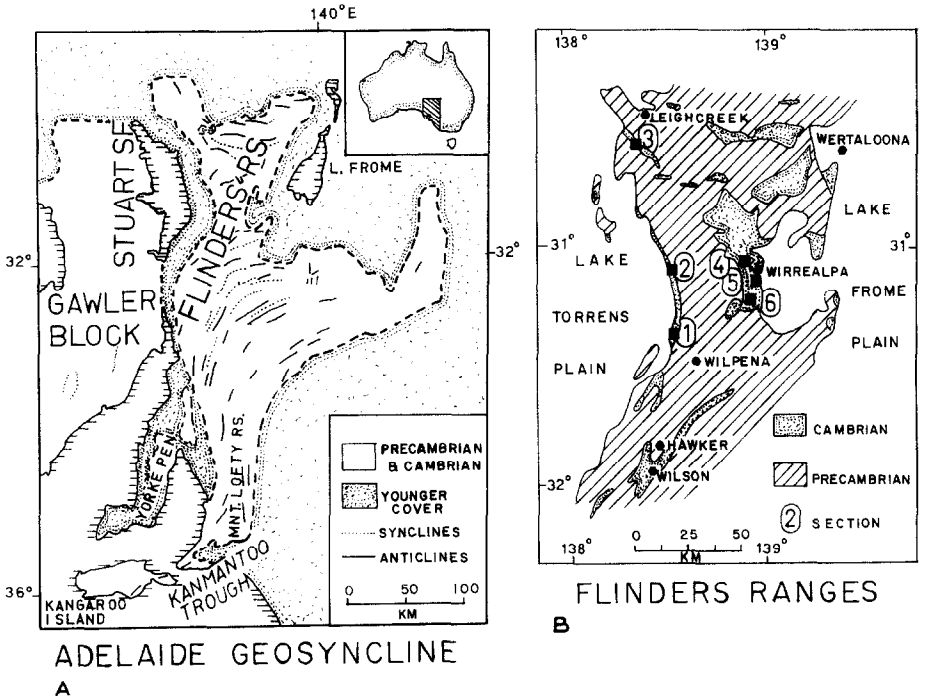


Fig. I.1. Geological sketch map of the Adelaide "Geosyncline" and the Flinders Ranges. *1—6* = sampling localities (see text).

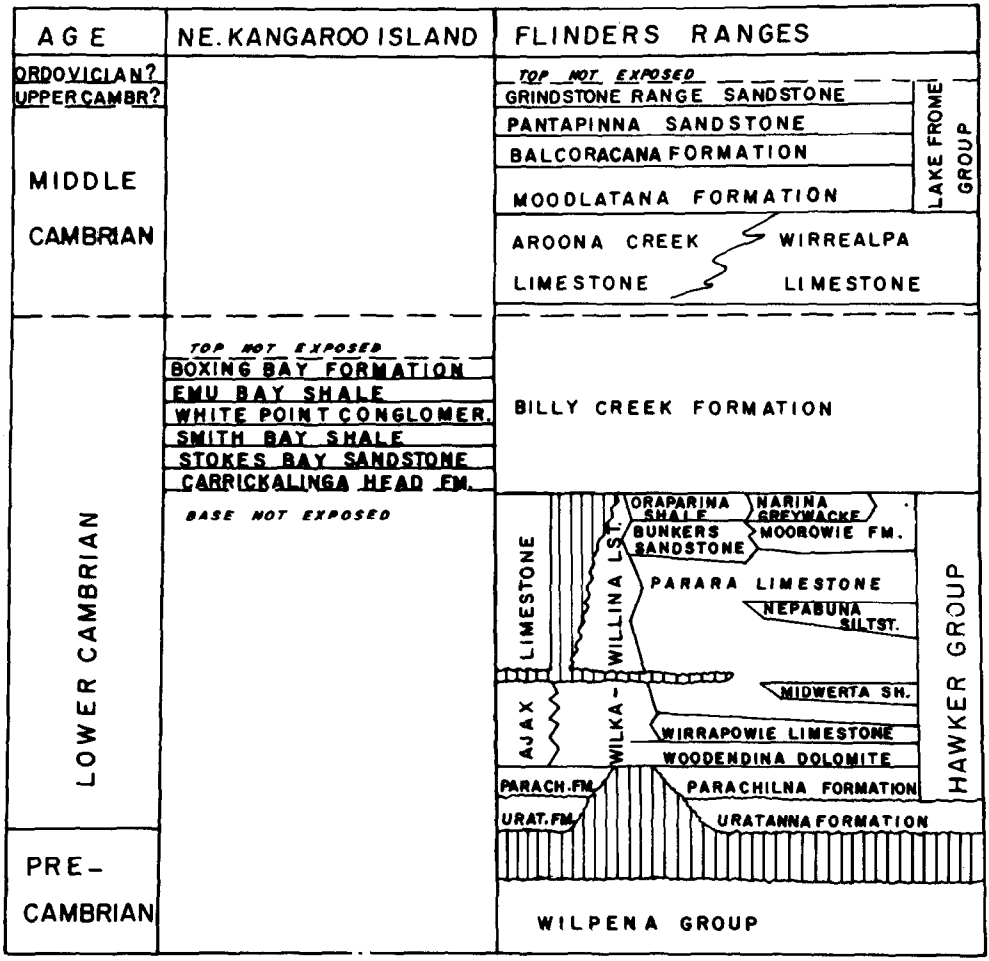


Fig. I.2. Sequence of Cambrian formations in the Flinders Ranges correlated with the Kangaroo Island sequence.

tinued intermittently with only minor volcanic activity until the onset of the Cambro-Ordovician Delamarian Orogeny (Fig. I.2).

The Adelaide "Geosyncline" is separated from the Gawler craton to the west by a north–south trending belt of fractures, the Torrens Hinge zone. The Adelaide "Geosyncline" is discordantly bounded to the southeast by the Cambrian Kanmantoo trough, but the extent of the Adelaide "Geosyncline" to the west (Wopfner, 1966, 1970) is not very well known because of cover rock.

During the Cambro-Ordovician Delamarian Orogeny, the contents of the Kanmantoo trough and the Adelaide "Geosyncline" were folded, faulted and metamorphosed. Pre- to syntectonic granite intruded the eastern part

of the Delamarian fold belt (515—506 m.y., Milnes et al., 1977), followed locally by post-orogenic granitic intrusion like in the Mount Painter region (460—430 m.y., Compston et al., 1966; Coats and Blissets, 1971).

Effects of regional metamorphism are confined mostly to the southern, eastern and extreme northern areas of the orogen, with the metamorphic grade generally decreasing in a northwesterly direction. Lower greenschist facies metamorphism is evident in the Mount Lofty Ranges, but the central part of the Flinders Ranges has not been visibly affected.

Northwest to northeast trending synsedimentary faults and recently disputed diapiric structures (Webb, 1961; Coats, 1965; Dalgarno and Johnson, 1968; Burns et al., 1977) controlled thickness and facies of the Adelaidean and Cambrian sediments. Cratonic areas to the west (Gawler craton), north (Mount Painter block) and east (a.o. the Willyama inlier) have acted as the major sediment sources.

The bulk of the sediments of the Adelaide "Geosyncline" is made up by middle to late Proterozoic rocks of the Adelaide System (Fig. I.2; Thompson et al., 1976), which were studied palaeomagnetically by McWilliams (1977). The present study concerns the well exposed Cambrian sequence of the Flinders Ranges and its stratigraphy will be described briefly (Fig. I.2).

A distinct erosional disconformity between the Ediacara fauna bearing Pound Quartzite, i.e. the topmost formation of the Wilpena Group (Adelaide System), and the overlying Uratanna Formation supposedly covers the Precambrian—Cambrian boundary (Daily, 1973, 1974). This late Adelaidean regressive period was followed by a transgression at the onset of the Cambrian, associated with gradual subsidence of the basin floor of the "Geosyncline" throughout Early Cambrian times. Continuing shallow water type sedimentation, generally dark coloured carbonates and shales, indicates that sedimentation and subsidence rates were similar. Pronounced synsedimentary faulting and diapirism resulted in heterotopical facies changes in the Lower Cambrian Hawker Group (Fig. I.2). At the end of the Early Cambrian, a regression associated with the Kangarooian movements (Daily and Forbes, 1969) led to the slightly evaporitic red-bed environment of the Billy Creek Formation and was followed by a short lived period of widespread carbonate deposition (Wirrealpa Limestone, Aroona Creek Limestone) at the beginning of the Middle Cambrian. Reversion to a high energy red-bed environment characterizes most of the Lake Frome Group sedimentation. These high-energy deposits are attributed to the end phase of the Kangarooian movements (Daily and Forbes, 1969) and herald the onset of the Delamarian Orogeny which started during Late Cambrian times. Continuity of sedimentation above the Lake Frome Group is unknown in the Flinders Ranges.

Trace fossils in the Uratanna and Parachilna Formations and a gastropod fauna in the Parachilna Formation indicate an early Early Cambrian age. Well developed faunal assemblages (Daily, 1956) in the Lower Cambrian

Hawker Group, with an abundant development of Archaeocyatha (Walter, 1967), allow correlation across the many local facies developments. Rare trilobite occurrences indicate a late-Early Cambrian age for the Billy Creek Formation, whereas abundant faunas in the overlying Wirrealpa Limestone and the correlated Aroona Creek Limestone indicate an early Middle Cambrian age. Sporadic trilobite finds in the two lowermost formations of the Lake Frome Group, i.e. the Moodlatana Formation and the Balcoracana Formation, indicate Middle- and late-Middle Cambrian ages, respectively. Fossils are seemingly absent in the overlying Pantapinna Sandstone and the Grindstone Range Sandstone, i.e. the topmost formations of the Lake Frome Group. A Late Cambrian age, possibly reaching into the Early Ordovician (B. Daily, in Thompson et al., 1976) has been suggested for these formations. As will be discussed later, comparison of the palaeomagnetic data presented here with new data from the Amadeus Basin [Section III] suggests that most, if not all, of the Lake Frome Group sedimentation was confined to the Middle Cambrian.

SAMPLING LOCALITIES

Samples, spanning the Cambrian sedimentation record, were collected at six localities in the central Flinders Ranges (Fig. I.1B).

(1) Parachilna Gorge (138.53° E 31.12° S)

Three sites (APRA—APRC) were taken in the more basal part of the Parachilna Formation at Parachilna Gorge in reddish to locally discoloured argillaceous sandstone beds with well developed vertical burrow tracks, about 9—15 m above the contact with the underlying Pound Quartzite.

(2) Bunyeroo Gorge (138.55° E 31.41° S)

The Hawker Group has been sampled in the gorge itself, and the overlying Billy Creek Formation—Wirrealpa Limestone sequence has been sampled along a profile to the southwest of the gorge.

Hawker Group. Eleven sites (AWIA—AWIE, APAB—APAG) have been taken over a total stratigraphic thickness of about 500 m through massive limestone banks of the Wilkawillina Formation, and interfingering sequences of dark grey nodular and thinly bedded limestones and shales of the Parara Formation. The Parara facies represents strongly reducing conditions (Daily, 1972b), which may have resulted in its poor palaeomagnetic characteristics. Another two sites (AORA—AORB) were taken in thinly bedded green shales in the basal part of the overlying Oraparina Formation, which is poorly exposed at the western entrance of the gorge.

Billy Creek Formation—Wirrealpa Limestone. Three sites (ABCE—ABCG) were taken throughout a flaggy dolomitic limestone bank about 3 m thick

chosen by Dalgarno (1964) as the base of the Billy Creek Formation. Two sites (ABCC—ABCD) were sampled about 175 m above the base of the Billy Creek Formation in thinly bedded dark red mudstones of the basal member. Another two sites (ABCA—ABCD) were taken at a small hillock further to the southwest, about 100 m stratigraphically below the top of the formation, in reddish to pink coloured micaceous sandstones which form the upper member of the formation.

Two sites (AWRA—AWRB) were taken in the Wirrealpa Limestone. Site AWRB was taken in a massive dolomitic limestone bank at the base of the formation, whereas site AWRA was taken in a rubbly limestone bank about 75 m higher stratigraphically.

(3) Mount Scott Area (138.35° E 30.61° S)

Ajax Limestone. Fifty-two block samples (BAJA1-52) were taken at regular intervals of 1—2 m through the top 80 m of the Ajax Limestone, along a prominent creek south of Aroona Dam. The sequence consists of light red to pink argillaceous limestone which becomes progressively more massive towards the higher part of the section. The Ajax Limestone is correlated with the carbonates of the Hawker Group in the central Flinders Ranges (Daily,1956; B. Daily, in Thompson et al., 1976).

Aroona Creek Limestone. Another 24 block samples (BACA1-24) were taken through the top 25 m of this formation, consisting of pale coloured well bedded dolomitic limestone, well exposed where the formation cuts the Aroona Creek. The basal part of the Aroona Creek Limestone has been correlated with the Wirrealpa Limestone (Daily, 1956; Daily, in Thompson et al., 1976) since both contain a Redlichia fauna.

(4) Mount Billy Creek (138.92° E 31.27° S)

The well described Cambrian sequence of Mount Billy Creek (Daily, 1956; Dalgarno, 1964) shows a peculiar facies development attributed to syn-sedimentary faulting associated with the Oraparina diapir. Diapiric activity is thought to have ended during deposition of the Wirrealpa Limestone (Coats, 1965; Dalgarno and Johnson 1968; Youngs, 1976). Stratigraphic profiles were sampled through the topmost beds of the Billy Creek Formation, the Wirrealpa Limestone and the Moodlatana Formation.

Billy Creek Formation. Four sites (BBCA—BBCD) were taken through prominent sandstone banks at the top 24 m of the Billy Creek Formation.

Wirrealpa Limestone. Eleven sites (BWIA—BWIK) were taken over a stratigraphic interval of 120 m in the more coherent banks of the formation, made up of mottled limestones with some massive limestone banks at the base.

Moodlatana Formation. Forty seven samples (BM001-47) were drilled at

regular intervals of about 1—1.5 m over a stratigraphic sequence of 75 m, starting about 125 m above the top of the Wirrealpa Limestone, in well-bedded red sandy shales and sandy siltstones.

(5) Red Hill Well (138.92° E 31.09° S)

One site (AWRC) was taken in well bedded massive oolitic limestones, in the upper part of the Wirrealpa Limestone. Another two sites (AMOC, AMOD) were drilled in the basal 120 m of the overlying Moodlatana Formation.

(6) Balcoracana Creek (138.92° E 31.15° S)

Lake Frome Group sediments are well exposed in this profile.

Moodlatana Formation. Two sites (AMOA, AMOB) have been taken about 40 m below the top of the formation in coarse-grained crossbedded and thick-banked rather soft feldspathic sandstones.

Balcoracana Formation. Four sites (APNA—APND) were taken through thick-banked micaceous siltstone and minor feldspathic sandstones in the middle part of the formation.

Pantapinna Formation. Four sites (APNE—APNH) were taken through thick-banked cross-bedded pink brown micaceous and feldspathic sandstones.

SAMPLING TECHNIQUE AND LABORATORY TREATMENT

Where local conditions permitted core samples were obtained with a portable 1-inch diamond drill. Otherwise block-samples were taken (Mount Scott area), which were cored in the laboratory with a drill press. The samples were orientated in the field with both a magnetic and a solar compass using a device similar to that described by Embleton and Edwards (1973).

Samples were taken either as separate sites generally consisting of 6—8 samples or, if conditions permitted, as stratigraphic profiles with each sample taken from a separate bed at regular 1—2 m intervals. Stratigraphic intervals were measured with a Jacob's Staff and an attached Abney level. Mostly two specimens, 2.5 cm diameter and 2.25 cm height, were sliced from each core.

Up to three specimens from each site, or one specimen from each third sample in case of stratigraphic profiles, have been subjected to progressive thermal demagnetization studies. The heatings were done in air in a furnace described by McElhinny et al. (1971). Because of the complexity of magnetization, all pilot specimens were progressively demagnetized in 12—18 steps with decreasing temperature intervals up to 685° C. The intensity of susceptibility was measured after each heating step on a Digico bulk susceptibility unit. The remaining specimens were subjected to partial progressive demag-

netization in 8—12 steps over a selected temperature interval. Care was taken to keep the specimens in a very much reduced magnetic environment since the specimens tended to acquire Viscous Remanent Magnetization (VRM) after the higher heating steps.

Two pilot specimens from each site from the Bunyeroo Gorge, Balcoracana Creek, Parachilna Gorge and Red Hill Well localities were subjected to a progressive alternating field demagnetization in 12—20 steps up to 210 mT in a two-axis specimens tumbler. Because of excessive development of spurious components and failure to isolate separate components completely, further AF-demagnetization was not pursued for the bulk of the samples.

All magnetic measurements were made on a SCT 2-axis super-conducting rock magnetometer which was interfaced to a Digico mini-computer for immediate data reduction.

Directional analysis of all specimen components was made by means of orthogonal projection figures (Zijderveld, 1967, 1975).

RESULTS

A limiting factor for remanence measurement of carbonate rocks in particular was a strong development of viscous magnetization after heating above 400—450° C. Because of the individual behaviour of the specimens during thermal demagnetization, laborious demagnetization series had to be run coupled with careful analysis of the orthogonal projection figures. Within a site and even within a sample, individual magnetic components often showed widely different blocking temperature ranges.

Throughout this study and especially in the lower intensity carbonate samples it was observed that the demagnetization paths often bypassed the origin of the projection figure during breakdown of what can be interpreted as the last remaining magnetic component. The reason for this bypassing is not completely understood. Various possible causes may be mentioned:

(a) Inhomogeneous magnetization (Collinson, 1977), which becomes more prominent after continued demagnetization.

(b) Presence of an unidentified additional magnetic component.

(c) Acquisition of a VRM due to a slight residual field within the super-conducting shield. It has to be mentioned, however, that similar observations have been made during earlier studies of carbonate samples on a Jelinek JR-3 spinner (Klootwijk and VandenBerg, 1975; VandenBerg et al., 1978) and during red-bed measurements with an astatic magnetometer (Klootwijk, 1974). So this observation cannot be attributed solely to supposed deficiences in measuring apparatus.

Clearly, orthogonal projection figures are a most useful tool for analyses of such samples. Component directions can be determined by reading the first derivative of straight trajectories of the vector graph, which is analogous to a smoothed analytical vector subtraction.

In general, three magnetic components could be distinguished (Tables I.1—I.3):

TABLE I. 1

FLINDERS RANGES PRIMARY DIRECTIONS

Formation	Site	x' Samples	x' Specimens	Initial Intensity mAm^{-1}	Decl (degrees)	Incl (degrees)	K	α95 (degrees)	Polarity	Remarks
A) BUNYEROO GORGE				Location 138.55°E, 31.41°S; Samples 144; Specimens 253						
WIRREALPA	AWRA	6	18	1.5-6.9	219.5	-24	14	9.6	R	
LIMESTONE	AWRB	6	20	1.5-3.2	241	+4	16	8.4	R	
BILLY CREEK	ABCA	6	19	6.7-14.2	214	-10	29	6.4	R	
FORMATION	ABCB	6	14	11.7-87.4	268.5	-64.5	15.5	12.8	R	Excluded
	ABCD	5	7	3.2-5.8	25	-29.5	3	41.7	N	
	ABCC	5	9	3.4-6.7	28.5	+49	11.5	15.9	N	
basalmost beds	ABCG	5	6	1.7-2.6	251	+10.5	12	20.1	R	
BILLY CREEK FM.	ABCF	5	6	1.7-2.8	251.5	-18	6	29.9	R	
	ABCE	5	5	3.0-6.1	242.5	-31.5	16	19.8	R	
ORAPARINNA	AORB	1	1	0.6-3.8	248	-43.5	-	-	R	Excluded
SHALES	AORA	4	4	0.3-0.6	258	-22	4	52.2	R	
PARARA	APAG	4	5	2.3-3.0	246	-18.5	10	25.3	R	
LIMESTONE	APAF	3	3	1.6-4.0	137.5	-48	56.5	16.6		Excluded pseudo stable
	APAE	3	3	0.7-3.6	148.5	-55.5	38	20.3		" " "
	APAD	2	2	2.1-2.5	145	-69.5	-	-		" " "
WILKA WILLINIA	AWIE	5	7	3.5-9.0	191	-67	115.5	5.6		" " "
LIMESTONE	AWID	5	5	6.8-16.0	228.5	-70.	37	12.8		" " "
	APAC	4	4	2.2-13.1	152	-51.5	37	15.3		" " "
	APAB	7	10	4.5-18.7	256.5	-55.	29	9.1		Excluded pseudo stable
	AWIC	5	7	3.5-4.8	209	-46.5	8	23.1	R	
	AWIB	7	9	1.6-2.7	242	-38.5	30.5	9.5	R	
	AWIA	5	6	0.9-2.9	214	-46	9	23.6	R	
B) PARACHILNA CREEK				Location 138.53°E, 31.12°S; Samples 17; Specimens 23						
PARACHILNA	APRC	4	4	3.0-6.5	48.5	23.5	3	-	N	
FORMATION	APRA	3	3	5.6-8.5	203	-29.5	7.5	48.4	R	
	APRB	-	-	1.1-12.0						Mostly recent field components
C) RED HILL WELL				Location 138.9°E, 31.09°S; Samples 21; Specimens 57						
MOODLATANA	AMOC	2	2	5.3-8.3	243	17.5	-	-	R	
FORMATION	AMOD	-	-	2.1-3.8						Only "Delamarian" components
WIRREALPA LIMESTONE	AWRC	7	14	0.6-3.2	330.5	-47.5	37.5	6.6	R	Excluded
D) BALCORACANA CREEK				Location 138.92°E, 31.15°S; Samples 67; Specimens 118						
PANTAPINNA	APNH	3	3	1.0-4.7	237.5	+3	29.5	23.0	R+N	
SANDSTONE	APNG	2	3	1.2-2.8	238	+14.5	7	49.8	R	
	APNF	4	5	1.1-3.2	229	-0	5	37.7	N+R	
	APNE	5	6	0.7-3.9	212	+16	5	34.4	R	
BALCORA CANA	APND	7	12	5.2-6.3	237	-15	10.5	14.1	R	
FORMATION	APNC	6	7	5.7-6.9	297.5	+57	27.5	11.7	R	Excluded
	APNB	5	6	2.0-3.1	68	+6	7.5	25.8	N	
	APNA	7	9	4.9-8.9	285.5	-19	95	5.3	R	Excluded
MOODLATANA	AMOA	7	11	3.8-10.2	230	-42.5	3	31.3	R	
FORMATION	AMOB	7	12	7.0-13.9	229	-11.5	9	15.4	R	
E) MOUNT BILLY CREEK				Location 138.92°E, 31.27°S; Samples 134; Specimens 196						
MOODLATANA	BMOA 51-75		6		218.5	-8	7	26.7	R	
FORMATION	BMOA 17-57		17(N)		99	-0.5	3	24.0	N	
			39(R)		239	+4	5	11.5	R	
			56(R+N)		250	+3	3.5	11.5	R+N	
	BMOA 1-17		32		197	+11.5	20	5.8	R	Excursion ?
	TOTAL	53	94	1.8-8.4	227.5	+6	4	8.7	R+N	
WIRREALPA LIMESTONE	AWIA-AWII	26	47	0.1-0.6	194.5	-7	6	9.4	R	Excursion ?
	AWIK	9	12	0.2-0.5	266	24	13.5	12.2	R?	Aberrant
BILLY CREEK	BBCD-		7(N)		59.5	3.5	9	34.7	N	
FORMATION	BBCA		22(R)		227	-5.5	6	13.6	R	
		22	29(R+N)	2.2-8.4	230	-5	5.5	12.4	R+N	
F) ARDONA CREEK				Location 138.35°E, -30.61°S; Samples 76; Specimens 149						
ARDONA CREEK LIMESTONE	BACA 1-24	13	22	0.2-2.9	213	+5	8.5	15.1	R+N	
AJAX LIMESTONE	BAJA 1-52	27	40	0.1-8.4	216	-24.5	4.5	14.9	R+N	

x' Number of samples (specimens) wherein a characteristic component could be determined

TABLE I. 2

FLINDERS RANGE SECONDARY DIRECTIONS

LOCALITY	SITE(S)	SPECIMENS[1]	Decl (degrees)	Incl (degrees)	K	α95 (degrees)	Polarity	S-Pole Positions Long. (degrees)	Lat (degrees)	dp (degrees)	dm (degrees)	Pole No. Fig. 8	Remarks
A) "DELAMARIAN" DIRECTION													
BUNYEROO GORGE	AWIA/ AORA/ AORB	9	310	+16.5	5	25.7	R	79.6E	27.6N	13.7	26.5	9	
RED HILL WELL	AMOC/ AMOD	25	303.5	-1.5	8	10.8	R+N	67.1E	28.4N	5.4	10.8	6	Not evident
BALCORACANA CREEK	AMOB												
	APNH	47	304.5	+32.5	5	10.2	R+N	83.5E	17.8N	6.5	11.5	10	
MOUNT BILLY CREEK		25	309.5	+7.5	4.5	15.8	R+N	75.4E	30.4N	8.0	15.9	8	Evident in BILLY CREEK formation only
ARDONA CREEK	ARDONA CREEK LST.	22	300.5	+13	6.5	13.4	R+N						Mostly reversed
	AJAX LST.	22	304	+4	4	17.4	R+N						Mostly reversed
	COMBINED	44	302.5	+8.5	5	10.6	R+N	70.1E	24.9N	5.4	10.7	7	
B) RECENT FIELD DIRECTION								PLFD[2]		PLFP[3]		DD[4]	
BUNYEROO GORGE		6	337.5	-46.5	36.5	11.3	N	7	-62	115.4E	-76.9S	0	-50
PARACHILNA CREEK		4	170	+51	28.	17.5	R						
		8	340.5	-59	13	16.0	N						
	COMBINED	12	344	-56.5	16	11.2	N+R	7	-62	115.4E	-76.9S	0	-50
RED HILL WELL		Not Evident											
BALCORACANA CREEK		22	10.5	-61.5	15	8.3	N	7	-62	115.4E	-76.9S	0	-50
MOUNT BILLY CREEK													
	MOODLATANA FM.	8	345	-58	14.5	15.1	N						
	WIRREALPA LST.	26	1.5	-65.5	16	7.4	N						
	BILLY CREEK FM.	4	2.5	-49	16	23.8	N						
	COMBINED	38	358	-62.5	15	6.2	N	7	-62			0	-50
ARDONA CREEK	ARDONA CREEK LST.	20	352.5	-51.5	25.5	3.0	N						
	AJAX LST.	92	358.5	-55.5	17	8.2	N						
	COMBINED	112	353.5	-52	23.5	2.8	N	7	-62	116.5E	-76.2S	0	-50

1 Number of specimens wherein the component concerned could be identified

2 Present local field direction

3 Corresponding pole position according to the axial geocentric dipole formulae

4 Expected direction at the sampling locality according to the axial geocentric dipole formulae

(a) A recent field component (RFC) of large intensity, up to 90% of the initial intensity in the carbonate rocks, could be identified in all rock samples (Fig. I.3A, I.7D) and was particularly dominant in the rock sequences from Mount Billy Creek and from the Mount Scott area (Fig. I.3B). This RFC was generally eliminated at the lower temperature ranges (Table I.4). During thermal analysis at the higher temperature ranges, generally two characteristic magnetic components could be distinguished.

(b) A SSW—SW directed component (A) with a moderate to low upwards dip. Both normal and reversed polarities are present.

(c) A WNW—NNW directed component (B) with a low downwards dip,

TABLE I. 3

FLINDERS RANGES MEAN PRIMARY DIRECTIONS

SITES COMBINED	Direction					S-Pole Position				Pole Number Figure 8
	Decl (degrees)	Incl (degrees)	K	α 95 (degrees)	N	long (degrees)	lat (degrees)	dp (degrees)	dm (degrees)	
PANTAPINNA SANDSTONE (i.e. Lake Frome Group-top)	229.4	+8.5	31.6	16.6	4	29.2E	36.4S	8.4	16.7	3
LAKE FROME GROUP - base	233.5	-4.2	14.6	13.1	10	26.1E	29.3S	6.6	13.1	5
LAKE FROME GROUP COMBINED	232.3	-0.5	16.6	10.1	14	26.9E	31.4S	5.1	10.1	4
BILLY CREEK FM., WIRREALPA LST, AROONA CREEK LST.	224.1	-1.2	11.0	14.4	11	20.1E	37.4S	7.2	14.4	2
HAWKER GROUP - top (APAG, AORA, ABCE, ABCF, ABCG)	249.9	-16.0	24.0	15.9	5	30.7E	12.6S	8.4	16.4	20
HAWKER GROUP - base (APRA) ARRC, AWIA, AWIB, AWIC, BAJA)	22o.7	-33.7	15.9	12.5	10	2.3E	26.7S	8.1	14.3	1
HAWKER GROUP total	233.4	-27.8	12.3	11.4	15	14.9E	21.3S	6.8	12.5	19

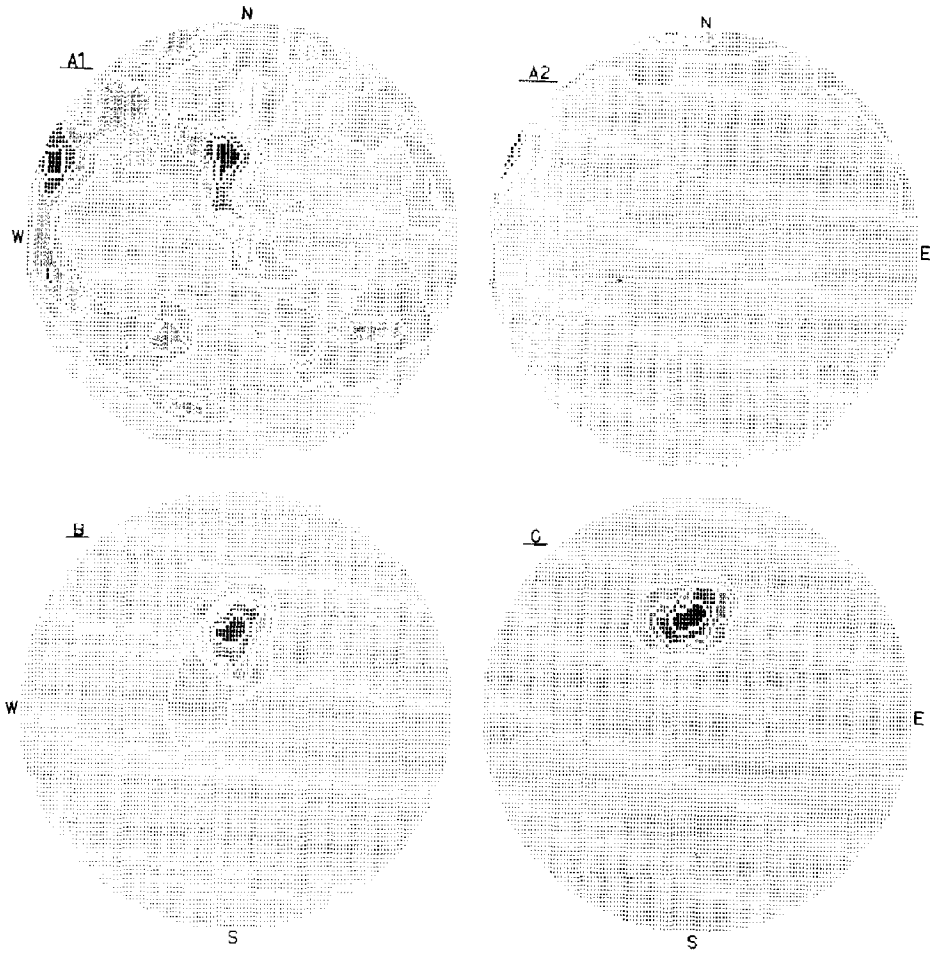


Fig. I.3. Frequency distribution of magnetic directions in equal area projection, counted with a 1% circle. Five frequency intervals of equal magnitude have been contoured. A1. NRM directions combined from the Bunyeroo Gorge, Parachilna Gorge, Red Hill Well and Balcoracana Creek, upper hemisphere projection, maximum frequency = 17. A2. As above, lower hemisphere projection, maximum frequency = 13. B. Initial NRM directions combined from the magnetostratigraphic profiles sampled in Mount Billy Creek and in Aroona Creek, upper hemisphere projection, maximum frequency = 64. C. Recent field directions observed in the Ajax Limestone profile at Aroona Creek, upper hemisphere projection, maximum frequency = 3.

and of both normal and reversed polarity. This component was particularly noticeable in samples from the eastern Flinders Ranges and from the Mount Scott area, but was less prominent in samples from the western Flinders Ranges.

Directional data are summarized in Tables I.1, I.2 and I.3. Some of the

TABLE I. 4

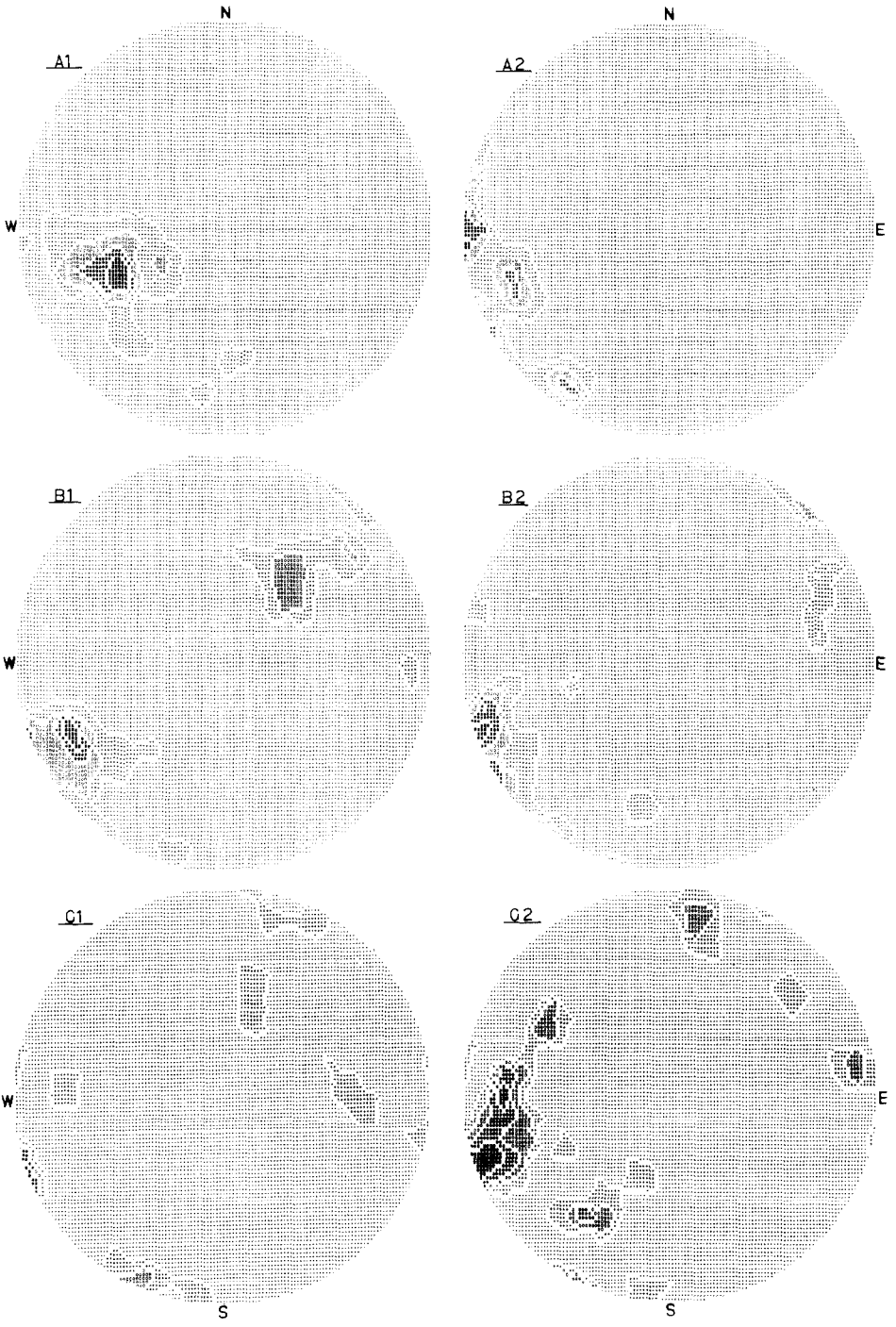
FLINDERS RANGES SUMMARY MAGNETIC PROPERTIES

	RECENT FIELD COMPONENT			"DELAMARIAN" OVERPRINT			PRIMARY COMPONENT				
FORMATION	INITIAL INTENSITY %	COMPLETE BREAKDOWN TEMPERATURE	REMARKS	POLARITY	BLOCKING RANGE	REMARKS	POLARITY	BLOCKING RANGE	REMARKS	VERY RAPID SUSCEPTIBILITY INCREASE	BAD SEPARATION RFC AND PRIMARY
PANTAPINNA Fm	25-60%	400-500°C		N+R	400-670°C	Pronounced	N+R	500-670°C	Weaker	Fairly constant	
BALCORACANA Fm	30-60%	400-500°C		R	630-670° (450-500°)	APNA, APNC Some samples	R+N	630-680°C (450-500)	APNB, APND Some	Above 400°C	
MOODLANTANA	50-60%	300°C	Red Hill Well, Balcoracana Creek	R	630-670°C	Pronounced	R+N	630-670°C	Weak	Above 400°C	
Fm	70-80%	300°C	Mount Billy Creek	N+R	610-670°C	Weak	N+R	630-670°C	Evident	Above 400°C	
ARDONA CREEK Lst	60-90%	250-400°C		N+R	250-650° (450-680°)	Some samples	N+R	250-650° (250-400°)	Some samples	Above 400°C	
WIRREALPA Lst	70-90%	230-260°C		Not pronounced Some indication of N polarities at Mount Billy Creek			R	230-450° (350°)	Some samples	Above 400°C	
BILLY CREEK Fm											
top member	40-70%	400°C		N+R	400-600°	At Mount Billy Creek only	N+R	500-600°C		Decrease above 400°C	
basal member	40-70%			N+R	Not pronounced		N+R	>600°C		Above 400°C	
basal dolomitic limestone beds	80-90%	230°C			Not pronounced		R	230-460°C		Above 460°C	
AJAX Lst	40-100%	200-275°C	In lower part, only component present	N+R	200-675°	Only in upper part	N+R	250-680°C	Only in upper half	Above 560°C	Lower p
ORAPARENNA SHALES	60-90%	250°C		N	250-480°	AORA, AORB	R+N	250-480°C	Mainly AORA	Above 480°C	AORB
PARARA Lst	80-90%	250°C			Not pronounced		R	250-480°C	APAG only	Above 400°C	APAE, AP
WILKAWILLINA Lst	80-90%	250°C		N	250-480°	AWIA only	R	250-480°C	AWIA, AWIB, AWIC	Above 480°C	AWID, AWI ADAB, AP
PARACHILNA Fm	40-95%	400°C			Not pronounced		R+N	600-675°C		Decrease above 400°C	

main magnetic characteristics of the various formations, and also those characteristics which need special comment are described below. Other characteristics, especially those concerning the RFC, are presented in Table 1.4.

Parachilna Formation (Parachilna Gorge)

A characteristic magnetic component (A) was gradually broken down (Fig. I.5A, B) with an evident blocking temperature range between 600°C and 675°C (Fig. I.6A). The susceptibility decreased noticeably at temperatures above 400°C.


A1
N
W
A2
N
E
B1
W
B2
E
C1
W
S
C2
E
S

Hawker Group (Bunyeroo Gorge)

In limestone samples of the Wilkawillina facies, (sites AWIA—AWIC and APAG) and also in site AORA of the Oraparina Shale, an A-component (reversed polarity) was separated at temperatures between 250° C and 480° C (Figs. I.4A, I.5C). Excessive build-up of VRM associated with a very rapid increase in susceptibility prevented further thermal treatment. A particular difficulty with the Parara-type limestones was an interpreted composite breakdown at constant ratio of the slight remaining RFC and the A-component at temperatures above 250° C. This resulted in an intriguing pattern of pseudostable directions which had to be rejected (Table I.1).

Ajax Limestone (Mount Scott area)

In the lower half of the sequence, a RFC was the only component (Fig. I.4, I.5D) which could be separated up to 680° C. In addition to this RFC, an A- and a B-component of both normal and reversed polarity were more dominantly present in the upper half of the section (Fig. I.4C, I.5E, F). The RFC then became completely broken down between 200° C and 450° C, with the A- and B-components broken down separately at higher temperatures up to 680° C. Individuality of the specimens was very apparent and no distinct temperature intervals could be distinguished for the A- and B-components. In a few samples only, an A- and a B-component were simultaneously present, the B-component was then broken down at the lower temperatures up to 600° C and the A-component at the higher temperatures. Mostly, however, only either an A- or a B-component could be isolated, with different specimens from the same sample containing different components. All samples showed a gradual breakdown of the characteristic magnetic component and no evident blocking range was visible (Fig. I.6L). Susceptibility increased very rapidly above 500° C (Fig. I.6V), but build-up of VRM was not so strong that it prevented further measurement, as was generally the case with the non-reddish carbonates.

Fig. I.4. Frequency distribution of magnetic directions in equal area projection, counted with a 1% circle. See description under caption to Fig. I.3. A1. Primary and pseudostable magnetic directions from the Wilkawillina Limestone, upper hemisphere projection, maximum frequency = 11. A2. As above, lower hemisphere projection, maximum frequency = 3. B1. Primary magnetic directions from the Aroona Creek Limestone, upper hemisphere projection, maximum frequency = 7. B2. As above, lower hemisphere projection, maximum frequency = 5. C1. Primary magnetic directions from the Ajax Limestone upper hemisphere projection, maximum frequency = 4. C2. As above, lower hemisphere projection, maximum frequency = 4.

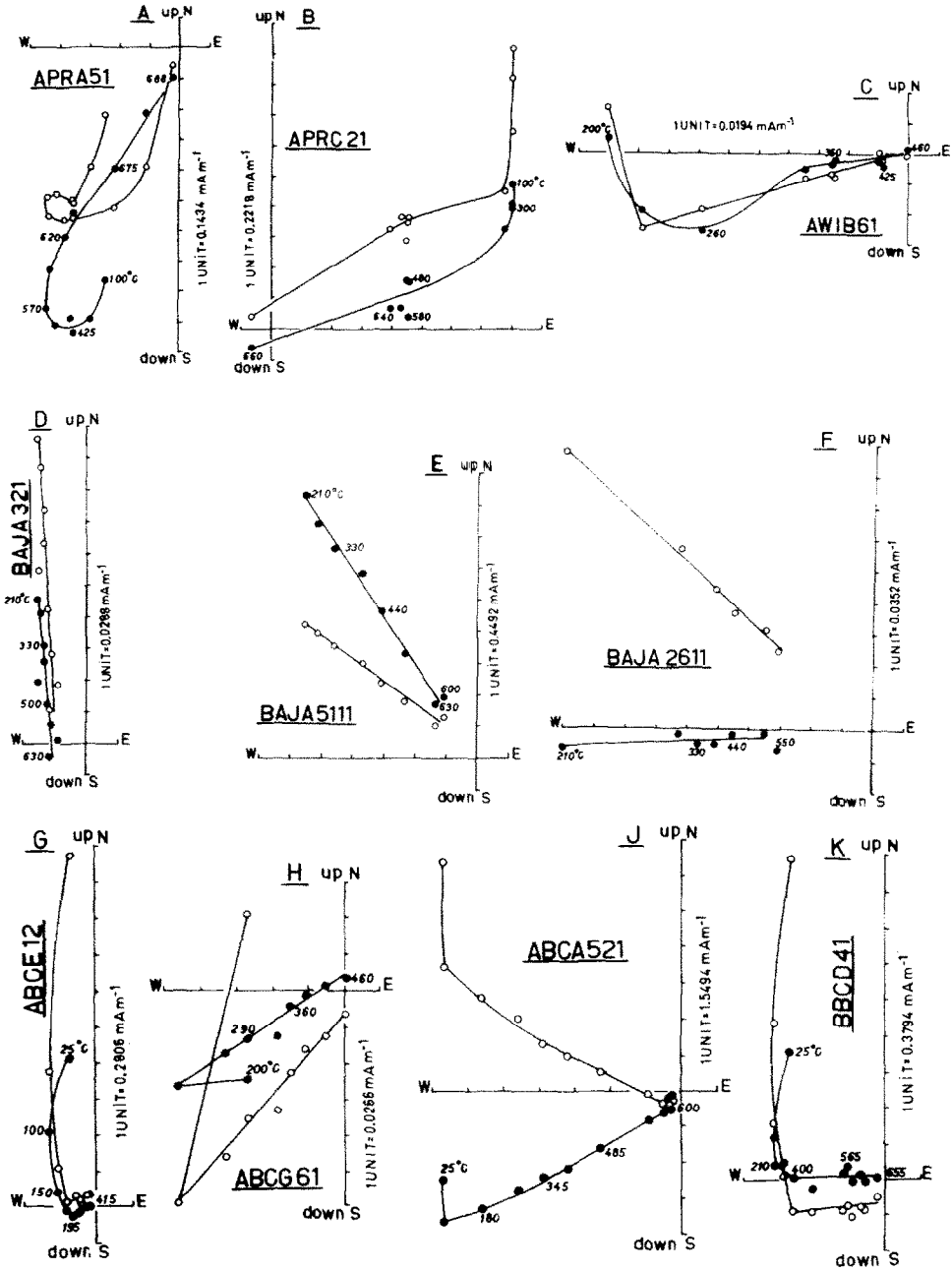
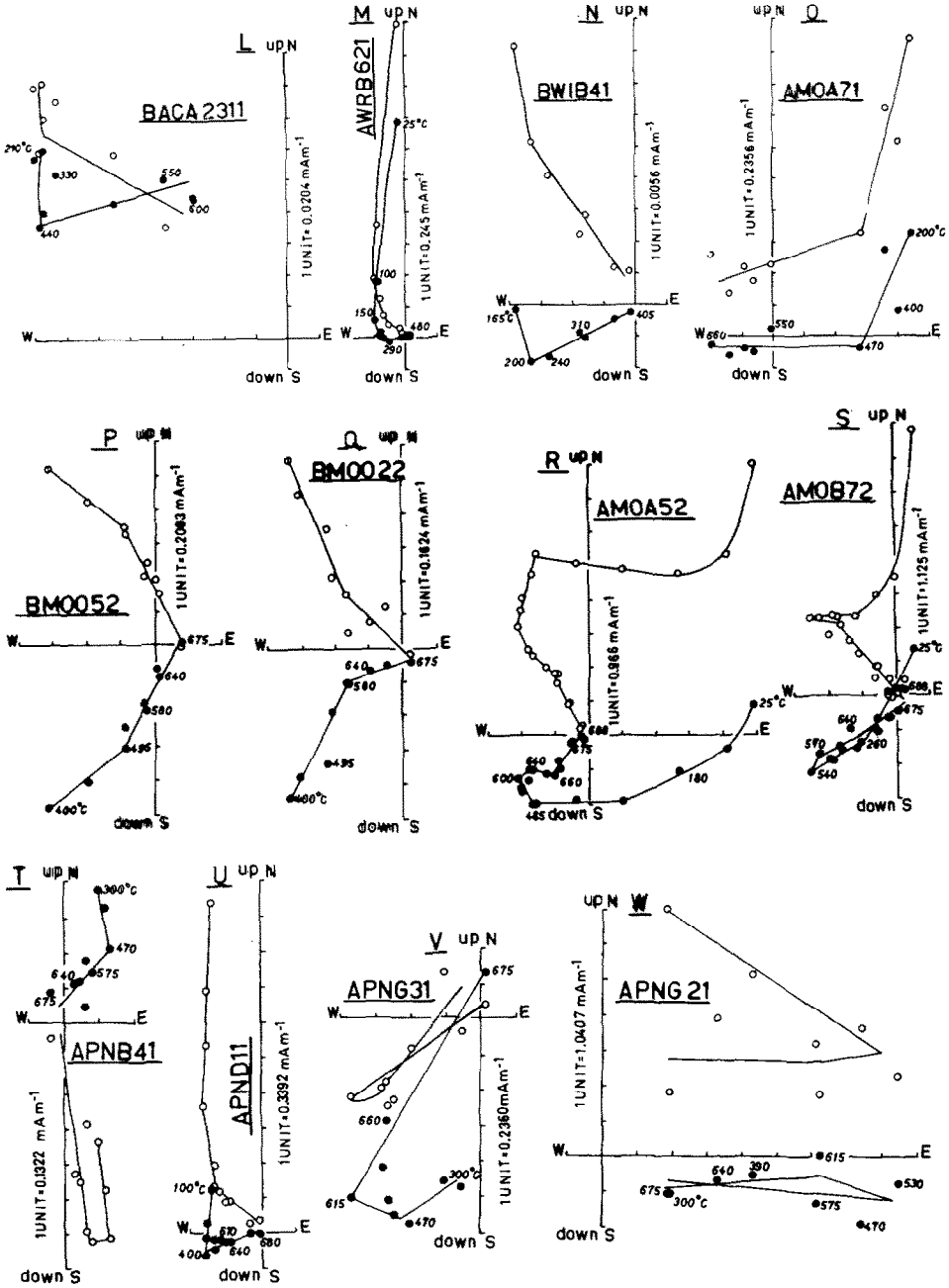


Fig. I.5. Demagnetization diagrams of specimens treated by thermal methods. The points denote successive positions — in orthogonal projection — of the end of the resultant magnetization vector during progressive demagnetization. Open circles denote projections on the vertical east—west plane. Full circles denote projections on the horizontal plane. Numbers denote successive peak values of the applied temperatures. Magnetic com-



ponents shown: 1 = recent field component (RFC), 2 = secondary Delamarian component (B), 3 = primary component (A), N = normal polarity, R = reversed polarity. A = 1N + 3R, B = 1N + 3N, C = 1N + 3R, D = 1N, E = 2R, F = 3R, G = 1N + 3R, H = 1N + 3R, J = 1N + 3R, K = 1N + 3R, L = 1N + 3R, M = 1N + 3R, O = 1N + 3R, P = 3N, Q = 3N, R = 1N + 3N + 3R, S = 1N + 3N + 3R, T = 1N + 3R, U = 1N + 3R, V = 3N + 3R, W = 3R + 3N. Results are not corrected for bedding.

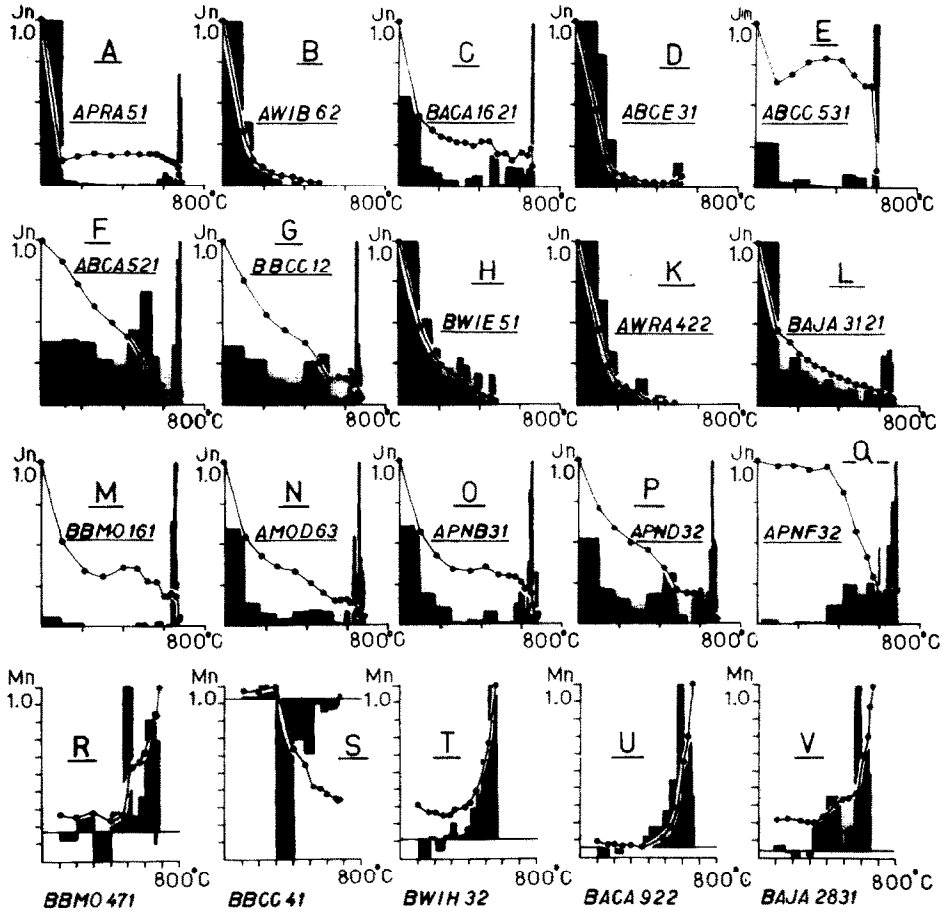


Fig. I.6. A—Q. Normalized intensity decay curves of the remanent magnetization during thermal treatment. The blocking temperature spectra are shown also. R—V. Normalized intensity decay curves of the induced magnetization showing the increase in susceptibility after higher temperatures. The histograms show the stepwise increase (decrease).

Billy Creek Formation (Bunyeroo Gorge, Mount Billy Creek)

(a) *Basal dolomitic limestone* (Bunyeroo Gorge). An A-component of reversed polarity was separately and gradually broken down from about 260° C till 460° C (Fig. I.5G, H). Excessive VRM build-up and strong increase in susceptibility became apparent only at more elevated temperatures and did not prevent the determination of the A-component.

(b) *Red-bed sequence* (Bunyeroo Gorge, Mount Billy Creek). An A-component of both normal and reversed polarity was broken down separately above 400° C (Fig. I.5J, K). Samples from the more sandy top member showed two distinct blocking temperature ranges, one around 500° C and

another above 600° C (Fig. I.6F, G). Only the latter blocking temperature range was present in the samples from the basal member with the higher carbonate content (ABCC, ABCD, Fig. I.6E). A notable exception were site ABCB samples which did not show any evidence of a distinct blocking temperature range. Therefore, these internally very consistent but aberrant directions have been discarded.

A B-component was not identified in samples from the Bunyeroo Gorge section but this component was present in some specimens from the Mount Billy Creek section, and became broken down gradually between 400° C and 680° C. In case both an A- and a B-component were present in the same specimen, the A-component then became broken down below 600° C and the B-component above 600° C. The susceptibility of samples from the sandy top member generally decreased above 400° C (Fig. I.6S), whereas the samples from the more calcareous basal member showed an increase in susceptibility at the higher temperatures.

Wirrealpa Limestone (Bunyeroo Gorge, Red Hill Well, Mount Billy Creek)

In samples from the Bunyeroo Gorge (AWRA, AWRB) only an A-component of reversed polarity was present, which became broken down between 260° C and 450° C (Fig. I.6K). Samples from site AWRC sampled at Red Hill Well, and also samples from a basal bank at Mount Billy Creek (BWII) revealed well determinable and internally consistent directions, but these were distinctly aberrant from the overall B-component direction of the Wirrealpa Limestone. These directions were discarded, although the reason for this aberrancy remains obscure. The remaining ten sites of the Wirrealpa Limestone at Mount Billy Creek (BWIA—BWIK) all showed well-determinable A-component directions of reversed polarity (Fig. I.5N), which were broken down between 260° C and 450° C (Fig. I.6H, T). There is some indication for a minor blocking temperature range around 350° C.

Aroona Creek Limestone (Mount Scott area)

An A-component and/or a B-component, each with both normal and reversed polarity, became broken down between 400° C and 650° C (Fig. I.4B, L, C). When both components were present, the A-component was broken down generally between 250° C and 400° C and the B-component between 450° C and 650° C. There was no distinct blocking temperature range noticeable (Fig. I.6C), apart from some weak indications between 350° C and 400° C.

Moodlatana Formation (Red Hill Well, Balcoracana Creek, Mount Billy Creek)

In the Red Hill Well sites (AMOC, AMOD) a B-component of reversed polarity became broken down at the higher temperatures up to 670° C (Fig.

I.6N). Only a few specimens from these sites showed the breakdown of the A-component of reversed polarity in the same interval. The Balcoracana Creek sites (AMOA, AMOB), in contrast, showed a pronounced A-component, which became broken down between 500° C and 675° C (Fig. I.5O, R, S,), but a B-component was not well identifiable. Both normal and reversed polarity directions of this A-component occur in the same specimen (Fig. I.5R, S).

The Moodlatana samples from Mount Billy Creek revealed an A-component only (Fig. I.5P, Q) of reversed polarity (through-out) and normal polarity (middle part only), which became broken down down above 400° C. All samples showed a distinct blocking temperature range between 630° C and 670° C with some indication for another less pronounced blocking temperature range around 500° C (Fig. I.6M, R).

Balcoracana Formation (Balcoracana Creek)

An A-component of normal (APNB) or reversed polarity (APND) was broken down separately above 500° C up to 680° C (Fig. I.5T). Site APNA and APNC samples showed well concentrating directions of reversed polarity only, broken down in the same temperature interval (Fig. I.5U), which streaked towards the B-component directions as determined from other formations. All samples revealed a distinct blocking temperature range between 630° C and 680° C and another less pronounced one at 450—500° C (Fig. I.6O, P).

Pantapinna Formation (Balcoracana Creek)

A- and B-components both of normal and reversed polarity occur (Fig. I.5V, W), the latter being the more prominent. The A-component generally becomes broken down at the higher temperatures in the range 500—670° C (Fig. I.6Q). In some specimens both a normal and reversed polarity of the B- or A-component were present. Most samples showed a fairly wide blocking temperature spectrum between 400° C and 670° C (Fig. I.6Q).

INTERPRETATION

A major problem during thermal analysis of especially the carbonate specimens was excessive increase in susceptibility and in VRM (Fig. I.6T, U, V,) at temperatures well above 400° C. This virtually prevented analysis of the data after heating above 450° C despite magnetic shielding.

As all heatings were done in air, this susceptibility and VRM increase is most probably due to production of new magnetic phases, e.g., magnetite, maghaemite and haematite, resulting from such mineralogical changes as (a) dehydration of hydrated iron oxides (Hedley, 1968; Strangway et al., 1968; Vlasov and Gornushkina, 1973; Collinson, 1974; Ryabushkin, 1976; Elston

and Bressler, 1977); (b) oxidation of pyrite (Challis, 1975) which might be a dominant process in the carbonates deposited under reducing environments (Parara Limestone); (c) decomposition of iron silicates (Dunlop, 1972; Larson and Walker, 1975).

It is well known (Dunlop, 1972) that newly formed magnetic phases may acquire a magnetization in absence of an ambient magnetic field, because of exchange coupling with the original magnetic phase or because of a resultant magnetostatic interaction field from nearby grains. In case of carbonate samples with a weak initial intensity which are subsequently further reduced during the demagnetization process, formation of such a secondary magnetic component may make it impossible to determine the primary magnetic component. Most red-bed samples showed an increase in susceptibility (Fig. I.6R) after prolonged heating at temperatures above 500—600°C. A notable exception were red-bed samples from the Parachilna Formation and from the top member of the Billy Creek Formation (Fig. I.6S, Table I.4), which showed a significant decrease in susceptibility after heating above 400°C. This may be due to inversion of maghaemite to haematite (Turner and Ixer, 1977).

The individual behaviour of the specimens during thermal demagnetization was variable (Fig. I.5). It was often found that specimens taken from the same sample showed widely different blocking temperature regions for magnetic components of similar direction. This may partly be due to initial presence of an intimate mixture of authigenic magnetic minerals (Larson and Walker, 1975) further enhanced by a large variability in thermomagnetic properties of the transformed phase upon oxidation (Ryabushkin, 1976). Therefore, I prefer to treat the specimen directions as the fundamental unit for calculation of mean directions, rather than basing this calculation on sample-mean directions.

As shown before, two characteristic magnetic components have been distinguished: (a) a SSW directed component (A) with medium to low upwards inclination, and (b) a WNW—NNW directed component (B) with a low downwards inclination. Normal and reversed polarities of both components occur. A feature observed throughout this series of studies (Sections I, II, III) was the appreciable scatter in individual specimen directions, representing one of the two characteristic magnetic components. Rather than attributing this phenomenon to inaccuracies in determining individual specimen directions, this scatter has to be seen as an intrinsic property of the samples, reflecting a complex magnetization history. Although orthogonal projection figures are a very powerful tool in interpretation, such an analysis necessarily fails in case of a multi-component magnetic system with partly overlapping stability spectra. Clearly, composite breakdown of multicomponent magnetizations with variable component ratios throughout a formation, necessarily (1) contributes to an enhanced scatter in directions and (2) yields streaks between the end directions of the various components. In the present studies (Sections I, II, III) this effect results in a non-Fisherian distribution

of specimen directions with most of the spread residing in the declination. The spreads of directions of the two characteristic magnetic components overlap. Because of the distinct behaviour of each specimen it was virtually impossible to group the component directions according to their magnetic characteristics. The best alternative was to subject the directions of individual specimen components to frequency analysis and to judge mean component directions from peaks evident in the frequency distribution. Component directions of the individual specimens were then grouped on the basis of their proximity to one of the dominant directions. Admittedly, this is a somewhat subjective approach which may statistically enhance the separation, yielding results farther apart than they are in actuality. However, being the sole option available, the method tends to be justified as the obtained mean directions could be interpreted in a meaningful way with respect to the mean early Palaeozoic palaeomagnetic data from Gondwanaland and the tectonic and thermo-chemical history of the region studied.

The occasional presence of both normal and reversed polarities of the same magnetic component in the same specimen (Fig. I.7A), points to a fairly long period of acquisition of the characteristic magnetic component spanning at least the period of one polarity reversal. Such a prolonged acquisition of magnetization may be due to the formation of authigenic oxyhydroxides and haematite during diagenesis, and later dehydration and oxidation due to changes in Eh—Ph conditions and/or tectonic and thermo-chemical effects. For instance, Dijksman (1977) observed in Miocene red clay beds from central Spain that the same magnetic direction resided in both magnetite and a haematite magnetic phase. The latter was acquired with a time lag of several 100.000 yrs due to the effect of changing Eh—Ph condition in the CRM formation process.

Dual polarity observations were also made in Billy Creek equivalent beds on Kangaroo Island and in some sequences in the Amadeus Basin (Section III). In the latter basin, specimens from red beds in a Middle Cambrian section near Areyonga showed components of dual polarity, whereas time equivalent rocks of a carbonate sequence near Ross River did not show this effect. Clearly, such a prolonged period of acquisition of magnetization, leading occasionally in red beds to a dual polarity, questions the suitability of such thermally demagnetized red beds results for magnetostratigraphic studies. Chemical demagnetization may prove to be more successful in those cases where early and late diagenetic magnetization components can be separated on the basis of distinct chemical properties (Roy and Park, 1974), but has not been undertaken in the course of the present study.

A comparison with the mean Gondwanaland data points to the SSW—SW directed magnetic component (A) as the primary magnetization. Its direction shows a minor gradual change through the stratigraphical sequence as will be discussed (Table I.1). I interpret the WNW—NNW aligned characteristic magnetic component (B) as a secondary magnetization which predates regional folding of the Adelaide "Geosyncline" sediments during the

Late Cambrian—Early Ordovician Delamarian Orogeny. This is evident from the positive fold test upon comparison of bedding corrected mean directions (Table I.2) from west dipping beds in the western Flinders Ranges (Hawker Group, Billy Creek Formation, Wirrealpa Limestone) with east dipping beds from the eastern Flinders Ranges (Billy Creek Formation) and with south—west dipping beds from the Mount Scott area (Ajax Limestone, Aroona Creek Limestone). Acquisition of this B-component most probably resulted from enhanced chemical activity, operating at slightly elevated temperatures. An early thermal phase predating the main deformative event of the Delamarian Orogeny is well reflected in pre- to syntectonically emplaced granites in the Mount Lofty Ranges and on southern Kangaroo Island during the Late Cambrian (±510 m.y.; Milnes et al., 1977), and in pre-deformative new growth of metamorphic minerals (Fleming and Offler, 1968; Offler and Fleming, 1968). Closing dates of the isotopic systems suggest that such an enhanced heat flux might have continued in the southern extreme of the Adelaide "Geosyncline" for about 50 m.y. into the Late Ordovician.

The interpretation of this B-component as a thermo-chemically induced secondary magnetic component of Late Cambrian—Early Ordovician age is further supported by observations of regional changes in metamorphic grade of especially the late Proterozoic Tapley Hill beds in the Adelaide "Geosyncline" by McKirdy et al. (1975). Their observations indicate a general increase in metamorphic grade from west to east across the "Geosyncline", with most evident metamorphism in the southern part of the Adelaide "Geosyncline" and in the northern Flinders Ranges, and a apparent absence of metamorphism in the central Flinders Ranges.

Particularly interesting is the magnetic anomaly pattern of the Tapley Hill beds, attributed by them to metamorphic generation of magnetite from haematite (Dunlop, 1972). From a detailed interpretation of the magnetic anomalies (Tucker, 1972), two stages of acquisition of secondary remanent magnetization were identified. Before folding, the beds acquired a remanent magnetic component close to the plane of sedimentary layering in a direction 135° ± 10. After folding another secondary magnetic component was imposed, again close to the bedding but in a general direction normal to the trace of outcrop. The agreement both in time of acquisition and in direction of McKirdy et al.'s (1975) earlier mentioned secondary magnetic component with the characteristic component B is striking. The other secondary magnetic component described by McKirdy et al., although not well recognizable in the present study, may have further attributed to the appreciable scatter in directions of the characteristic magnetic components. Although the grade of metamorphism in the central Flinders Ranges proved to be beyond the resolving power of McKirdy et al.'s method, the present study indicates the general trend of an increase in metamorphic grade from west to east over the "Geosyncline" also to be present in the central Flinders Ranges. The secondary magnetic component B was hardly noticeable in the sections from the western side of the central Flinders Ranges (Bunyeroo Gorge, Parachilna

Gorge), but appeared the more evident at the eastern side of the central Flinders Ranges (Balcoracana Creek, Mount Billy Creek) and in the northern Flinders Ranges (Aroona Creek).

As will be elaborated in the following sections (II, III), similar overprints are present also in Billy Creek equivalent red beds on Kangaroo Island and in Middle—Late Cambrian sequences from the Amadeus Basin. In the latter region, there are no indications for Late Cambrian—Early Ordovician orogenic activity (Wells et al., 1970; Wells, 1976), so the evident magnetic overprinting has to be assigned to an as yet unnoticed thermo-chemical event contemporaneous with the one which affected the Adelaide "Geosyncline".

Finally, the primary magnetic component A needs some discussion. The few earlier studies on Cambrian palaeomagnetism of the Adelaide "Geosyncline" (Briden, 1967; Giddings, 1974; Embleton and Giddings, 1974; Giddings and Embleton, 1976) also point to problems similar to those encountered in the present study: individuality of the specimens upon thermal treatment, remagnetization during the Late Cambrian—Early Ordovician times and frequently a high scatter in directions. This prevented Briden (1967) from obtaining conclusive results for the Billy Creek Formation. Results mentioned by Embleton and Giddings (1974) for red-bed sequences of late-Early Cambrian age (Billy Creek Formation), Middie Cambrian age (lower Lake Frome Group) and of a presumed Late Cambrian age (upper Lake Frome Group, probably the Grindstone Range Sandstone) show some similarity to the results obtained during this study. However, the difficulty in removing the thermo-chemically overprinted component B, not separated by them, and the large scatter in the few data that remain after rejections, urge a careful attitude to the numerical data presented in these studies.

The mean data for the primary magnetization of the Cambrian sequence of the Flinders Ranges as obtained in this study shows some gradual change in direction (Figs. I.7A, B, C) and in pole position (Fig. I.8). The bulk of the polar motion recorded in the Flinders Ranges sequences must have taken place in Early Cambrian times, but unfortunately acceptable data from the Early Cambrian Hawker Group (Table I.1) are rare. The few low accuracy data from the earliest Cambrian Parachilna Formation and from the basal Wilkawillina Limestone, supported by the Ajax Limestone results, indicate a gradual move in pole position away from the Late Precambrian—Early Cambrian quasi-static pole position as recorded in Amadeus Basin sequences (Kirschvink, 1978a, b), towards a more stable Middle Cambrian pole position as indicated by the Billy Creek red beds, the Wirrealpa Limestone—Aroona Creek Limestone sequences and the Lake Frome Group red beds. This pattern of movement accords with the Middle—Late Cambrian data from the Amadeus Basin and from the Billy Creek equivalent of Kangaroo Island (Sections II, III), and fits with the general trend of a revised APW path for Gondwanaland (cf. Figs. III.13, 15).

The mean result (Table I.1) for carbonates of the top Hawker Group

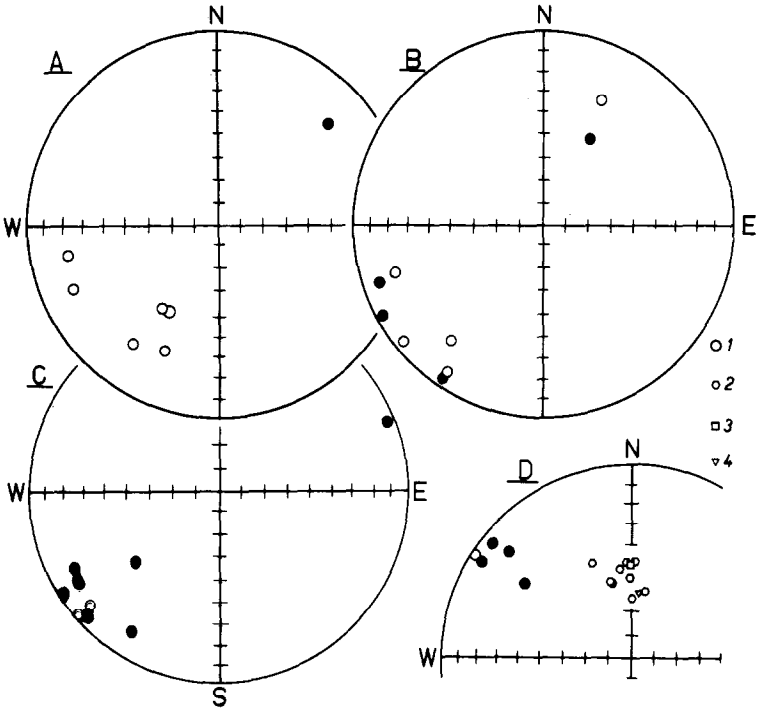
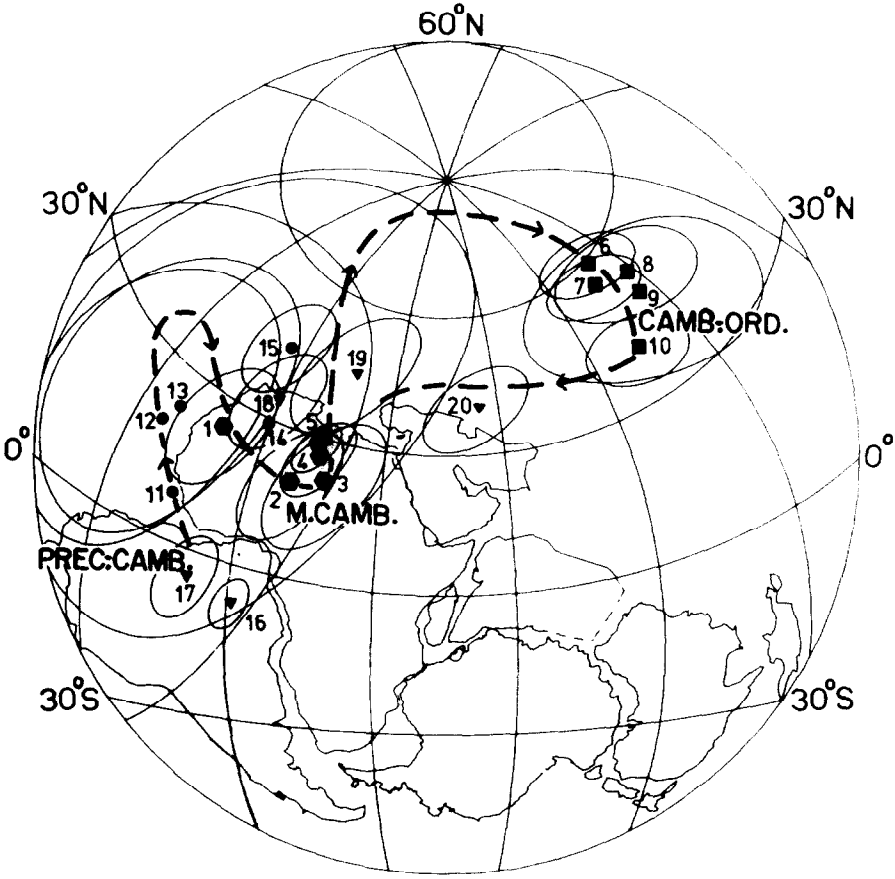


Fig. I.7. Equal-area projection of mean directions (*1*) from the Early Cambrian Hawker Group and the Ajax Limestone. B. As above, for the late-Early—early-Middle Cambrian Billy Creek Formation, Wirrealpa Limestone and Aroona Creek Limestone. C. As above, for the Middle? Cambrian Lake Frome Group. D. Mean secondary "Delamarian" directions from various localities, *2* = Mean recent field directions from various localities, *3* = virtual direction at the sampling locality according to the axial geocentric dipole model, *4* = present local field direction.

(sites APAG, AORA) and the basal Billy Creek beds (sites APCE, ABCF, ABCG), is notably out of sequence. The pole position seems to be deflected towards the Late Cambrian—Early Ordovician pole position corresponding with the overprinted B-component. It is quite possible that this result represents a pseudo-stable component due to a composite breakdown of the SW directed primary component (A) and the NW directed secondary component (B). Like the results from the underlying carbonate beds of the middle Hawker Group, which show composite breakdown of a primary magnetization component (A) and a recent field component (RFC) (Table I.1), this result has been discarded.

Another notable observation is the southward deflected declination of the primary magnetic component in the upper Wirrealpa Limestone—lower Moodlatana succession sampled at Mount Billy Creek (Table I.1). The corresponding pole positions (Fig. I.8) are near to late Precambrian pole positions



FLINDERS RANGES RESULTS

Fig. I.8. Newly obtained pole positions for the Cambrian of the Flinders Ranges, superimposed on a revised early Palaeozoic polar wander path for the combined Gondwanaland continents. (Fig. III.13). Hexagons: Supposedly well determined Middle Cambrian positions, *1* = mean basal Hawker Group pole, *2* = mean Billy Creek Formation-Wirrealpa Limestone-Aroona Creek Limestone pole, *3* = Pantapinna Sandstone pole, *4* = combined Lake Frome Group pole, *5* = basal Lake Frome Group pole. Squares: Mean pole positions corresponding with the "Delamarian" overprint as obtained from various localities, *6* = Red Hill Well, *7* = Aroona Creek, *8* = Mount Billy Creek, *9* = Bunyeroo Gorge, *10* = Balcoracana Creek. Full dots: less well determined pole positions from the basal Hawker Group, *11* = Parachilna Formation, site APRA, *12* = Wilkawillina Limestone, site AWIC, *13* = Idem, site AWIA, *14* = Parachilna Formation, site APRC, *15* = Wilkawillina Limestone, site AWIB. Inverted triangles: aberrant pole positions from part of the Mount Billy Creek section, *16* = basal sampled part of the Moodlatana Formation, *17* = top part of the Wirrealpa Limestone, *18* = basal part of the Wirrealpa Limestone, site BWIK, *19* = mean pole position from the Hawker Group, *20* = mean pole position from the top part of the Hawker Group only.

as obtained by Kirschvink (1978a, b) for the Amadeus Basin (cf. Figs. III.13, 14). The origin of this well observed aberrant set of directions is not clear, they may possibly represent an excursion of the geomagnetic field.

The upper age limit of the Lake Frome Group sedimentation is not very well established. According to Daily (in Thompson et al., 1976), sedimentation in the Adelaide "Geosyncline" ceased with the onset of the Delamarian Orogeny. Daily (1956) originally suggested a Middle- to possible Late Cambrian age for the topmost beds of the Lake Frome Group, successively revised (Daily, in Thompson et al., 1976) to a possible continuation of sedimentation into the Early Ordovician. A comparison of Lake Frome Group Data here presented with data of Giddings (1974) and Embleton and Giddings (1974) for the upper Lake Frome Group beds, with Middle to Late Cambrian results for the Amadeus Basin (Section III) suggests that acquisition of magnetization of the Lake Frome Group beds pre-dated acquisition in the upper part of the Shannon Formation of the Ross River section. The latter Shannon Formation is of early-Late Cambrian age (Wells et al., 1970). In the absence of indications that the acquisition of the primary magnetization of the Shannon beds lagged appreciably behind sedimentation, there is thus fair reason to assume that the Lake Frome Group sedimentation did not continue into Late Cambrian times.

CONCLUSIONS

A major problem encountered in this study was the directional interpretation of individual magnetic components. Aberrant directions present, in particular, in carbonate rocks of the Early Cambrian Hawker Group, have been interpreted as pseudo-stable directions caused by composite breakdown of various magnetic components.

Three magnetic components have been indentified:

(1) A well recognizable recent field component (RFC) of appreciable intensity.

(2) A WNW—NNW directed secondary magnetic component (B) of low downwards inclination.

(3) SSW—SW directed component (A) of medium to low upwards inclination, interpreted as representing the primary magnetization.

Both latter components were present as normal and reversed polarity directions. Because of the distinctive behaviour of the specimens during thermal demagnetization treatment and the appreciable scatter in component directions, a frequency analysis had to be applied for assignment of component directions to either or both groupings.

The secondary magnetic components (B) is interpreted as resulting from thermo-chemical activity during a period of regional upheating pre-dating the diastrophistic phase of the Cambro-Ordovician Delamarian Orogeny.

The primary magnetic component indicates reduced polar motion during the Middle Cambrian (quasi static interval). The few early Cambrian data

available suggests a period of rapid polar motion, during which the pole moved from its Late Precambrian—Early Cambrian quasi-static position to its Middle Cambrian quasi-static position.

A comparison of Lake Frome Group palaeomagnetic data with newly obtained and well dated results from the Amadeus Basin indicates that the Lake Frome Group sedimentation probably did not continue into Late Cambrian times.

II. Late Early Cambrian results from Kangaroo Island, South Australia

INTRODUCTION

The peculiar shape of the Adelaide "Geosyncline" (cf. Fig. I.1), sigmoidal with an added northwards spur of the Flinders Ranges has inspired a number of widely different genetic models.

Crawford and Campbell (1973a) suggested possible large-scale horizontal displacement within the Australo-Antarctic assembly during Late Cambrian and mainly Ordovician times. They postulated two deformation phases to explain the southwestern bend of the Adelaide "Geosyncline", supposed by them to have been aligned originally in a meridional direction. Compression during the Late Cambrian caused by movement of the Gawler Block, followed in Early Ordovician times by large scale fault drag of Kangaroo Island and the Mount Lofty Ranges. This drag was caused by at least 300 km of dextral movement along a postulated NW—SE aligned shear zone, passing from the southern extreme of the Gawler Block along northern Tasmania. By contrast Harrington et al. (1973), supported by Veevers (1976), postulated that sinistral and not dextral movement along a rather similar fault zone occurred during the middle Palaeozoic, followed by dextral movement up to 900 km in the late Mesozoic. Harrington et al. agree with convicing arguments put forward by Daily et al. (1973) that the southwestern bend of the Adelaide "Geosyncline" is not due to fault drag, but represents primary depositional trends, around the margin of the stable Gawler Block. Likewise, large-scale rotations in the northern part of the Adelaide "Geosyncline" also postulated by Crawford and Campbell (1973a) were refuted by Glen et al. (1977) on the grounds of similarity in deformation trends between the Willyama block and the Gawler Block. Scheibner (1974), like Crawford and Campbell (1973a, b) was less impressed by the stability of the Gawler Block, and suggested deformation of the southern Adelaide "Geosyncline" by a southeasterly rotation of the Gawler Block during the Late Cambrian—Early Ordovician Delamarian Orogeny. Other models put forward by Stewart and Mount (1972), Katz (1976), Dunnet (1976), Coward (1976) and Stewart (1976) suggest formation of the Adelaide "Geosyncline" through horizontal displacements for which various magnitudes, various shear zones and various timings of movement are advanced. Stewart and Mount (1972), Coward (1976) and Stewart (1976) in particular, suggest a

relation between a shear zone in the Adelaide "Geosyncline" and a zone of seismic activity (Doyle et al., 1968; Doyle, 1971; Clearly and Simpson, 1971; Stewart and Mount, 1972), which follows the Adelaide "Geosyncline" and may be continuable through the Australian continent and southwards into the Indian Ocean, ultimately reflected in the offset of the Antarctic Ridge. From electrical conductivity studies, Lilley (1975) describes an evident conductor zone in the northern Flinders and mentions ductile shear as a possible cause for enhanced electrical conductivity. Although this conductor is displaced eastwards with respect to the zone of seismic activity, it may well be related to the various N—NNW trending shear zones suggested by Stewart and Mount (1972), Coward (1976), Stewart (1976) and Glen et al. (1977).

Palaeomagnetic studies may provide some constraints to the many widely divergent hypotheses for the development of the Adelaide Orogenic zone, with most of them postulating some kind of rotational or translational movements. Small differential movements along a shear zone which may have been slightly oblique to the Cambro-Ordovician palaeolatitude pattern (cf. Fig. III.13) are beyond the current resolution of the palaeomagnetic method, and are also difficult to test on geological ground (Glen et al., 1977). Associated rotations, however, can often be detected palaeomagnetically.

Only a few palaeomagnetic results for the southern part of the Adelaide "Geosyncline" have previously been published. McWilliams (1977) noted a discrepancy between results by Karner (1975) from the Brachina Formation (Wilpena Group, late Proterozoic) collected near Hallett Cove, south of Adelaide and his own data from the Brachina Formation of the central Flinders Ranges, but he was unable to interpret this discrepancy in accordance with existing models for deformation of the Adelaide "Geosyncline". Unnoted by McWilliams, there is a marked similarity between results obtained by Briden (1967) from Marinoan sediments sampled near Hallett Cove (decl. = 191°, incl. = +39°), and a result obtained by McWilliams (1977) for the Yerelina subgroup (decl. = 184°, incl. = +33°) sampled in the northern Flinders Ranges. Such a similarity implies no rotational deformation of the Adelaide "Geosyncline" since the late Proterozoic. McWilliams preferred, however, to interpret hos own results as a secondary overprint.

Briden's (1967) results from a late-Early Cambrian sequence on Kangaroo Island, similar to the sequence studied in this paper, are not useful for a comparison with other areas of the Adelaide "Geosyncline" because of the reported difficulty in isolating the primary magnetization. Giddings and Embleton (1974, 1976) obtained comparable palaeomagnetic directions for Proterozoic dyke systems from Yorke Peninsula and from Eyre Peninsula, and therefore refuted Crawford's hypothesis of large-scale horizontal displacements and associated faultdrag in southern Australia. Crawford and Campbell (1973b) subsequently acknowledged the stability of Yorke Peninsula with respect to the Gawler Block. It can be argued, however, that

Gidding and Embleton's palaeomagnetic results from Yorke Peninsula may not be representative for the Kangaroo Island—Mount Lofty Range segment of the Adelaide "Geosyncline", as these areas are separated by the Torrens Lineament which divides the Gawler Block from the more mobile Adelaide "Geosyncline".

Further palaeomagnetic testing of genetic models for the Adelaide "Geosyncline" is clearly desirable. Well documented data from the Cambrian of the Flinders Ranges (Section I) and from the Amadeus Basin (Section III) have been obtained in the course of this project, and palaeomagnetic research on the Mount Lofty Range—Kangaroo Island arc was initiated as a natural extension to these studies.

The late-Early Cambrian red-bed sequences on northeastern Kangaroo Island is correlated by Daily (in Thompson et al., 1976) on faunal evidence with the Billy Creek Formation of the Flinders Ranges. The former sequence was only mildly affected during the Delamarian Orogeny. Therefore, a study of this sequence has been undertaken and the results are reported here.

GEOLOGICAL SETTING

Shallow-water-type rocks of late-Early Cambrian age, which are correlated with paralic deposits of the Billy Creek Formation of the Flinders Ranges (Fig. I.2) are exposed over the central northern part of Kangaroo Island (Fig. II.1), and are known in the subsurface of adjacent Yorke Peninsula (Stuart and Von Sanden, 1972). The main southern part of the island is made up by metasediments of the Kanmantoo Group and the underlying Precambrian. The Flysch-type Kanmantoo sediments form the deep-water facies equivalent of the shelf type sediments to the north. Deposition of both facies took place in response to the episodic Kangarooian movements (Daily, 1956; Daily and Forbes, 1969; Wopfner in Parkin, 1969), i.e., the Waitpingan subsidence in the Kanmantoo trough and the associated Cassinian uplift in the region far to the northeast.

Recent subsurface data have indicated continuation of this sedimentary basin (Stuart and Von Sanden, 1972; Smith and Kamerling, 1969) along the southern Gawler Block. An eastwards continuation of the east—west trending belt of Kanmantoo Group sediments is well recognizable on Fleurieu Peninsula and in the Mount Lofty Ranges (Daily, 1969). Subsurface data (B.P. Thompson, in Parkin, 1969) suggest the presence of Kanmantoo Group sediments also in the Murray Basin and in western Victoria. From Kangaroo Island and the continental shelf to the south of it, these Kanmantoo Group sediments may continue in WNW direction through the offshore Duntrone basin (Smith and Kamerling, 1969), which follows the southern margin of the Gawler Block. To the south and west of the Gawler Block beds were identified which are tentatively correlated with the Billy Creek Formation and the Lake Frome Group of the Flinders Ranges.

Major fault zones, like the Torrens Lineament, appear to follow the trend

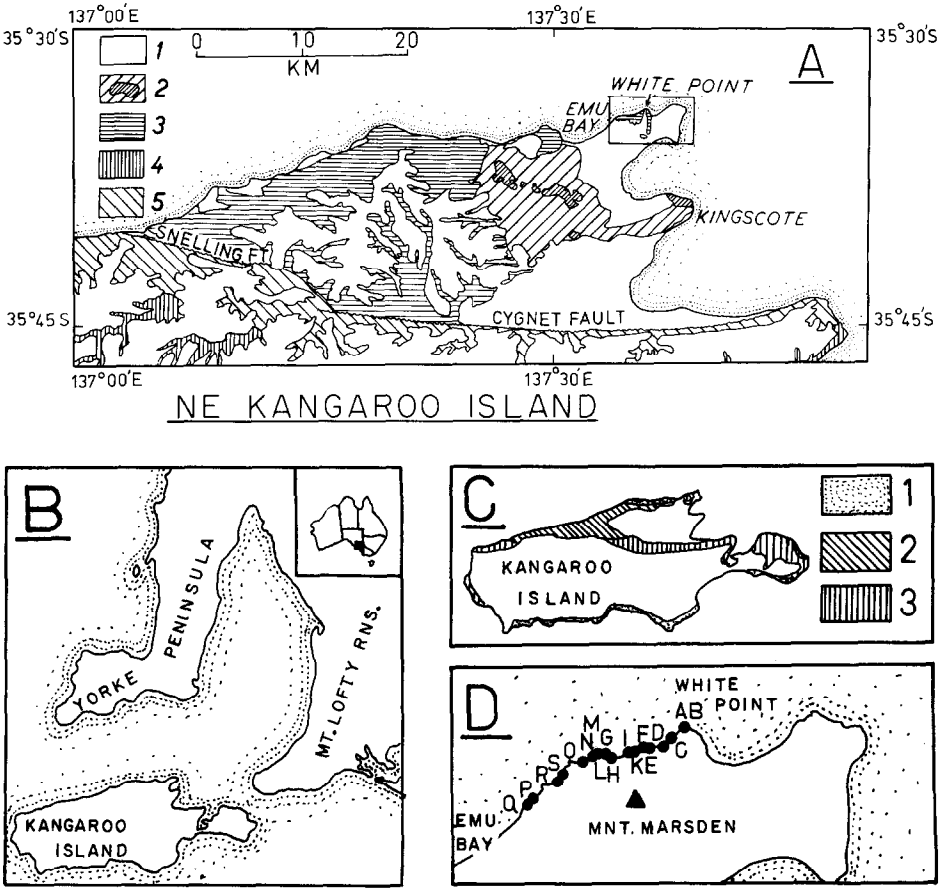


Fig. II.1. A. Geological sketchmap of the northeastern part of Kangaroo Island. *1* = Cainozoic deposits, *2* = Permian glacial till and Jurassic basalt flows, *3* = Cambrian-undeep marine, *4* = Precambrian, *5* = Kanmantoo Group sediments. B. Regional location map. C. Metamorphic zonation of Kangaroo Island after Offler and Fleming (1968). *1* = chlorite zone, *2* = biotite zone, *3* = andalusite—staurolite zone. D. Site locality map.

of this sedimentary basin and seem to sweep around the Gawler Block (Stuart and Von Sanden, 1972; Daily et al., 1973). This contrasts with an earlier interpretation (Tectonic map of Australia and New Guinea, 1971) that the N—S aligned Torrens Lineament abuts against the E—W trending Cygnet—Snelling fault zone on Kangaroo Island (Fig. II.1A). This Cygnet—Snelling fault zone abruptly separates the undisturbed Billy Creek equivalent beds of Northern Kangaroo Island from the Kanmantoo Group metasediments to the south and can be interpreted (Stuart and Von Sanden, 1972) as the westward continuation of a fault zone swinging from the Mount Lofty

Ranges along the north coast of the Fleurieu Peninsula towards Kangaroo Island, and possibly beyond.

The trend of the Cygnet—Snelling fault is well reflected in the metamorphic zonation of Kangaroo Island (Offler and Fleming, 1968, Fig. II.1C) which originated during the Late Cambrian—Early Ordovician Delamarian Orogeny. Metamorphism up to the andalusite—staurolite facies is recognizable to the south of this fault zone only. Lower-grade metamorphism in biotite facies occurs further to the north, whereas the well exposed Cambrian sequence along the north coast has been affected up to the chlorite facies only. Sampling has been restricted, therefore, to this region only (Fig. II.1D).

TABLE II. 1

KANGAROO ISLAND (Location 137.6°E, 35.6°S; samples 108; specimens 203)

SITE	Relative Stratigraphic Level	Formation [x1]	Samples [x2]	Specimens [x2]	Initial NRM Intensity mAm⁻¹	PRIMARY MAGNETIZATION SITE MEAN DIRECTION Decl (degrees)	Incl (degrees)	κ	α95 (degrees)	Polarity	Remarks
AKIA	1190	BBF	2	2	6.4-11.0	253	-13.5	-	-	R	
AKIB	1187	BBF	2	2	5.7-14.0	212	- 3	-	-	N+R	
AKIC	890	BBF	5	7	5.1-9.8	226.5	-24	5.5	27.8	N+R	
AKID	887	BBF	4	5	4.0-8.6	222.5	-24	3.5	42.3	R+N	
AKIE	700	BBF	5	6	4.5-11.5	217.5	-7	7.5	26.4	N+R	
AKIF	632	BBF	2	2	1.6-8.9	201	-12	-	-	N	
AKIG	545	EBS	-	-	0.6-3.2	-	-	-	-	-	Recent field component only
AKIH	526	EBS	-	-	0.9-5.9	-	-	-	-	-	Recent field component only
AKII	485	EBS	1	2	2.8-9.3	204	+17	-	-	N	
AKIK	440	EBS	1	3	2.6-5.0	203.5	+30.5	-	-	R	
AKIL	328	WPC	2	3	2.7-8.9	224.5	-18.5	7.5	47.7	N	
AKIM	324	WPC	5	6	5.1-8.4	257	-15.5	8	25.4	N+R	
AKIN	233	WDC	4	6	5.7-8.7	267	-14	10.5	21.5	N+R	
AKIO	162	WPC	6	8	7.4-13.6	233.5	+33	6.5	23.9	R+N	
AKIP	132	WPC	6	8	8.5-12.3	244	-17.5	8	20.9	N+R	
AKIQ	158	WPC	6	11	7.1-12.3	218	-22	7.5	17.6	N+R	
AKIR	0	SBS	6	8	9.4-12.3	211	-3	5.5	26.2	N	
AKIS	155	WPC	4	5	5.1-6.9	238.5	-17.5	6.5	32.3	N+R	

	MEAN DIRECTIONS N	decl	incl	κ	α95	S-POLE POSITIONS long	lat	dp	dm	POLE NO. FIG. 7
PRIMARY MAGNETIZATION (sites)	16	224.5	-4.4	9.4	12.3	15.1E	33.9S	6.2	12.3	
PRIMARY MAGNETIZATION (specimens)	77	228.7	-10.2	5.3	7	16.2E	28.9S	4.0	7.9	21
"DELAMARIAN" OVERPRINT (specimens)	54	301.0	+21.5	7.6	7.5	75.8E	17.4N	4.2	7.9	22
RECENT FIELD [x3] COMPONENT (specimens)										
N-polarity	59	8.3	-49.4	12.2	5.5					
R-polarity	79	178.2	+53.7	18.0	3.9					
N+R	138	3.1	-51.7	15.6	3.2					

[x1] BBF = BOXING BAY FORMATION; EBS = EMU BAY SHALE; WPC = WHITE POINT CONGLOMERATE; SBS = SMITH BAY SHALE

[x2] Number of samples (specimens), wherein a primary component could be determined.

[x3] Present local field direction; decl = 7°, incl = 68; corresponding pole position according to the axial geocentric dipole for Adelaide 121.7°E - 73.7°S

SAMPLING LOCALITIES

A conformable sequence of late-Early Cambrian strata, about 1500 m thick, is well exposed in the cliff section north of Mount Marsden (Fig. II.1D), between Bald Rock on the eastern side of Emu Bay and White Point at the western extreme of Boxing Bay. A total of seventeen sites (108 block samples) of at least six samples each has been sampled through the lower 1200 m of this sequence (Table II.1). All samples were orientated in the field with both a magnetic and a solar compass, using a device analoguous to that described by Embleton and Edwards (1973).

Four formations have been sampled along this profile (Fig. II.1D, Table II.1):

(A) One site (AKIR) has been collected in the lowermost sampled formation the Smith Bay Shale in a red to purple quartzitic sandstone (AKIR) and micaceous shale sequence which forms the 120 m thick upper member of this 150 m thick formation.

(B) Eight sites (AKIL—AKIQ and AKIS) have been sampled in reddish shale and sandstone intercalations (occasionally quartzitic) in the overlying about 400 m thick White Point Conglomerate.

(C) Another four sites were taken in the conformably overlying Emu Bay Shale Formation, about 100 m thick. Two sites (AKIG, AKIH) were sampled in silicified dark greenish thin bedded micaceous shales, silts and sandstones, which form the upper part of this formation. Two other sites (AKII, AKIK) were sampled in medium to coarse-grained reddish sandstone banks in the basal part of the formation.

(D) Six sites (AKIA—AKIF) were taken through the basal 550 m of the overlying Boxing Bay Formation, in red to reddish feldspathic medium to coarse-grained quartzitic sandstones with evident cross bedding.

LABORATORY TREATMENT AND RESULTS

Treatment in the laboratory was essentially the same as described in the Flinders Ranges study (Section I). At least three cores were taken from each block sample using a drill press. Two specimens from each site were subjected to progressive thermal demagnetization in fifteen steps up to 685° C. Remaining specimens, two from each sample, were subjected to partial progressive thermal demagnetization in nine steps between 300° C and 675° C. Such an elaborate demagnetization procedure was necessary because of the individual behaviour of specimens upon thermal cleaning as had been noted also for comparable rocks of the Flinders Ranges and the Amadeus Basin (Sections I and III, respectively). Care was taken to keep the specimens in low field magnetic environments to prevent excessive build up of viscous remanent magnetization (VRM). All magnetic measurements were carried out on a computerized ScT 2-axis superconducting rock magnetometer, and susceptibility measurements were taken on a Digico bulk susceptibility unit. Direc-

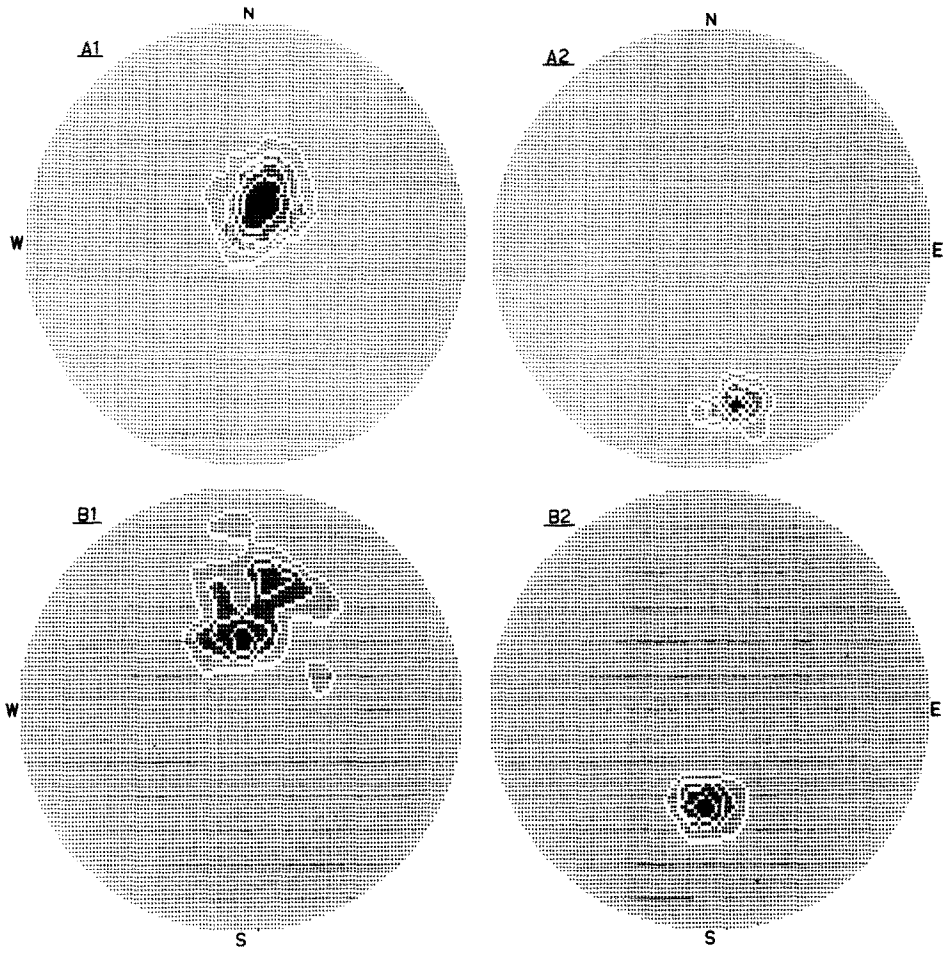


Fig. II.2. Frequency distribution of magnetic directions in equal area projection, counted with a 1% circle. Five frequency intervals of equal magnitude have been contoured. A1. NRM directions combined, upper hemisphere projection, maximum frequency = 40. A2. As above, lower hemisphere projection, maximum frequency = 6. B1. Recent field component directions combined, upper hemisphere projection, maximum frequency = 11. B2. As above, lower hemisphere projection, maximum frequency = 36.

tional analysis of all specimen components was made by means of orthogonal projection figures (Zijderveld 1967, 1975).

Three magnetic components have been identified: A recent field component (RFC) and two characteristic magnetic components (A and B), which both predate the main folding phases of the Late Cambrian—Early Ordovician Delamarian Orogeny. These directions are similar to those reported from the Flinders Ranges (Section I).

The frequency distribution plots (Fig. II.2A) show a recent field com-

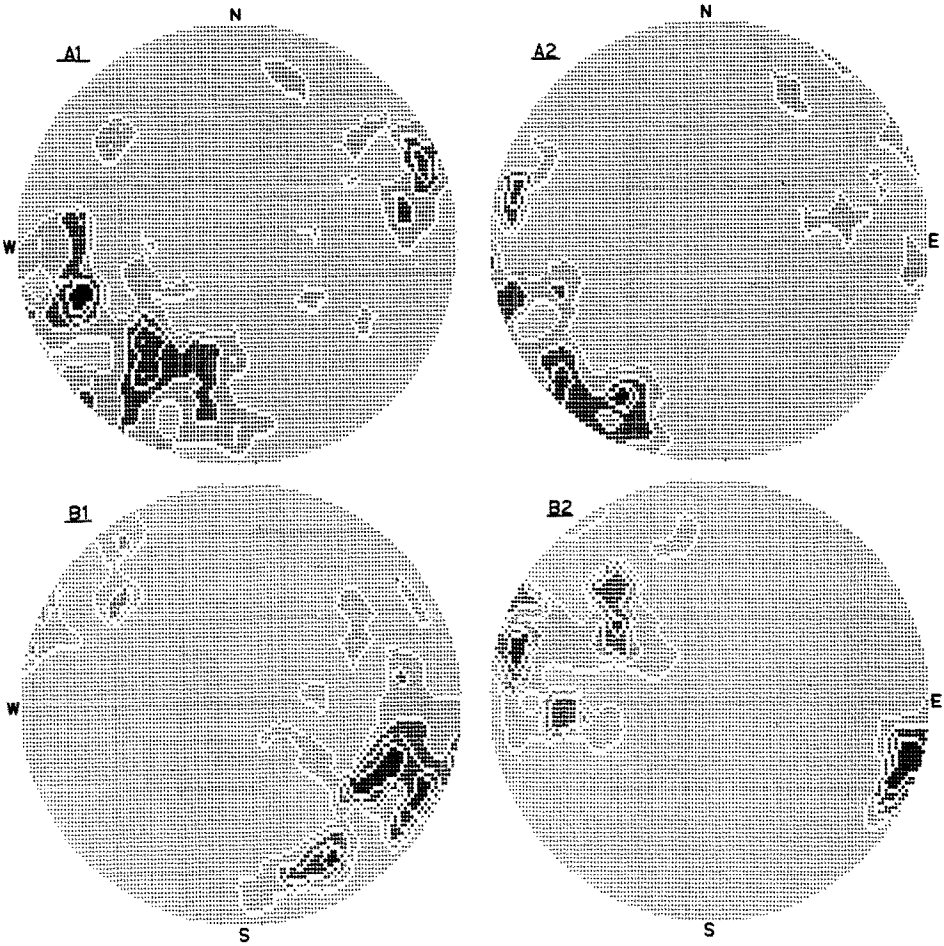
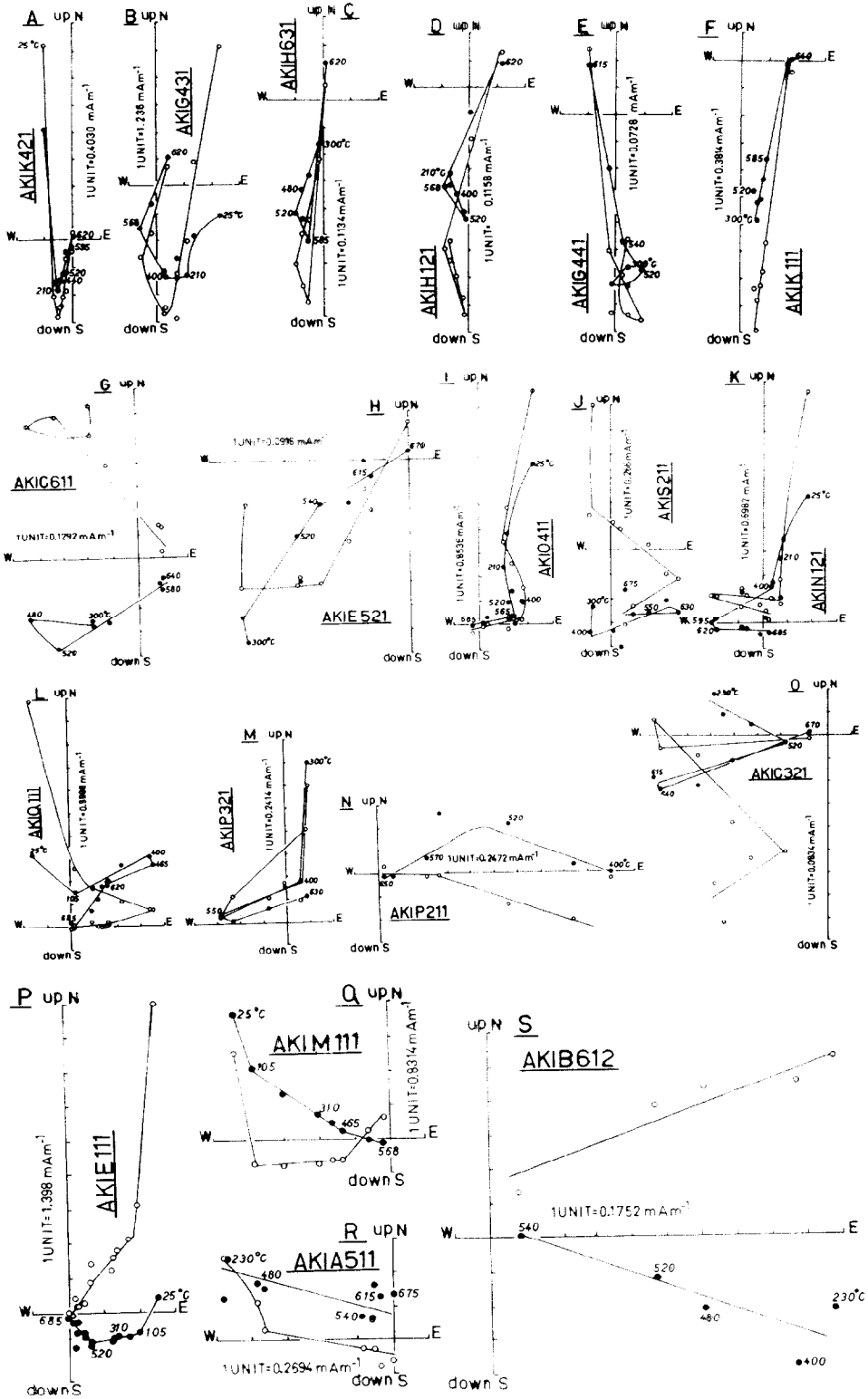


Fig. II.3. Frequency distribution of magnetization directions in equal area projection, counted with a 1% circle. See description under caption to Fig. II.2. A1. Combined primary magnetic component directions, upper hemisphere projection, maximum frequency = 5. A2. As above, lower hemisphere projection, maximum frequency = 4. B1. Combined secondary "Delamarian" component directions, upper hemisphere projection, maximum frequency = 4. B2. As above, lower hemisphere projection, maximum frequency = 3.

ponent (RFC) of 60—80% of the initial intensity (Figs. II.4, 5) as the dominant magnetization. This RFC generally is completely removed between 200°C and 500°C (Fig. II.3), except in the case of the rather aberrant sites AKIG and AKIH, in which this RFC was the sole constituent. Both normal and reversed polarities of this RFC are present throughout the sample collection. Various samples from the same site, but also various specimens from the same sample contain both normal and reversed polarity directions. This


A up N
AKIK421
1UNIT=0.4030 mAm⁻¹
down S
B up N
AKIG431
1UNIT=1.238 mAm⁻¹
down S
C
AKIH631
1UNIT=0.1134 mAm⁻¹
down S
D up N
AKIH121
1UNIT= 0.1158 mAm⁻¹
down S
E up N
AKIG441
1UNIT=0.0728
down S
F up N
AKIK111
1UNIT=0.384 mAm⁻¹
down S
G up N
AKIC611
1UNIT=0.1292 mAm⁻¹
down S
H up N
AKIE521
1UNIT=0.2996 mAm⁻¹
down S
I up N
AKIO411
1UNIT=0.853 mAm⁻¹
down S
J up N
AKIS211
1UNIT=0.26 mAm⁻¹
down S
K up N
AKIN121
1UNIT=0.5982 mAm⁻¹
down S
L up N
AKIQ111
1UNIT=0.5988 mAm⁻¹
down S
M up N
AKIP321
1UNIT=0.2414 mAm⁻¹
down S
N up N
AKIP211
1UNIT=0.2472 mAm⁻¹
down S
O up N
AKIC321
1UNIT=0.0934 mAm⁻¹
down S
P up N
AKIE111
1UNIT= 1.398 mAm⁻¹
down S
Q up N
AKIM111
1UNIT=0.9314 mAm⁻¹
down S
R up N
AKIA511
1UNIT=0.2694 mAm⁻¹
down S
S up N
AKIB612
1UNIT=0.1752 mAm⁻¹
down S

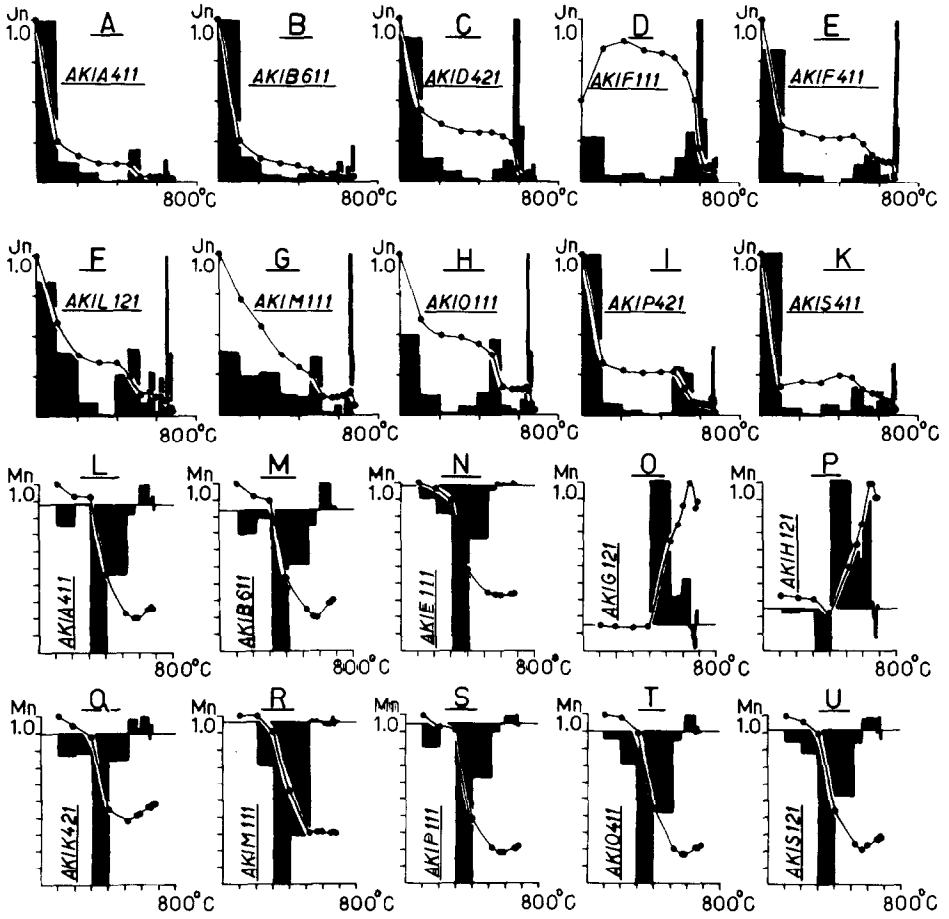


Fig. II.5. A—K. Normalized intensity decay curves of the remanent magnetization during thermal treatment. The blocking temperature spectra are shown also. L—U. Normalized intensity curves of the induced magnetization showing the general decrease in susceptibility after treatment at higher temperatures. The histograms show the stepwide decrease.

Fig. II.4. Demagnetization diagrams of specimens treated by thermal methods. The points denote successive positions in orthogonal projection, of the end of the resultant magnetization vector during progressive demagnetization. Open circles denote projections on the vertical east—west plane. Full circles denote projections on the horizontal plane. Numbers denote successive peak values of the applied temperatures. Magnetic components shown: 1 = recent field component directions (RFC), 2 = secondary Delamarian component directions (B), 3 = primary magnetization component directions (A), N = Normal polarity, R = Reversed polarity. A = 1N + 3R, B = 1N + 1R, C = 1N + 1R, D = 1N + 1R, E = 1N + 1R, R = 1R, G = 3R, H = 3R, I = 1N + 3N, J = 1N + 3R + 3N, K = 1N + 3N + 3R, L = 1N + 3R + 3N, M = 1N + 3N + 3R, N = 2N + 3N, O = 2R + 3N + 3R, P = 1N + 2N, Q = 1N + 2R, R = 2R, S = 2R. Results are not corrected for bedding.

phenomenon was noticed particularly in samples from the dark grey shale horizons of sites AKIG and AKIH (Fig. II.4A—F).

The intensity decay curves in general show two distinct blocking temperature ranges (Fig. II.5A, B, E—K). A lower one between 500°C and 600°C, probably associated with a magnetite phase, and a higher one between 650°C and 675°C, characteristic of haematite. However, a number of samples show a wider distribution of blocking temperatures (Fig. II.5C, D). All samples except those from the aberrant Emu Bay Shales (AKIG, AKIH), decreased in susceptibility between 300°C and 500°C (Fig. II.5L—N, Q—U), which is probably due to inversion of maghaematite into haematite (Turner and Ixer, 1977). Both this dual blocking temperature range and the decrease in susceptibility upon heating are found also in the correlated Billy Creek Formation of the Flinders Ranges (Fig. I.6S).

Similar to the general magnetization pattern of the Flinders Ranges samples, two characteristic magnetic components have been identified:

(1) A SSW—WSW directed magnetic component (A) with low upwards dip (Fig. II.3A).

(2) A WNW—NNW directed magnetic component (B) with low downwards dip (Fig. II.3B).

Normal and reversed polarity directions of both components are present. Similar to the Flinders Ranges samples (Section I) different specimens showed a highly variable behaviour of the two characteristic magnetic components during thermal demagnetization. No correlation at all could be established between the two prevalent blocking temperature ranges and the two characteristic magnetic components. For these reasons and also because of the dual polarity representation of both characteristic components, it was occasionally not possible to attribute somewhat deviating magnetic directions, though very evident on the orthogonal demagnetization graphs, to either the B-component or the A-component grouping. It is possible that such directions represent pseudo-stable artefacts resulting from composite breakdown, at constant ratios, of two or more magnetic components. Consequently such directions have been discarded from the statistical analysis, which was based on analysis of frequency distribution plots (Fig. II.3A, B) as detailed elsewhere (Section I).

In a number of specimens throughout the sampled section (Fig. II.4, J, K, L, M, O), both a normal polarity direction (cleaned at 400—600°C) and a reversed polarity direction (cleaned at 600—680°C) of the A-component were present. This peculiar phenomenon, also observed in red-bed sequences from the Flinders Ranges (Section I) and the Amadeus Basin (Section III), points to a prolonged period of magnetization acquisition upon diagenesis. Findings of both normal and reversed polarity directions of the RFC in especially the Emu Bay Shale Formation suggests a similar acquisition of more recent origin.

Clearly this observation seriously limits the use of these thermally demagnetized red beds results for magnetostratigraphic purposes.

INTERPRETATION

Figure II.6B shows that both the normal and the reversed polarity direction of the (RFC) are in good agreement with the virtual direction at the sampling locality according to the axial geocentric dipole field model, but not with the present local field direction (Table II.1). This RFC must have been acquired during a prolonged period of fairly recent weathering, probably of latest Cainozoic age, which spanned at least one polarity reversal. Extensively laterized surfaces are present on Kangaroo Island and on adjacent areas of South Australia, and widely different ages ranging from Early Mesozoic (Daily et al., 1974) to Plio-Pleistocene (Northcote, 1946; Sprigg, 1955; Major and Vitals, 1973) have been suggested for this laterization period. Schmidt (1976, also in Schmidt et al., 1976) discerned a magnetization component of both normal and reversed polarity in some laterite samples from Kangaroo Island, which he correlated with secondary magnetic components found in the Springfield Basin of the Flinders Ranges and in the Perth Basin (Schmidt, 1976). Comparison of their pole positions with the Tertiary apparent polar wander path for Australia (Wellmann et al., 1969;

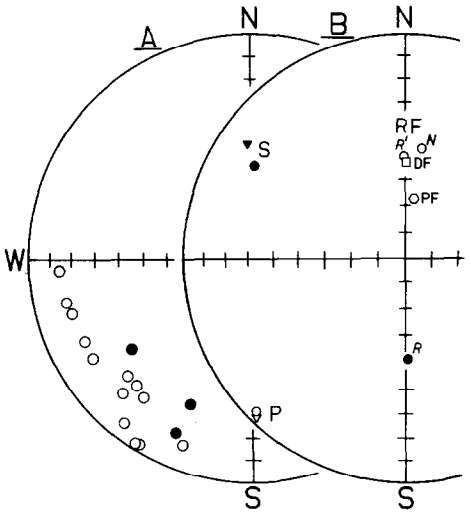


Fig. II.6. Equal area projection of mean directions, see Table II.1. A. Site mean direction of the primary magnetic component (A). B. Mean directional results from the Kangaroo Island samples (circles) for the primary magnetic component (P) and the secondary magnetic component (S), compared with the mean late-Early—early-Middle Cambrian results of the Flinders Ranges (inverted triangles). Square (*DF*) is the virtual direction at the sample locality according to the axial geocentric dipole field model. Hexagon (*PF*) is the present field direction at the sampling locality. Recent field component directions: N = normal polarity, R = reversed polarity, R' = swapped reversed polarity.

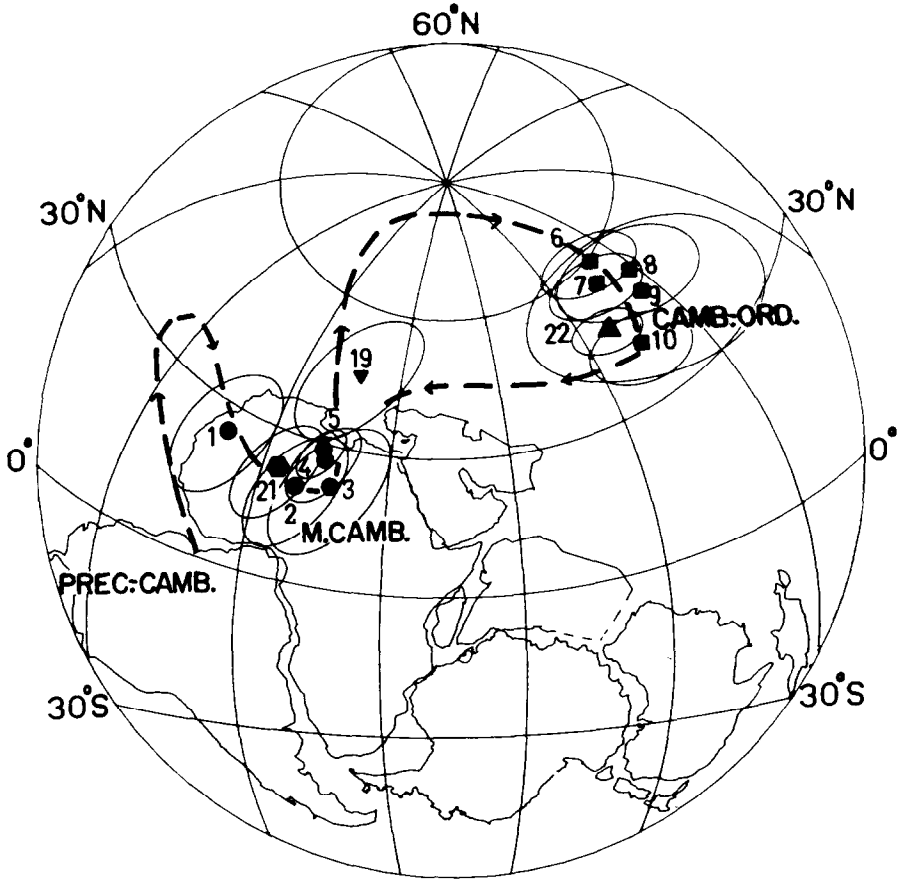
McElhinny et al., 1974b) led him to interpretation of his results (Schmidt, 1976; Schmidt et al., 1976) in terms of a dominant period of lateritic weathering during Late Oligocene—Early Miocene times which affected a large area of Australia (Woolnough, 1927). It should be mentioned, however, that Schmidt's interpretation of a Middle Tertiary remagnetization component, if correct, merely points to one specific episode in a possible complex remagnetization history and does not at all nullify suggestions for an older, possible early Mesozoic, phase of laterization as proposed by Daily et al. (1974).

The RFC directions found in the present study correspond with a pole position which is removed from the middle Tertiary segment of Australia's APW, and Schmidt et al's. (1976) laterite results from Kangaroo Island are in fact similarly removed (Fig. III.11). In contrast, the agreement of this RFC with the virtual "dipole field" direction (Fig. II.6A, Table II.1), suggests a latest Cainozoic period of acquisition.

The northwestwards directed and slightly downwards dipping B-component of both normal and reversed polarity (Fig. II.4N—S, II.6B) can be correlated after correction for bedding with a similar secondary magnetic component found in the Flinders Ranges (Section I) and in the Amadeus Basin (Section III). As detailed before (Section I), this secondary magnetic component must have been induced by thermo-chemical activity prior to the main folding phases of the Delamarian Orogeny. Detailed studies of metamorphic crystallization in the Mount Lofty Ranges (Fleming and Offler, 1968; Offler and Fleming, 1968; Milnes et al., 1977) identify three folding phases during the Delamarian Orogeny. Although intensive metamorphism and widespread granite intrusion, attributed to an unusually high heat flow from the mantle, commenced before the first folding phase (Offler and Fleming, 1968): syntectonic metamorphism and the non-closure of isotopic systems (Milness et al., 1977) indicate a maybe only locally prolonged period (at least 50 m.y.) of elevated temperatures. It is possible that a minor part of this B-component was acquired during a later stage of the Delamarian Orogeny and thus postdates one or more of the three folding phases. In the absence of any detailed knowledge about the magnetization—deformation relationship for the sampled Kangaroo Island beds, it was only possible to correct for the total accumulated dip. This simple correction procedure might be one of the reasons for the appreciable scatter in component directions.

The southwards directed and slightly upwards inclined A-component, of both normal and reversed polarity (Fig. II.4A, G—O, II.6A, Table II.1), agrees after correction for bedding with the mean primary magnetic direction of the correlated Billy Creek Formation of the Flinders Ranges, and is thought to represent the primary magnetization direction. B. Daily (in Thompson et al., 1976) correlated the studied sequence with the late-Early Cambrian Billy Creek Formation of the Flinders Ranges on the grounds of similarity in lithology and in fauna. This correlation is supported by the

observed similarity in magnetic characteristics, similar dual blocking temperature range, decrease of susceptibility upon heating, and an apparently prolonged period of acquisition of magnetization as evidenced by the presence of both normal and reversed polarities in some specimens. This points to an identical source of sediment and a similar depositional evironment. Although the late-Early to Middle Cambrian has been identified as a period of rather slow polar motion (Sections I and III), which reduces the resolution of any attempted palaeo-directional correlation, the good agreement in primary magnetic directions from the studied Kangaroo Island



ADELAIDE "GEOSYNCLINE" RESULTS

Fig. II.7. Newly obtained palaeomagnetic S-pole positions for the Cambrian of Kangaroo Island, compared with new results from the Flinders Ranges (Section I), superimposed on a revised early Palaeozoic combined polar wander path for Gondwanaland (Fig. III.13). *21* = mean primary magnetic direction, Kangaroo Island. *22* = mean Delamarian overprint, Kangaroo Island.

sequence and from the Billy Creek Formation of the Flinders Ranges adds further support to this correlation (Fig. II.6A, B).

Neglecting the danger of circular reasoning, it may equally well be stated that the agreement of both the primary magnetic direction (late-Early Cambrian) and the slightly younger secondary magnetic directions from the Flinders Ranges with those from Kangaroo Island (Fig. II.6B, 7), virtually exclude any rotational movement between both areas since late-Early Cambrian times. On the basis of available palaeomagnetic data it is not possible, however, to exclude the possibility of rotational movements in other areas of the Adelaide "Geosyncline". The agreement between Briden's (1967) results for Marinoan sediments from Hallett Cove (Mount Lofty Ranges) and McWilliams (1977) result for the Yerelina subgroup from the northern Flinders Ranges on the one hand, and the discrepancy between Karner's (1975) results for the Brachina Formation from Hallett Cove and McWilliam's results for the Brachina Formation from the Central Flinders Ranges on the other hand, needs further testing in this respect.

Further palaeomagnetic data are needed also to test various hypotheses suggesting translational movement along megashear zones in the Adelaide "Geosyncline" and along the southeast Australian shelf. The suggested magnitude of some of these movements and their oblique position with respect to the palaeolatitude pattern (cf. Fig. III.14) possibly falls within the resolution of the palaeomagnetic method.

CONCLUSIONS

The magnetization pattern of the late-Early Cambrian red-bed sequence on northeastern Kangaroo Island appeared to be similar to the magnetization history of penecontemporaneous sequences in the Flinders Ranges (Section I) and in the Amadeus Basin (Section III).

Three magnetic components have been identified:

(a) A recent magnetic component (RFC) of both normal and reversed polarity, with a direction very near to the virtual "dipole field" direction at the sample locality.

(b) A northwestwards directed and slightly downwards inclined secondary magnetic component (B) of both normal and reversed polarity. This magnetization originated largely before the main folding phases of the Delamarian Orogeny and most probably was induced by enhanced thermo-chemical activity which commenced before the diastrophistic phases of the Delamarian Orogeny. This activity may have persisted for a fairly long period and acquisition of an unquantified but probably minor part of this magnetization may have continued during the later folding phases. The correction applied here for total accumulated dip of the beds may have attributed to the appreciable scatter in directions.

(c) A southwestwards directed and slightly upwards inclined magnetic component (A) of both normal and reversed polarity, which has been

identified as the primary magnetization. A prolonged period of acquisition of magnetization upon diagenesis is indicated by the presence of both normal and reversed polarities of this component in the same specimens. Similar observations were made also on the primary magnetic content of Cambrian red beds from the Flinders Ranges (Section I) and from the Amadeus Basin (Section III) and may limit the applicability of these thermally demagnetized red beds for magnetostratigraphic purposes.

The directional agreement of both the primary magnetic component (A) and the secondary magnetic component (B) in the beds under study with a similar magnetization pattern in the correlated Billy Creek Formation of the Flinders Ranges, indicates absence of relative rotational movement between Kangaroo Island and the northwestern part of the Adelaide "Geosyncline", since late-Early Cambrian times. More palaeomagnetic data are needed, however, to further test various models postulating shear in the Adelaide "Geosyncline" and along southeast Australia's shelf.

III. Middle to early-Late Cambrian results from the Amadeus Basin (Northern Territory)

INTRODUCTION

Palaeomagnetic studies on Cambrian rocks from the Adelaide "Geosyncline" (Sections I and II) have met with considerable difficulties in separating the primary magnetic component from a dominant recent field component and from another thermo-chemically induced secondary component of Cambro-Ordovician age. Interpretation of results was not straightforward, and complementary information from other sequences of Cambrian age in Australia was needed. Therefore, a study on the Amadeus Basin succession was undertaken. Additional objectives were:

(1) Better definition of the Cambrian part of the Australian apparent polar wanderpath (APW), for which rapid polar motion has been inferred before (Embleton, 1972a, b; McElhinny and Embleton, 1974, 1976; McElhinny et al., 1974b).

(2) Study of possible intraplate tectonics as suggested amongst others by Harrington et al. (1973).

(3) Identification of palaeomagnetic constraints on age ranges of unfossiliferous formations through intracontinental comparison of magnetostratigraphic records and pole positions.

(4) Comparison of palaeomagnetic characteristics in a heterotopical facies environment as typical for the Amadeus Basin.

(5) Intercontinental correlation of polarity sequences.

The Amadeus Basin (Fig. III.1) forms a natural object for a palaeomagnetic study. Outcrop is good, and the structural and stratigraphical history of the basin is well known from extensive regional mapping as compiled by Wells et al. (1970) and Wells (1976). It is therefore not surprising that

Australian palaeomagnetic workers paid special attention to a study of the Amadeus Basin. Embleton (1972a, b) initiated palaeomagnetic research in this basin with a study of some formations from the Ellery Creek succession (Fig. III.3).

More extensive studies were recently finalised by McWilliams (1977) on late Proterozoic rocks and by Kirschvink (1978a, b) on several successions spanning the Precambrian—Cambrian boundary period. So the earlier part of the sedimentation record of the Amadeus Basin seems fairly well established.

The present study of two Middle—early-Late Cambrian successions in the Amadeus Basin was carried out in cooperation with Kirschvink's (1978a, b) magnetostratigraphic analysis of the Precambrian—Cambrian boundary period. To enhance mutual compatibility and moreover to detail a suggested rapid polar motion (Embleton, 1972a, b; McElhinny and Embleton, 1974, 1976; McElhinny et al., 1974b) a detailed sampling at regular intervals throughout the formations was pursued in this study.

GEOLOGICAL SETTING

The Amadeus Basin in Central Australia is an east—west aligned intracratonic depression stretching over about 800 km from the southern part of the Northern Territory into West Australia (Fig. III.1). A 9000-m thick sedimentation record spanning a period from late Proterozoic to late Palaeozoic is preserved in the basin (see, for a description, Wells et al., 1970 and Wells, 1976). Sedimentation started no more than 1000 m.y. ago with blanket deposits, and became more localized with the onset of three late Proterozoic diastrophistic movements. The latest Proterozoic—earliest Cambrian Petermann Ranges Orogeny was the most severe of these diastrophistic movements as is evident from large scale folding, nappe movements and metamorphism along the southern basin margin. These three movements resulted in uplift of the areas in the south which became the main provenance for Cambrian sediments that poured into the subsiding area in to the north. Variable ratios of subsidence in the basin and the localized sediment source in the southwest resulted during Cambrian times in heterotopical facies conditions. Coarse-grained sediments were deposited close to the uplifted area, grading in north and northeast direction into sandstone, shale (some marine) and carbonate rocks (i.e. the Eninta Sandstone, Tempe Formation, Illara Sandstone, Deception Formation and Petermann Sandstone, Fig. III.2). This sequence of predominantly clastic sediments became much thinner developed in a less subsided north—south zone near the centre of the basin. In the northeastern part of the basin, carbonate sedimentation became dominant after deposition of the Arumbera Sandstone; the basal formation of the Pertaoorta Group (Fig. II.2). This Group further includes (Fig. III.2) carbonate rocks (Todd River Dolomite) tectonically reduced evaporites (Chandler Limestone), penesaline and marine stromatolitic carbonates (Giles Creek Dolomite, Shannon Formation) and calcareous silt

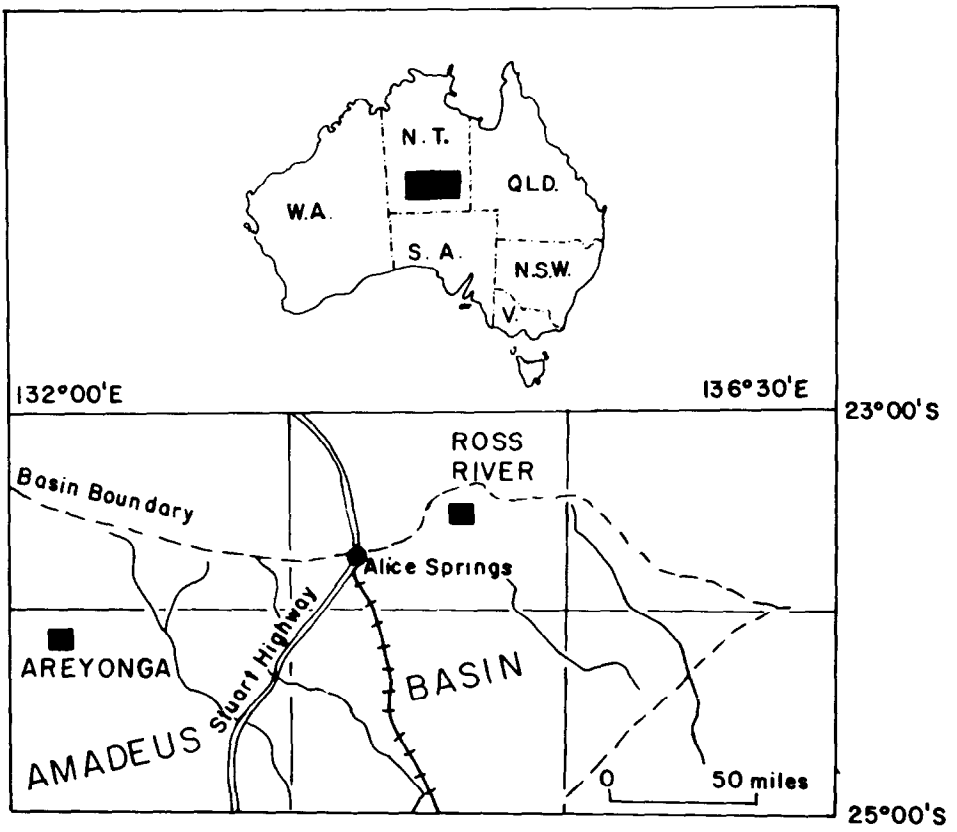


Fig. III.1. Locality of the Amadeus Basin.

and silty sands (Goyder Formation). The Arumbera Sandstone (which spans the Precambiran—Cambrian boundary period (Daily, 1972a; Kirschvink, 1978a)) and the Goyder Formation show minor facies changes only and are well continuable throughout the basin.

Age control on the sampled parts of the Cambrian successions is rather well established in the carbonate sequence of the eastern Amadeus Basin (Ross River section), but is sparse in the arenitic sequence (Areyonga section) from the central and western part of the basin. In the Areyonga section, the upper part of the Tempe Formation is dated as early-Middle Cambrian on the basis of a brachiopod—trilobite—hyolithid—gastropod fauna. No fauna is known from the underlying Eninta Sandstone, nor in the overlying Illara Sandstone and the Deception Formation, but Middle Cambrian ages have been inferred for both latter formations on the basis of their stratigraphical position underneath the Petermann Sandstone. The latter is tentatively dated as a late-Middle—early-Late Cambrian on the basis of a sparse gastropod fauna. In the Ross River section, the Giles Creek Dolomite is

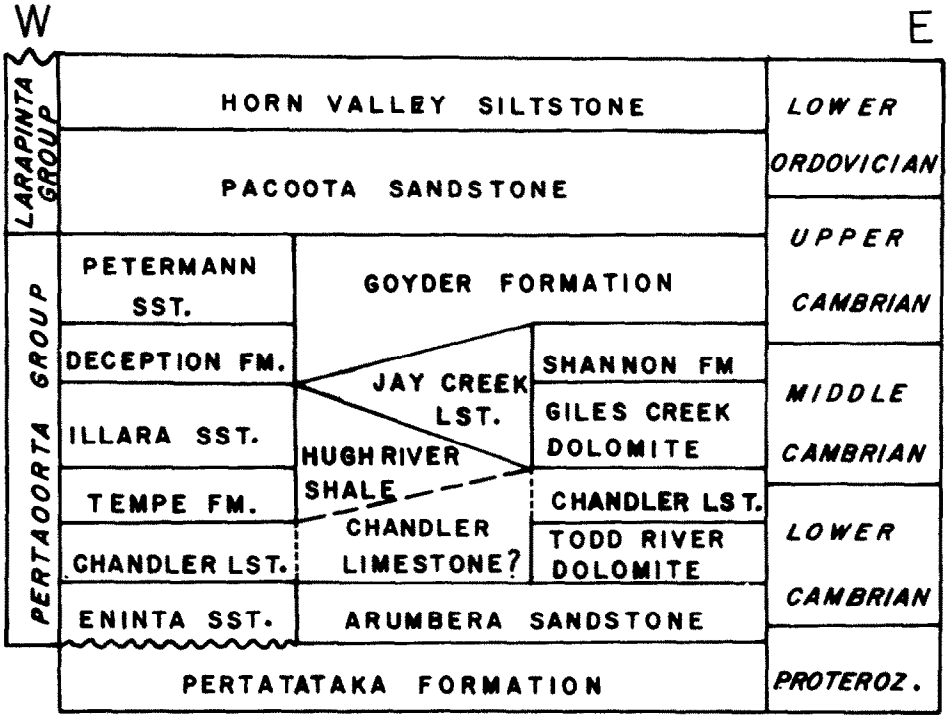


Fig. III.2. Stratigraphic ideal column for the Cambrian of the central and eastern areas of the Amadeus Basin.

dated as early-Middle Cambrian on basis of a hyolithid—brachiopod—gastropod—trilobite fauna, whereas the underlying Todd River Dolomite is dated as Early Cambrian mainly on basis of archaeocyathans and brachiopods. The overlying Shannon Formation contains a stromatolite fauna, characteristic for a late Mindyallan age (early-Late Cambrian). Three Palaeozoic movements have occurred (Wells et al., 1970; Wells, 1976) which mainly affected the northern part of the Basin, in contrast to the late Proterozoic movements, which mainly affected its southern part. This resulted from the Cambro-Ordovician boundary period onwards in southwards transgressive sedimentation, intermittent with successive uplifts (the Rodingan Movement of Silurian age and the Late Devonian Pertnjara Movement), which came to an end with the onset of the Late Devonian—Early Carboniferous Alice Springs Orogeny.

Most of the folding of the Amadeus Basin occurred during the Alice Springs Orogeny, which resulted also in large-scale Nappe structures and resetting of isotopic systems in the area directly to the northeast of the basin. The Amadeus Basin remained stabilized after this deformation. Epeirogenic movements brought the area close to submergence in Permian

and in Mesozoic times. Minor crustal movements occurred during the Tertiary with erosion and weathering attributing to the present relief.

SAMPLED SECTIONS

Core samples, obtained with a portable drill, were oriented in the field with both a magnetic and a solar compass, using a device described by Embleton and Edwards (1973). Samples were taken in the form of magneto-stratigraphic profiles, either with each sample taken from a separate bed at regular intervals of generally less than 2 m (Deception Formation, Giles Creek Dolomite), or in the case of very thick formations, with a greater interval after each sixth sample (Tempe Formation, Illara Sandstone, Shannon Formation). Stratigraphic intervals were measured with a Jacob's Staff and an attached Abney level.

For comparison of palaeomagnetic characteristics in time-equivalent sections of different lithology, a late-Early—Middle Cambrian red-bed sequence has been sampled at Areyonga (Fig. III.3, central Amadeus Basin) and a Middle—early-Late Cambrian carbonate sequence at Ross River (eastern Amadeus Basin, Fig. III.3).

(A) Areyonga section

This section was sampled at the northern head of the Areyonga Gorge (132.27°E 24.10°S, Fig. III.3) which dissects the WNW—ESE trending Gardiner Range anticline; a north-vergent structure with a locally reduced northern limb. An en-echelon set of north—south trending transverse faults is well developed at the Areyonga Gorge, but the various formations can easily be followed in the field.

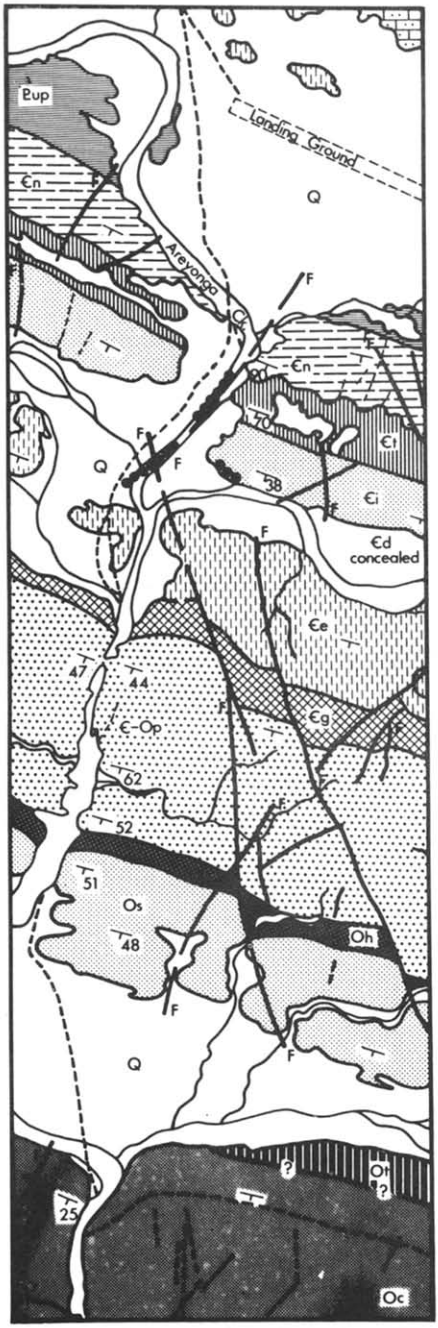
A Middle Cambrian sequence (Tempe Formation, Illara Sandstone, Deception Formation), possibly starting already in the latest-Early Cambrian, has been sampled in the steeply southwards dipping (50—80°) southern limb of the anticlinal structure.

Tempe Formation. In the streambed of the Areyonga Creek, a total of 77 samples (13 sites) has been taken over a stratigraphic sequence of 228 m, starting about 60 m above the top of the underlying Eninta Sandstone. Samples were mainly taken in red to reddish coloured siltstones and in addition in some red dolomites and greyish greenish glauconitic sandstone.

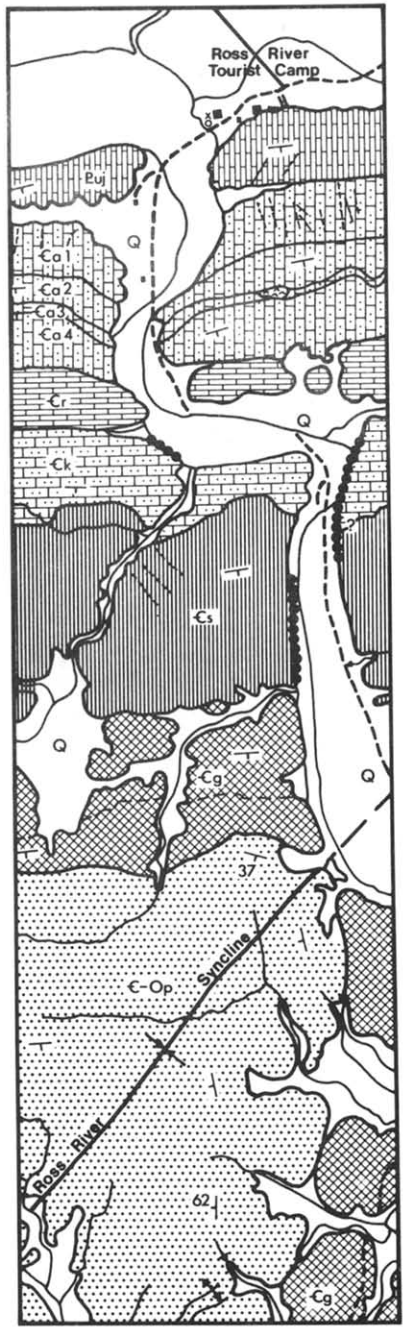
Illara Sandstone. The uppermost part of this formation has been sampled in a small ridge to the east of the streambed, as the streambed section itself seemed reduced by transverse faulting. Twenty-six samples (4 sites), spanning a stratigraphic interval of 22 m, have been taken in crossbedded reddish sandstones.

Deception Formation. This formation consists mainly of reddish micaceous crossbedded sandstones, siltstones and shales and becomes easily eroded. A total of 63 samples has been taken in the streambed at regular intervals through the top 148 m of the formation.

AREYONGA



ROSS RIVER



Geology compiled by A.T. Wells and J. Kennard, 1976

Bureau of Mineral Resources, Canberra A.C.T.

(B) Ross River section

The eastern closure of a WSW—ENE trending brachy-syncline, made up by the Pertaoorta Group, is well exposed in the Ross River Gorge (134.49° E 23.62° S, Fig. III.3). The Alice Springs Orogeny has had a major effect on the structure of this area as is well reflected in the development of the Arltunga Nappe Complex directly to the north (Wells et al., 1970; Wells, 1976). Evaporite sequences from the late Proterozoic Bitter Springs Formation and from the Early Cambrian Chandler Limestone may have acted as décollement horizons with a complicated thrust zone connecting both. Mass movement was southwards directed, but no palaeomagnetic indications have been found so far (Kirschvink, 1978a, b) that such a translational movement was accompanied by noticeable rotational movement.

A section through the Giles Creek Dolomite and the overlying Shannon Formation was sampled in the northern part of the Ross River Gorge in the WNW dipping northern limb of the Ross River syncline.

Giles Creek Dolomite. This formation which consists of dolomite with interbeds of limestone, siltstone and shale, has been sampled in two overlapping sequences. A total of 41 samples (AGCC1-41) has been taken through mottled to nodular dark grey dolomitic limestones over a stratigraphic interval of 71 m, from the basalmost exposed beds till about 18 m below the base of a massive dolomitic limestone member which forms a very prominent ridge in the terrain. An overlapping sequence of twelve samples (AGCC42-53) have been drilled through another 23 m thick section of mottled dolomitic limestone at the base of the massive cliff, and a further 61 samples (AGCC54-102, ASHA1-5, ASHB1-7) have been cored higher up over a stratigraphic interval of 168 m through massive light coloured biostromal dolomitic limestone, up to the top of the formation.

Shannon Formation. Detailed sampling of this formation, which comprises siltstones, limestones and shales, was impractical because of its thickness; so a total of 48 samples (8 sites) were spread over a total stratigraphic interval of 347 m up to the top of the formation. All cores were drilled in predominantly massive white bluish dolomitic limestone beds and in yellow coloured dolomite beds.

Fig. III.3. Geological detail maps of the sections sampled south of Areyonga and to the south of Ross River. Geology after Wells (1976). Areyonga section: *Pub* = Pertatataka Fm; *En* = Eninita Sst.; *Et* = Tempe Fm.; *Ei* = Illara Sst.; *Ed* = Deception Fm.; *Ee* = Petermann Sst.; *Eq* = Goyder Fm.; *E-op, oh, os, ot, oc* = Larapinta Group; *Q* = Quaternary. Ross River section: *Puj* = Pertatataka Fm.; *Ea* = Arumbera Sst.; *Er* = Todd River Dolomite; *Ek* = Giles Creek Dolomite; *Es* = Shannon Fm.; *Eg* = Goyder Fm.; *E-op* = Larapinta Group.

LABORATORY TREATMENT AND RESULTS

Laboratory treatment and directional analysis was essentially the same as described in the Flinders Ranges study (Section I). At least two specimens from each site, or in the case of a detailed stratigraphic profile one specimen from each third sample, were subjected to progressive thermal demagnetization studies. Red-bed specimens were demagnetized in fifteen steps up to 685° C and carbonate specimens in thirteen steps up to a maximum temperature of 565° C. All remaining red-bed specimens, except those from the Illara Sandstone (to be discussed) were partially progressively demagnetized in nine steps between 300° C and 675° C, whereas all remaining carbonate specimens were demagnetized in six to eleven steps between 150° C and 460° C. Such an elaborate demagnetization procedure was necessary as each specimen reacted individually upon thermal treatment. This feature has been noted also in the accompanying studies on rocks from the Adelaide "Geosyncline" (Sections I and II). Care was taken to keep the specimens in very low magnetic fields in order to reduce excessive build up of viscous remanent magnetization (VRM) after treatment at the higher temperature steps. The VRM nevertheless prevented continued thermal analysis of the carbonate specimens above 450° C.

In general, four magnetic components could be identified. A recent field component (RFC) and three characteristic magnetic components (ASO, B and A) which are interpreted as predating the main folding phase of the Early Carboniferous Alice Springs Orogeny.

The recent field component (RFC) was present in all specimens and dominated the initial directions (Fig. III.4.1). The ASO components in SSW directed with a steep downward dip. Reversed polarity directions of the component were dominant in the Ross River section (Fig. III.4.1.), but both normal and reversed polarity directions of this ASO-component were present in the Areyonga section, although of a less dominant occurrence (Fig. III.4.1). The B-component is NW directed with a low downward (Areyonga section) to low upward (Ross River section) inclination. Both normal and reversed polarity directions occur (Fig. III.4.3). The A-component is SW (Giles Creek Dolomite and Areyonga section) to E (Shannon Formation) directed with a low upwards inclination, and is also present with normal and reversed polarities (Fig. III.4.2).

As will be detailed, the ASO component is thought to be associated with the Alice Springs Orogeny, and the A-component is identified as the primary magnetic component. The B-component is very similar to a secondary magnetic component consistently present in samples from the Adelaide "Geosyncline" (Sections I and II), which was interpreted before as a result of thermo-chemical activity during a Late Cambrian—Early Ordovician period of regional heating.

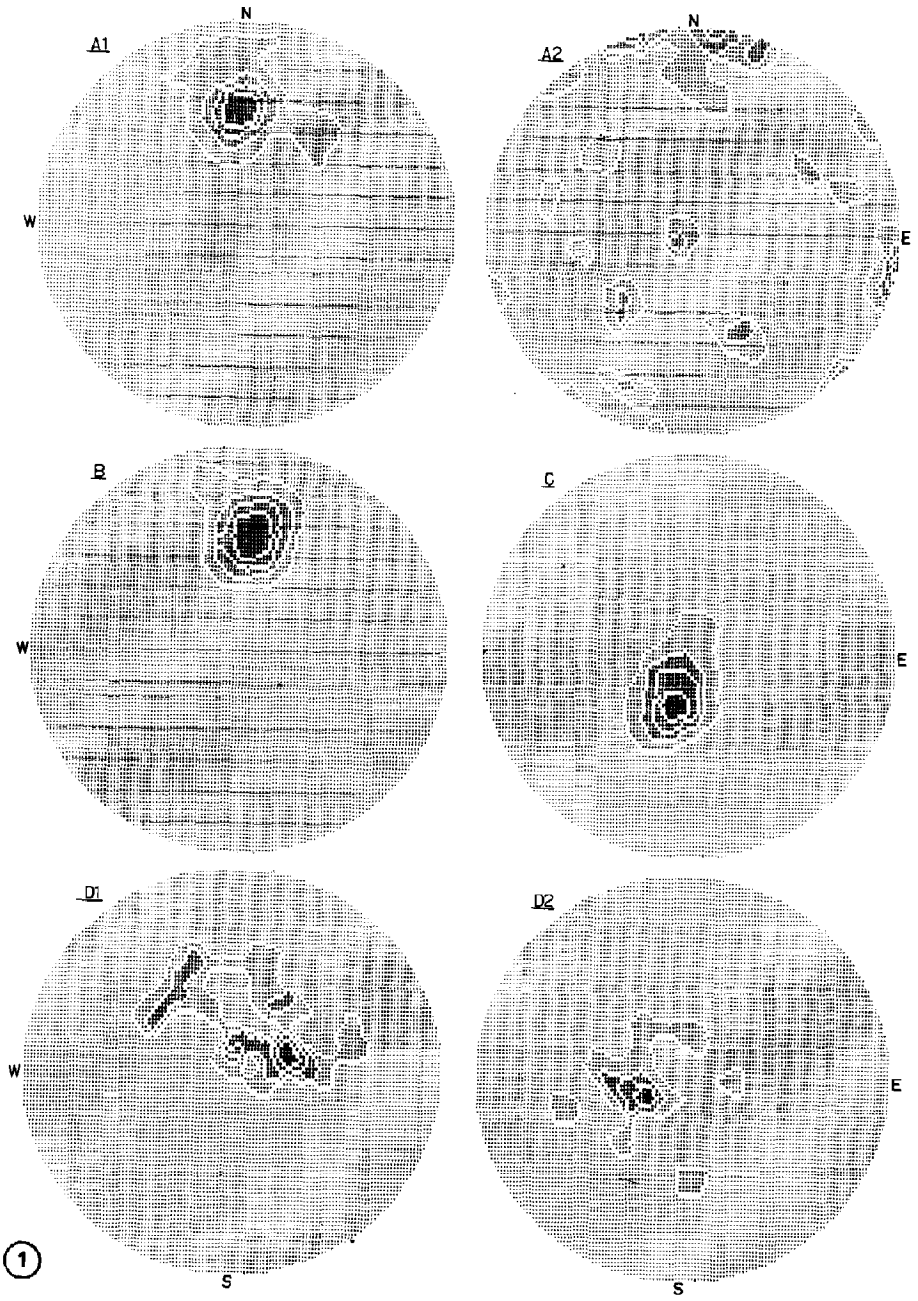


Fig. III.4(1). Frequency distribution of magnetic directions in equal area projection, counted with a 1% circle. Five frequency intervals of equal magnetitude have been contoured. A1. NRM directions for the Areyonga section, upper hemisphere projection, maximum frequency = 34. A2. As above, lower hemisphere projection, maximum frequency = 6. B. NRM directions combined for the Ross River section, upper hemisphere projection, maximum frequency = 37. C. Alice Springs Orogeny component, Ross River section, lower hemisphere projection, maximum frequency = 36. D1. Alice Springs Orogeny component, Areyonga section, upper hemisphere projection, maximum fre-

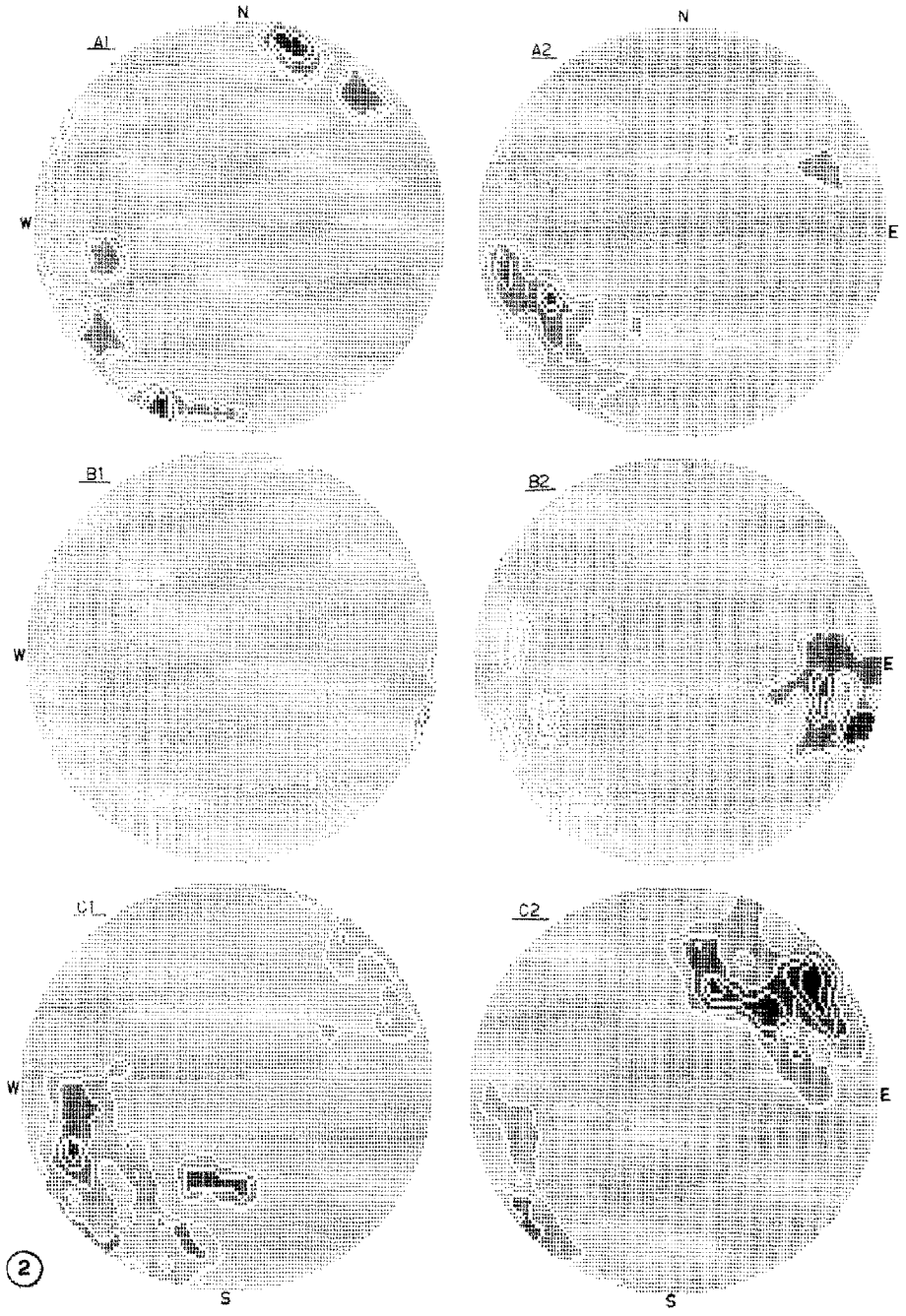


Fig. III.4(2). A1. Primary magnetization component Giles Creek Dolomite, upper hemisphere projection, maximum frequency = 2. A2. As above, lower hemisphere projection, maximum frequency = 5. B1. Primary magnetic component Shannon Formation, upper hemisphere projection, maximum frequency = 2. C1. Primary magnetic component for the Areyonga section, upper hemisphere projection, maximum frequency = 8.

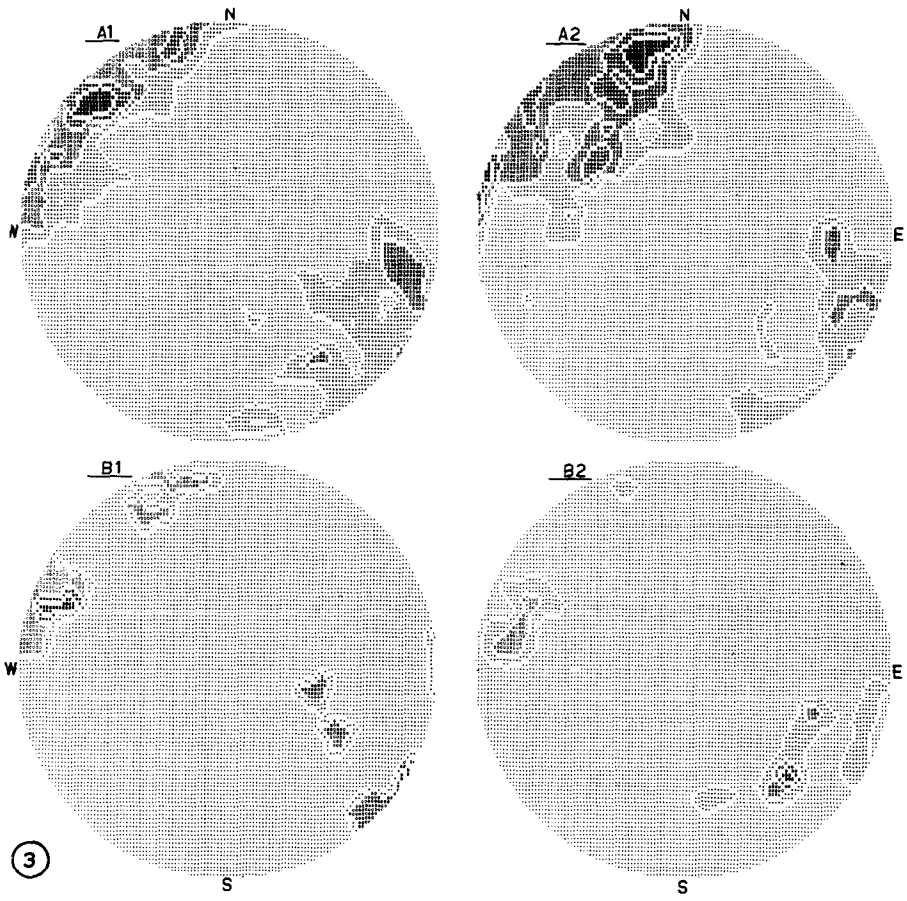
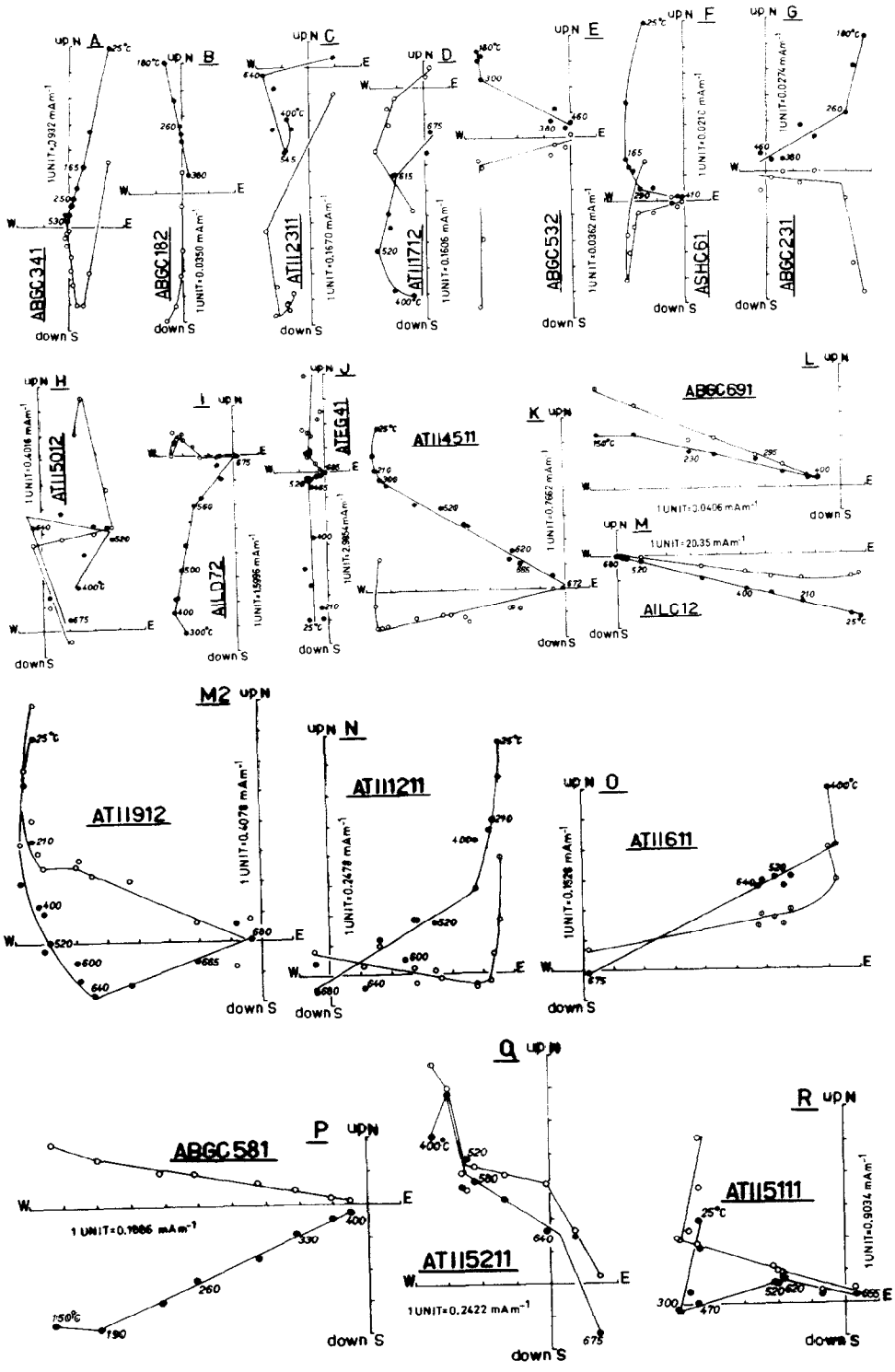


Fig. III.4(3). A1. Secondary "Cambro-Ordovician" component for the Areyonga section, upper hemisphere projection, maximum frequency = 10. A2. As above, lower hemisphere projection, maximum frequency = 9. B1. Secondary "Cambro-Ordovician" component for the Ross River section, upper hemisphere projection, maximum frequency = 2. B2. As above, lower hemisphere projection, maximum frequency = 6.

(A) Areyonga section

Tempe Formation. An RFC of 39—90% of the initial intensity was broken down completely at 300°C (Fig. III.5, 7). In some specimens another recent field component resided in magnetic domains with blocking temperatures above 600°C (Fig. III.6). An ASO component of both normal and reversed polarity was present in a few samples only (Fig. III.9B). The normal polarity ASO component was broken down between 470°C and 640°C, whereas the reversed polarity ASO component became eliminated at slightly lower


A
B
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D
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P
Q
R
M2
ABGC341
ABGC182
ATII2311
ATII1712
ABGC532
ASHC61
ABGC231
ATII5012
AILD72
ATEG41
ATII4511
ABGC691
AILC12
ATII1912
ATII1211
ATII1611
ABGC581
ATII5211
ATII5111
up N
down S
W.
E
1 UNIT=0.2932 mA m⁻¹
1 UNIT=0.0350 mA m⁻¹
1 UNIT=0.1670 mA m⁻¹
1 UNIT=0.1606 mA m⁻¹
1 UNIT=0.0362 mA m⁻¹
1 UNIT=0.021C mA m⁻¹
1 UNIT=0.027₄ mA m⁻¹
1 UNIT=0.4016 mA m⁻¹
1 UNIT=1.5995 mA m⁻¹
1 UNIT=2.9854 mA m⁻¹
1 UNIT=0.7662 mA m⁻¹
1 UNIT=0.0406 mA m⁻¹
1 UNIT=20.35 mA m⁻¹
1 UNIT=0.4078 mA m⁻¹
1 UNIT=0.2478 mA m⁻¹
1 UNIT=0.1838 mA m⁻¹
1 UNIT=0.1886 mA m⁻¹
1 UNIT=0.2422 mA m⁻¹
1 UNIT=0.9034 mA m⁻¹

temperatures. The secondary B-component is ubiquitous, mostly as reversed polarity directions. The primary A-component is less well developed; normal polarity directions predominate at the basal part of the sequence and reversed polarity directions do so higher up. An evident blocking temperature range between 500°C and 570°C, attributed to a magnetite phase, and a less evident blocking range at 650—670°C, attributed to haematite, are well visible on the normalized intensity decay curves (Fig. III.7). A number of specimens, however, show a wide blocking temperature spectrum with gradual magnetic breakdown above 500°C. It was not possible to identify breakdown of the A- or B-component with either of these blocking ranges, although the A-component becomes mainly broken down at the lower temperature range and the B-component at the higher temperatures. Specimens from the lower part of the section showed a susceptibility decrease between 300° and 500°C (Fig. III.8). With the onset of glauconite higher up in the sedimentation record, this characteristic changed to a distinct susceptibility increase after heating above 550°C.

Illara Sandstone. Initial intensities of these samples (Table III.1) were up to two orders of magnitude higher than those from samples of the Tempe Formation and the Deception Formation, sampled in the same Areyonga section. Initial directions were widely spread and did not reveal the dominant influence of an RFC, being so prominent in the other formations studied. The normalized intensity decay curves showed an evident blocking temperature range between 500°C and 570°C with another less pronounced blocking temperature range between 650°C and 670°C (Fig. III.7). In contrast, the orthogonal demagnetization graphs showed breakdown of an essentially mono-component magnetic system with hardly any indication for a RFC, so dominant elsewhere. As the samples were taken on a ridge, lightning effects were suspected to be present. For this reason, the remainder of the specimens were subjected first to alternating field demagnetization in seven steps up to 125 mT, followed by partial progressive thermal demagnetization in eleven steps between 200°C and 675°C. In most specimens, except from state AILC, the AF-demagnetization resulted in breakdown, largely below 30 mT, of a distinct magnetic component with

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Fig. III.5. Demagnetization diagrams of specimens treated by thermal methods. The points denote successive positions, in orthogonal projection of the end of the resultant magnetization vector during progressive demagnetization. Open circles denote projections on the vertical east—west plane. Full circles denote projections on the horizontal plane. Numbers denote successive peak values of the applied temperatures. Magnetic components shown: 1 = Recent field component; 2 = "Alice Springs Orogeny" component; 3 = "Cambro-Ordovician" component; 4 = Primary component; N = normal polarity; R = reversed polarity. A = 1N + 2N; B = 2N; C = 2N; D = 2N + 4R; E = 2N + 3R; F = 1N + 2N + 4R; G = 2N + 4N; H = 1N + 3N + 2N; I = 2R + 4R; J = 2R + 4R; K = 1N + 4N; L = 3R; M = 3N; S = 1N + 4R; N = 1N + 4N; O = 4N; P = 4R; Q = 1N + 3N + 1N; R = 1N + 4R + 3R.

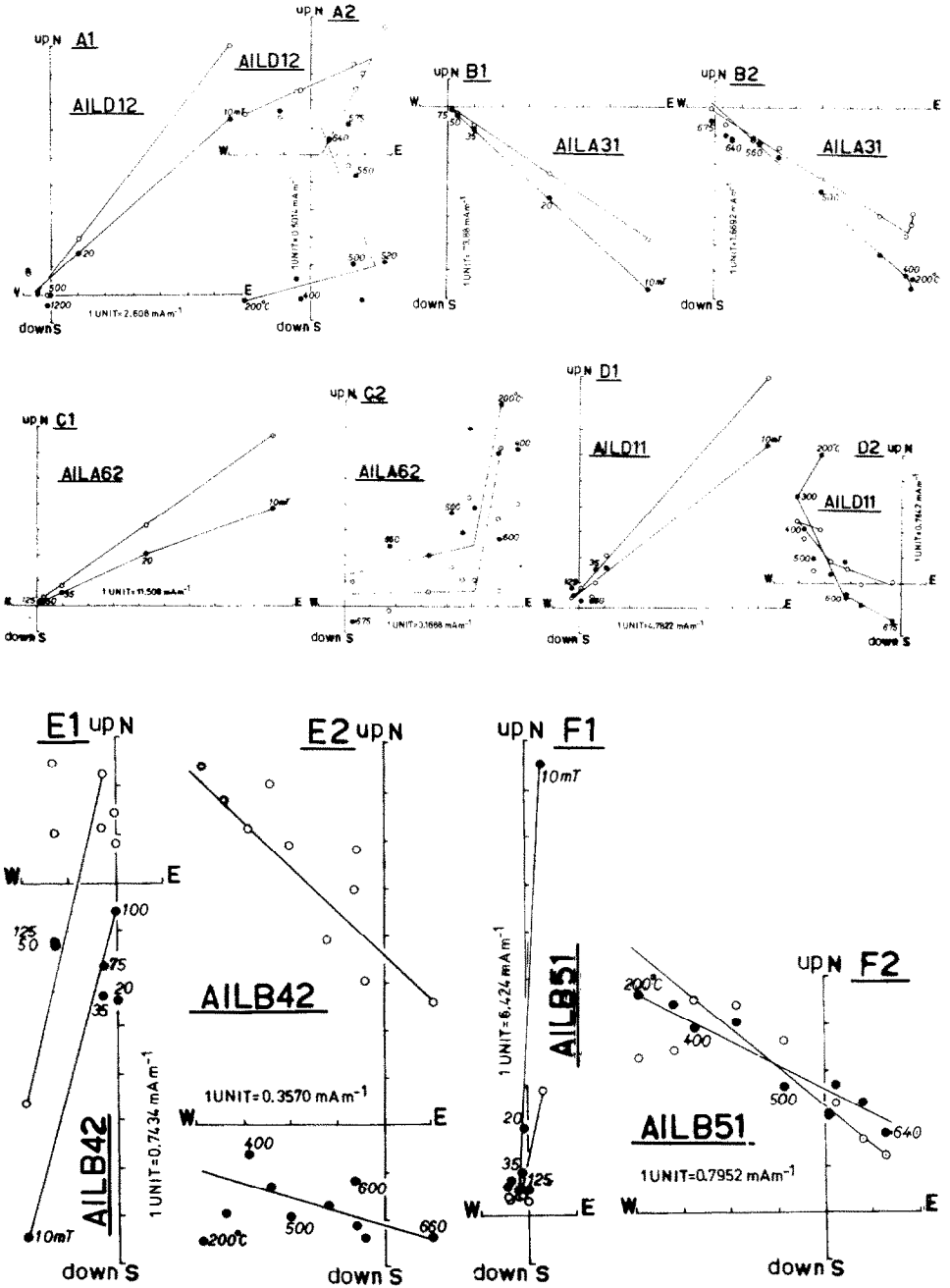


Fig. III.6. Demagnetization diagrams of specimens subjected to AF demagnetization up to 120 mT and subsequently to thermal demagnetization; see caption to Fig. III.5. A1 = 4N; A2 = 4R + 2N?; B1 = 3N; B2 = 3N; C1 = 4N; C2 = 1N + 4N; D1 = 4N?; D2 = 3R; E1 = ?; E2 = 3R; F1 = 1N; F2 = 3R.

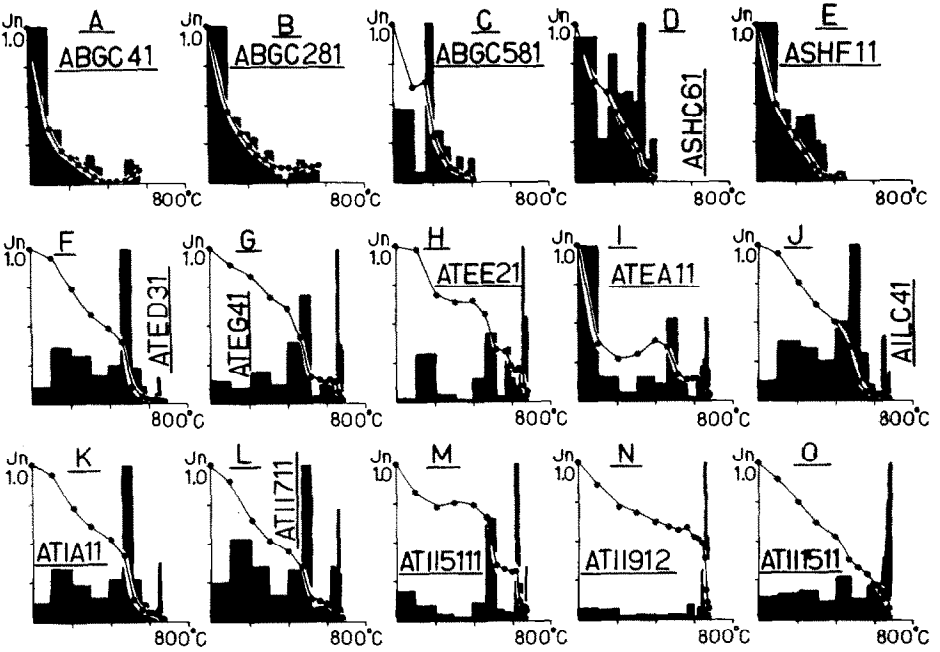


Fig. III.7. Normalized intensity decay curves of the remanent magnetization during thermal treatment. Blocking temperature spectra are shown also.

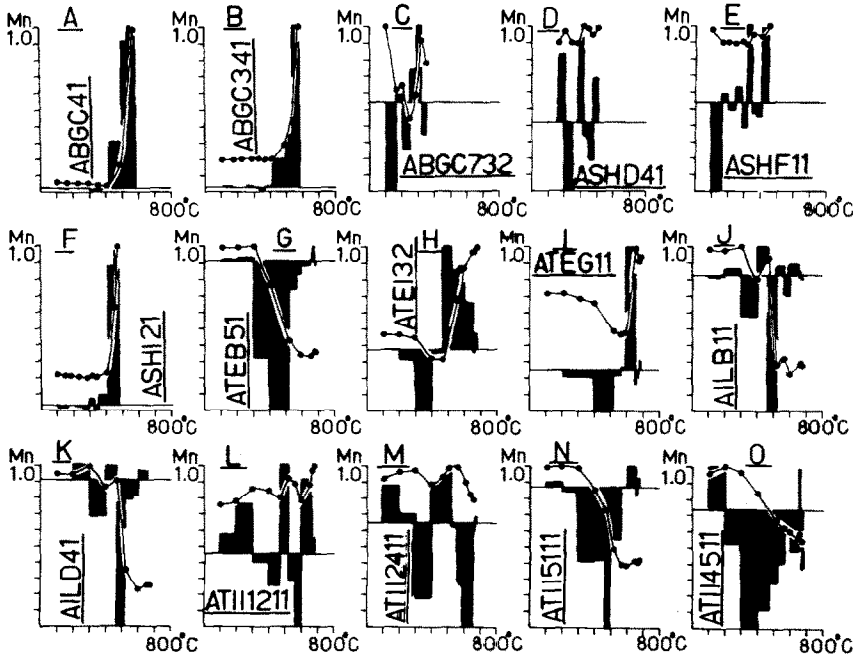


Fig. III.8. Normalized intensity decay curves of the induced magnetization showing the increase (decrease) in susceptibility after treatment at the higher temperatures. Histograms show the stepwise changes.

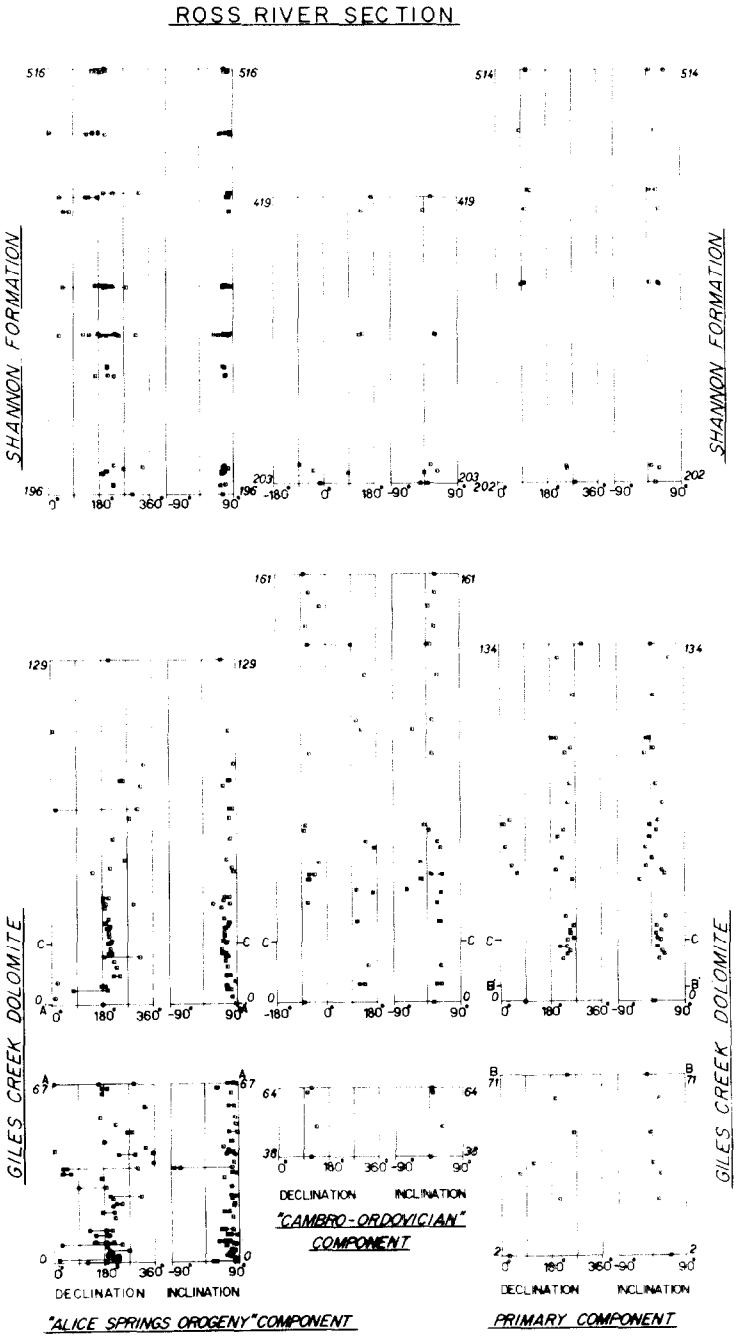


Fig. III.9A. Magnetostratigraphic profile of the Ross River section. Separate plots for the "Alice Springs Orogeny" component, the "Cambro-Ordovician" component and the primary component. Stratigraphic distances in meters.

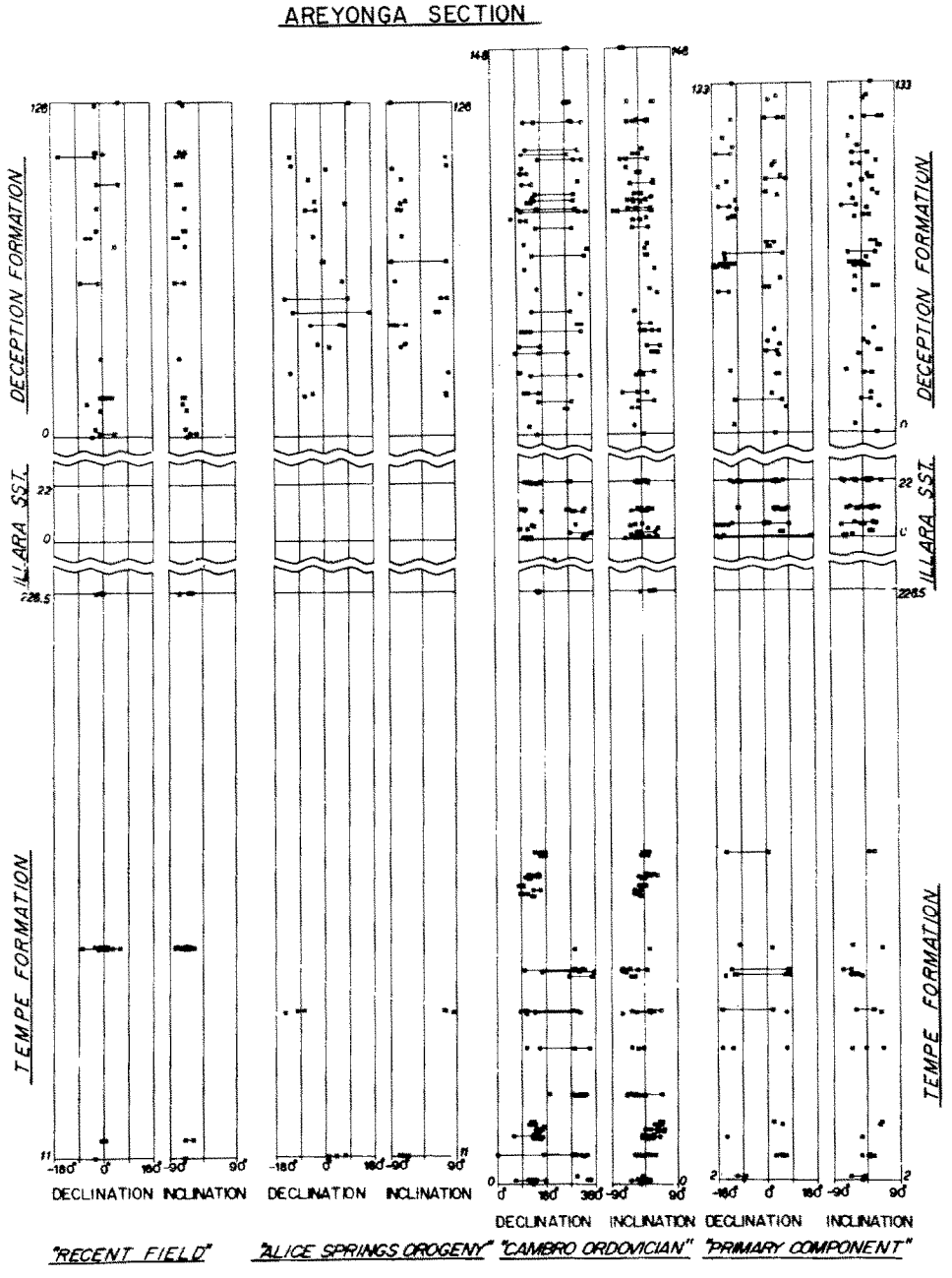


Fig. III.9B. Idem, for the Areyonga section, showing the recent field component in addition.

TABLE III. 1

AMADEUS BASIN

AREYONGA SECTION: location 132.27°E - 24.10°S

FORMATION	Samples (specimens)	Specimens x1 selected	Initial NRM intensity mAm-1	Decl (degrees)	Incl (degrees)	R	α95 (degrees)	Polarity	S-Pole Positions: long (degrees)	lat (degrees)	dp (degrees)	dm (degrees)	Pole No. fig. 12A	Remarks
A) PRIMARY DIRECTIONS														
DECEPTION FORMATION	63(113)	67	0.8-10.4	227	-21	8	6.5	N+R	10.6E	32.2S	3.6	6.8	24	
		31		228	-14	8.5	9.5	R						
		36		46.5	+27	8.5	8.7	N						
ILLARA SANDSTONE	26(42)	43	0.8-1357	229.	-23.	5	10.8	N+R	11.1E	29.5S	6.1	11.5	23	
		19		227	-32	4	19.7	R						
		24		51	+14	6	13.2	N						
TEMPE FORMATION	77(143)	26	0.7-403	225.5	-14	5	14.1	N+R	13.0E	35.9S	7.4	14.4	25	
		10		223.5	-19.5	11	15.3	R						
		16		47	+10	3.5	22.3	N						
B) "CAMBRO-ORDOVICIAN" OVERPRINTING														
TEMPE FM.		127		315	+10	5	6.1	R+N						
ILLARA SST		36		314.5	+3	7.5	9.5	R+N						Most reversed polarity
DECEPTION FM.		43		313	+9	6	9.9	R+N						
COMBINED		206		314.5	+8.5	5.5	4.6	R+N	68.6E	37.3N	2.3	4.6	30	
C) "ALICE SPRINGS OROGENY" OVERPRINTING														
all R-polarity		14		239	+72	12.5	11.6	R						
all N-polarity		22		25	-70.5	10	10.3	N						
COMBINED N+R		36		217.5	+72	10.5	7.7	N+R	102.6E	47.5S	11.9	13.6	32	After bedding-correction
COMBINED N+R		36		23.5	+38	7	9.9	N+R						before bedding-correction
D) RECENT FIELD COMPONENT														
DECEPTION FM.		24		353.5	-53	12	8.9	N						
TEMPE FM.		22		358	-48	12.5	9.0	N						Present local direction decl = 4°; incl = -56°
COMBINED		46		356	-50.5	12.5	6.2	N						Vertical direction according to the axial geocentric dipole decl=0° ; incl=-41.5°

x1 Number of specimens wherein the component concerned could be identified

strong affinity to either a normal or reversed polarity RFC component or to the primary A-component (Fig. III.6). Continued thermal demagnetization generally resulted in breakdown of a normal or reversed polarity B-component, whereas especially in sites AILC and AILD a normal and reversed A-component became broken down in addition (Fig. III.6). It can thus be concluded that the abnormally high intensities and stray directions are at least partly attributable to lightning effects which become largely removed at low field AF-demagnetization. Continued thermal demagnetization resulted in breakdown of either an A- or a B-component. There was a slight preference noticeable for the B-component to reside in the magnetic fraction with the haematite-like blocking temperature range, and for the A-component to reside in the magnetite-like fraction. In general, however, the Illara Sand-

stone specimens conformed to the trend observed throughout this series of studies, of a distinctive behaviour during thermal demagnetization. The susceptibility decreased sharply above 450° C (Fig. III.8). Only in some specimens an ASO component could be identified (Fig. III.5).

Deception Formation. A normal and reversed polarity RFC constituting 20—80% of the initial intensity and prominent throughout the whole sampled section, was broken down completely between 300° C and 500° C. Pilot demagnetization studies revealed a blocking temperature range between 630° C and 670° C with a less evident range between 500° C and 570° C (Fig. III.7). Some specimens displayed a disperse blocking temperature range between 450° C and 670° C. A normal and reversed polarity ASO component, present throughout the section, became generally broken down between 400° C and 570° C and resided apparently in a magnetite phase. In some specimens, however, the reversed polarity ASO component became broken down only at higher temperatures up to 675° C. A- and B-component directions of both normal and reversed polarity were equally well present throughout the section. Notable was the occasional occurrence of both a normal and a reversed B-component in the same specimen. More common was the simultaneous occurrence of normal and reversed polarity B- and to a lesser extent A-component directions in various specimens from the same sample (Fig. III.5). It was not possible to correlate the A- or the B-component with a particular temperature interval in the range 400—670° C. Specimens with a dual polarity B-component invariably showed a breakdown of the reversed component between 400° C and 580° C and of the normal component between 580° C and 675° C. Throughout the lower sandy part of the section, susceptibility remained fairly constant upon repeated heatings, but the specimens from the upper more shaly part showed a susceptibility decrease above 300° C (Fig. III.8).

(B) Ross River section

Giles Creek Dolomite. The magnetic intensity in specimens from the upper massive member was generally an order of magnitude lower than in specimens from the lower mottled member, and was near the accuracy limit of our cryogenic magnetometer. Application of orthogonal projection figures, however, facilitated directional analysis of these specimens. An RFC, constituting 40—60% of the initial intensity in samples from the lower mottled sequence and up to 80% in samples from the upper massive sequence, is evidently present in the NRM directions (Fig. III.4.1). This RFC component is generally completely broken down at 200° C (Figs. III.5, 7). An ASO component of reversed polarity is dominant throughout the whole Giles Creek Dolomite section. In most samples of the lower part of the section, this ASO component was the only component which could be identified in the temperature range 180—360° C, with no other components identifiable at higher temperatures. The primary magnetic component (A)

was dominantly of reversed polarity, except for a probable interval of mixed polarity at the base of the Giles Creek Dolomite and another well documented mixed polarity interval in the middle of the massive dolomitic limestone member (Fig. III.9A). In contrast, the B-component showed a more frequent alternation of normal, reversed and mixed polarity intervals. Breakdown of the A- and the B-component was gradual with no indication whatsoever for a discrete blocking temperature range (Fig. III.7). Samples from the lower mottled member of the Giles Creek Dolomite show a very rapid increase in susceptibility above 400° C (Fig. III.8). In contrast, samples from the upper massive member reveal a gradual susceptibility decrease untill 300° C followed by an increase at higher temperatures.

Shannon Formation. Magnetic characteristics of the Shannon Formation were rather similar to those from the Giles Creek Dolomite. An RFC of up to 80% of the initial intensity (up to 95% in the uppermost site ASHK) dominates the NRM directions (Fig. III.4.1) and is broken down completely at 200° C Figs. III.5, 7). An ASO component of reversed polarity which is present throughout the formation, is eliminated in the temperature interval 200—320° C. Most of the specimens contain either an A-component or more often a B-component which is broken down between 300° C and 450° C. In the basal sites, the B-component is present with both normal and reversed polarity directions, whereas the few B-component directions identified higher up in the formation are of normal polarity only. A distinct reversal pattern is present in the directions of the primary component. Reversed polarity directions occur only in the basal sites, whereas the upper half of the

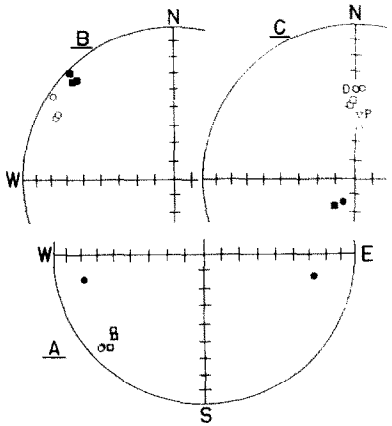


Fig. III.10. Equal area projection plots of formation mean direction. Squares = Areyonga section, circles = Ross River Section. A. Primary magnetization component. B. "Cambro-Ordovician" component. C. Recent field component (open symbols) and "Alice Springs Orogeny" component (full symbols). Hexagon (*D*) = virtual direction at the sampling locality according to the axial geocentric dipole model; inverted triangle (*P*) = present local field direction at the sampling locality.

formation reveals normal polarity directions only (Fig. III.9A). The declination of this E—W directed primary A-component of early-Late Cambrian age is markedly different from the NW—SE declinations of the A-component found in older formations of Middle Cambrian age in the Amadeus Basin and in the Adelaide "Geosyncline" (Fig. III.9A, 10A). Similarly to the Giles Creek Dolomite, all characteristic magnetic components became gradually broken down, and no discrete blocking temperature range could be identified (Fig. III.7). Susceptibility remained fairly constant throughout the thermal treatment except for sites ASHH—ASHI (stratigraphically high in the formation) which showed a very rapid increase in susceptibility after treatment above 400° C (Fig. III.8).

INTERPRETATION

Clearly, separation and identification of the four magnetic components present in the specimens would have been impossible without the application of orthogonal projection figures (Zijderveld, 1967, 1975). A limiting factor in the interpretation was the individual behaviour of the specimens upon thermal demagnetization. No overall magnetic characteristics, other than the appreciable difference in mean directions, were discernible to separate the primary magnetic component (A) from the secondary magnetic component (B). The spread in specimen directions attributed to either of both groupings is fairly large and both distributions do in fact overlap. Similar observations were made also in the accompanying studies on Cambrian sequences from the Adelaide "Geosyncline" (Sections I and II) to which is referred for a more detailed discussion. By analogy to these studies all specimen directions were subjected to a frequency analysis, the most prominent directions were identified, and individual specimen directions were attributed to one or the other of these groupings on the basis of their proximity.

Four magnetic components, i.e., a recent field component (RFC) and three characteristic components (ASO, B and A) have been identified throughout both the red-bed sequence at Areyonga Gorge and the carbonate sequence at Ross River (Figs. III.9A,B,10A).

(A) Recent Field component (RFC)

In the Areyonga section this RFC is present as a normal polarity direction only. This direction is intermediate between the present local field direction and the expected local field direction according to the geocentric axial dipole field model (Table III.1, Figs. III.10C, 11), and is thus most probably of a recent origin. In the Ross River section, in contrast, both normal and reversed polarity RFC directions are present (Fig. III.11). Again the normal polarity direction can be interpreted as being of recent origin (Figs. III.10C, 11, Table III.1). The few reversed polarity directions, however, show a steeper than expected inclination.

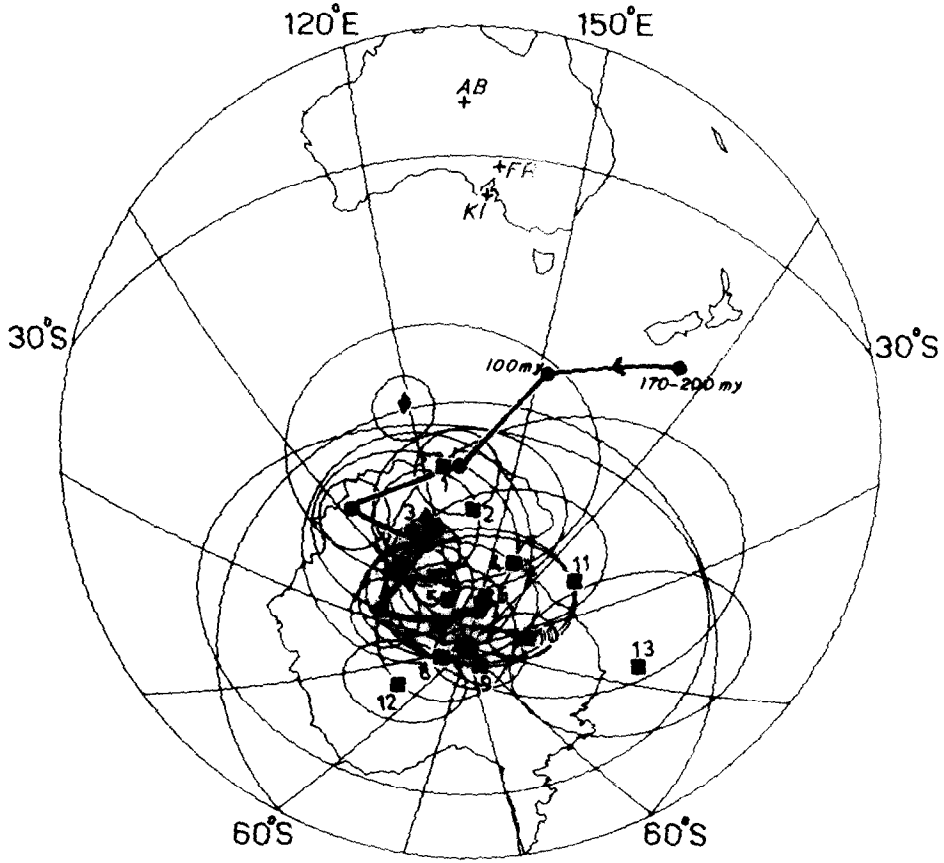


Fig. III.11. Pole positions (squares) and ellipses of confidence corresponding to the various recent field component directions obtained in the present studies from the Adelaide "Geosyncline" (Sections I and II) and the Amadeus Basin. The Mesozoic (Schmidt, 1976) and Cenozoic (McElhinny et al., 1974a) part of the apparent polar wander path for Australia (dots) is represented by the stages: 170—200 m.y., 100 m.y., 40—60 m.y., 25—35 m.y., 20—25 m.y., 15—20 m.y., 5—10 m.y., present. The inverted triangles represent secondary magnetization components obtained by Schmidt et al. (1976) and Luck (1970), attributed by Schmidt to a mid-Tertiary period of extensive remagnetization. Diamonds represent a Maastrichtian—Early Eocene and another Late Oligocene pole position recently obtained by Idnurm and Senior (1978). Large hexagon: "virtual" geomagnetic pole position corresponding with the present local field direction at the sample localities (crosses: *AB* = Amadeus Basin, *FR* = Flinders Ranges, *KI* = Kangaroo Island) Large dot: S-Pole position according to the axial geocentric dipole model. *1* = Ross River section, reversed polarity; *2* = Wirrealpa Limestone (Mount Billy Creek), normal polarity; 3 = Balcoracana Creek, normal polarity; 4 = Deception Formation, normal polarity; 5 = Ross River section, normal polarity; 6 = Areyonga section, normal polarity; *7* = Tempe Formation, normal polarity; *8* = Billy Creek Formation (Mount Billy Creek), normal polarity; 9 = Kangaroo Island, reversed polarity; *10* = Ajax Limestone, normal polarity; *11* = Moodlatana Formation (Mount Billy Creek), normal polarity; *12* = Kangaroo Island, normal polarity; *13* = Bunyeroo Creek, normal polarity.

Schmidt (1976) and Schmidt et al. (1976) report observations of secondary magnetic components, mostly of reversed polarity and of an obviously fairly recent origin, in sediments from the Perth Basin (W. Australia), from the Springfield Basin and from Kangaroo Island (Adelaide "Geosyncline", S. Australia) and quotes fairly similar observations for some Northern Territory Formations as reported by Luck (1970).

Comparison of corresponding pole positions, with the Tertiary apparent polar wander path for Australia (Wellman et al., 1969; McElhinny et al., 1974a) led Schmidt to the interpretation, well established by his reversed porality data, that these secondary magnetic components reflect a dominant phase of lateritic weathering in Late Oligocene—Early Miocene times (Woolnough, 1927), which incurred a synchronous remagnetization event over a large part of Australia. The reversed polarity RFC directions (Fig. III.1) obtained in the present Amadeus Basin study can be interpreted in support of Schmidt's hypothesis, but most of the RFC directions of both normal and reversed polarity (Section II) observed in this series of studies (I, II, III) are evidently — or may be interpreted in terms — of a more recent origin. The "virtual" pole position corresponding with the present local field direction at the sampling areas (Fig. III.11), coincides with the Oligocene—Miocene pole for Australia as obtained by McElhinny et al. (1974a). So, obviously the streaking in RFC pole positions as shown in Fig. III.11 (*2—13*) may equally well represent a sequence of remagnetization events spanning the later half of the Cainozoic, as well as reflecting a remagnetization of very recent origin, whose directions streak between the present local field direction at the sampling locality and the virtual direction expected according to the geocentric axial dipole field model.

(B) Alice Springs Orogeny component (ASO)

This component is most dominantly present in the Ross River section, where it occurs almost exclusively with a reversed polarity (Fig. III.9A), but is less dominant in the Areyonga section, where both normal and reversed polarity directions occur (Fig. III.9B). The pole position (Figs. III.12A, 13) corresponding to this ASO-component, after bedding correction has been applied (Tables III.1, 2, 3; Fig. III.10C), fits nicely on the Devono-Carboniferous segment of the mean polar wander path for Gondwanaland. Therefore, this ASO-component can be well attributed to thermo-chemical activity associated with the Alice Springs Orogeny which climaxed in the Early Carboniferous (Wells, 1976). At least moderate reheating occurred during this orogeny. This is clear from resetting of isotopic mineral ages in the Arunta Complex, to the north and northeast of Ross River. Its proximity to the more intensely affected areas may well explain the greater dominancy of the ASO component in the Ross River section than in the Areyonga section.

The much improved grouping of the ASO component directions when a

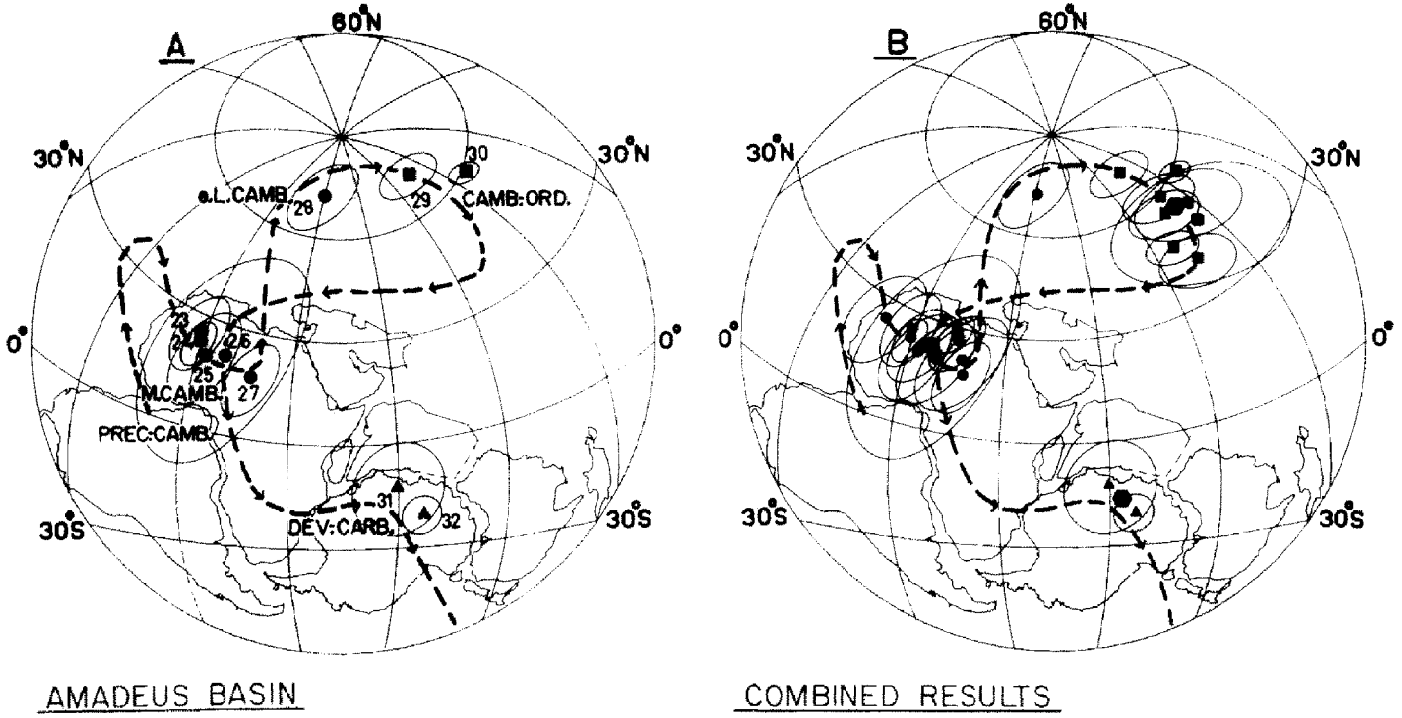


Fig. III.12.A. Newly obtained pole positions for the Cambrian of the Amadeus Basin corresponding to: (1) Primary magnetization (dots); (2) "Cambro-Ordovician" overprint (squares); (3) "Alice Springs Orogeny" overprint (triangles). *23* = Illara Sandstone; *24* = Deception Formation; *25* = Tempe Formation; *26* = basal Giles Creek Dolomite; *27* = top Giles Creek Dolomite; *28* = top Shannon Formation; *29*, *31* = Ross River section; *30*, *32* = Areyonga section. The revised early Palaeozoic combined APW for the Gondwanaland continents is also shown (see Fig. III.13). B. As above, supposedly reliable pole positions from the Flinders Ranges, Kangaroo Island and the Amadeus Basin have been superimposed on the revised early Palaeozoic APW for Gondwanaland (Fig. III.13). The large hexagons denote the mean pole positions for the primary Middle Cambrian grouping, the secondary "Cambro-Ordovician" grouping and the secondary Devono-Carboniferous "Alice Springs Orogeny" grouping.

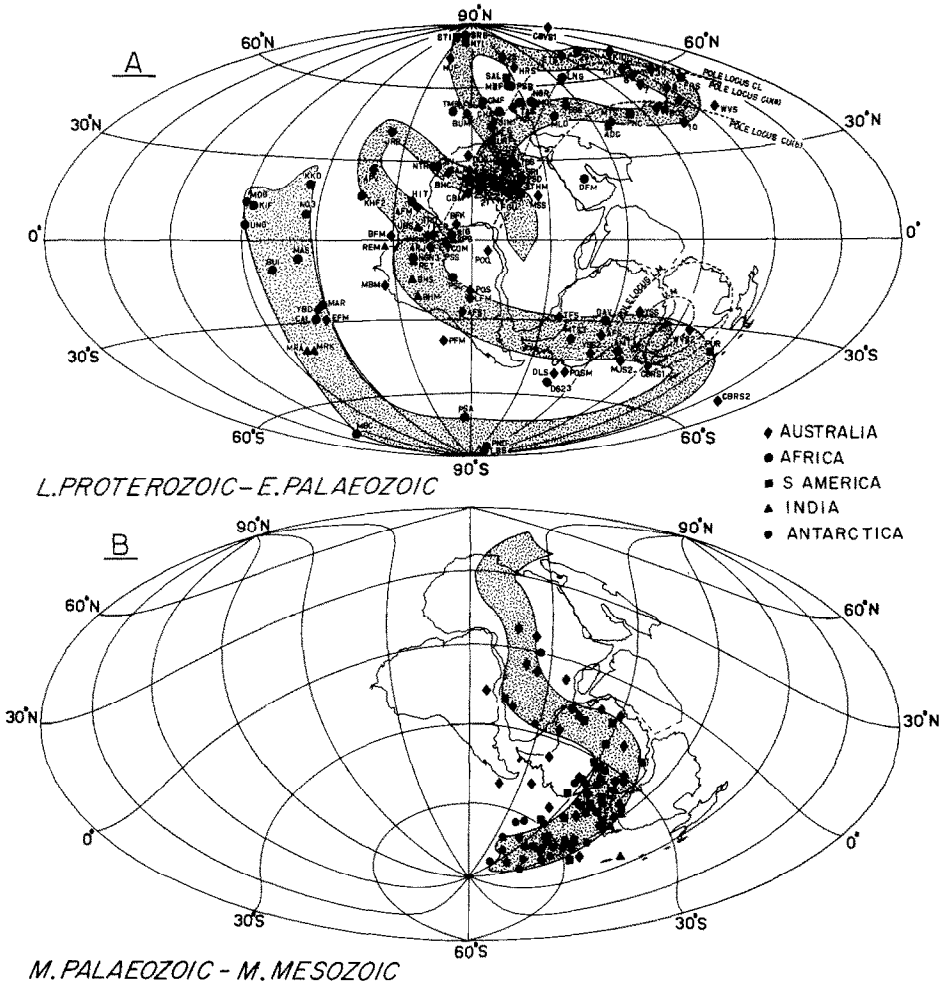


Fig. III.13. Pole positions for the Gondwanaland continents from about 750 m.y. ago to the early Palaeozoic (A) and from the middle Palaeozoic to the middle Mesozoic (B), plotted according to the Smith and Hallam (1970) reconstruction. The pole positions represent results listed by McElhinny and Embleton (1976), Schmidt (1976), McWilliams (1977), Klootwijk (1979) and in the present paper. For reason of clarity, the Gondwana APW has been shown in two separate plots A and B. Initials of the various pole positions have been added in Fig. 13A (list available from the author on request), but this was not possible in Fig. 13B. Aitoff projection.

correction for bedding is applied (Table III.1) clearly indicates acquisition of this magnetic component before the actual folding phase of the Alice Springs Orogeny. Similarly overprinted directions, also interpreted as to be associated with the Alice Springs Orogeny, were already reported by Kirschvink (1978b) for samples from the Ross River section and by McWilliams (1977)

TABLE III. 2

AMADEUS BASIN

ROSS RIVER SECTION; location 134.49°E, 23.62°S

FORMATION	Samples (specimens)	Specimens selected x¹	Initial intensity emu	Decl (degrees)	Incl (degrees)	K	α95 (degrees)	Polarity	S-Pole Positions: long (degrees)	lat (degrees)	dp (degrees)	dm (degrees)	Pole No. Fig. 12A	Remarks
A) PRIMARY DIRECTION														
SHANNON FM.	48(61)		0.001-18.1											
top		9		100	+27.5	28	10.6	N	34.9E	14.9N	6.3	11.6	28	Pronounced directional and polarity change
base		3		258	+20.5	18	29.9	R						
GILES CREEK DOLOMITE														
UPPER SECTION	73(93)	27	0.001-25	227.5	+9.5	6.5	11.7	R+N	30.0E	40.4S	6.0	11.8	27	Mostly reversed polarity
LOWER SECTION	41(58)	6	0.6-3.4	227.5	-9	5	32.6	R+N	19.7E	35.7S	16.6	32.9	26	Mostly reversed polarity
B) "CAMBRO-ORDOVICIAN" OVERPRINTING														
SHANNON FM.		19		297	-13	6	15.3	N+R						Mostly reversed polarity
GILES CREEK DOLOMITE-UPPER		27		304	-3.5	5.5	12.9	R+N						
GILES CREEK DOLOMITE-LOWER		4		298	-13.5	19.5	21.5	N						
ROSS RIVER SECTION COMBINED		50		301	-8.5	6	9.0	N+R	53.7E	29.9N	4.6	9.1	29	
C) "ALICE SPRINGS OROGENY" OVERPRINTING														
ROSS RIVER SECTION COMBINED		178		206.5	+76.5	14	2.9	R	118.1E	45.3S	5.0	5.4	31	(only 2 N-polarity specimens) after bedding correction
AS ABOVE		178		1	+66	13.5	3.0	R						before " "
D) RECENT FIELD COMPONENT														
ROSS RIVER SECTION COMBINED		22		3	-50.5	20	7.1	N+R						
		17		2.5	-47.5	19	8.4	N						present local field direction decl = 40°; incl = -56°
		5		184.5	+63	38.5	12.5	R						expected direction according to the axial geometric dipole model decl = 0°; incl = -41°

x¹) Number of specimens wherein the component concerned could be identified

TABLE III. 3

PERIOD	COMBINED STUDIES Mean S-Pole Position long (degrees)	lat (degrees)	k	α95 (degrees)	N	
ALICE SPRINGS OROGENY COMPONENT						
DEVONO - CARBONIFEROUS (AMADEUS BASIN)	110.5E	46.9S	-	-	2	(localities)
CAMBRO-ORDOVICIAN COMPONENT						
(AMADEUS BASIN)	60.8E	33.8N	-	-	2	"
"DELAMARIAN" COMPONENT, FLINDERS RANGES	75.3E	26.0N	10.8	7.4	5	"
"DELAMARIAN" COMPONENT, KANGAROO ISLAND	75.8E	17.3N	7.6	7.5	54	(specimens)
MEAN CAMBRO-ORDOVICIAN COMPONENT	72.0E	27.0N	52.9	7.3	8	(localities)
LATE EARLY-MIDDLE CAMBRIAN (KANGAROO ISLAND/BILLY CREEK FM., AROONA CREEK LST, WIRREALPA LST/ TEMPE FM./basal GILES CREEK DOLOMITE)	17.2E	34.5 S	301.5	5.3	4	(localities)
MIDDLE CAMBRIAN PROPER (ILLARA SST/DECEPTION FM./top GILES CREEK DOLOMITE/LAKE FROME GROUP)	19.3E	33.7 S	69.1	11.1	4	"
MIDDLE CAMBRIAN COMBINED	18.3E	34.1S	128.4	4.9	8	"

for samples from the Ellery Creek section and from the northeastern part of the Amadeus Basin. Kirschvink also interpreted this ASO-component to be of pretectonic origin. In contrast, McWilliams suggested a syn- to post-tectonic origin. The reason for this discrepancy is not well understood. A geographical differentiation in the timing of the main folding phase, as suggested by J.L. Kirschvink (personal communication, 1978) possibly may explain this observed disparity in results.

(C) The secondary B-component

This component of both normal and reversed polarity is very evident in the Areyonga section, but less so in the Ross River section (Figs. III.4C, 9, 10, Tables III.1, 2). Comparison with the main Gondwanaland APW (Figs. III.12, 13) indicates that this component was acquired prior to the Alice Springs Orogeny. However, apart from some middle Palaeozoic periods of minor tectonic instability in the Amadeus Basin, i.e. the Rodingan Movement and the Perntjara Movement, and the Bloodwood Movement in the adjacent Ngalia Basin, the latter not better defined than the period early-Middle Cambrian to Late Ordovician (A.T. Wells, personal communication, 1978), there is nothing in the tectonic record which could be associated with acquisition of this component. The pole positions corresponding with this

component are in remarkably good agreement with those from a similar secondary magnetic component noted in the Adelaide "Geosyncline" (Figs. III.9, 12A, B). Both in the Flinders Ranges (Section I) and in the Kangaroo Island (Section II) studies, acuisition of this secondary magnetic component was attributed to thermo-chemical activity associated with a phase of regional heating which predated the main folding phases of the Late Cambrian—Early Ordovician Delamarian Orogeny. Although not noted in the Amadeus Basin, Williams (1975) suggested that metamorphic effects of this orogeny extend far to the north of the Adelaide "Geosyncline" into the Pedirka Basin. Therefore, acquisition of this B-component due to thermo-chemical activity during a period of similarly enhanced heat flux may be envisaged for the Amadeus Basin also. The blanketing effect of the later Alice Springs Orogeny may well have obscured any traces from such a Late Cambrian—Ordovician heating in the Amadeus Basin. The presence of both normal and reversed polarities of this secondary component points to a prolonged thermo-chemical activity.

Such an acquisition mechanism for this B-component seems the more likely as a strongly increased heat flux during Late Cambrian—Early Ordovician times characterizes the Pan-African orogenic belts of Gondwanaland (Krylov, 1972). During this period, widespread isotopic resetting occurred in such zones as the Damaran—Katangan belt in Africa (Clifford, 1970), along the Pacific border of Gondwanaland (Craddock, 1977) and in particular the Transantarctic Mountains (Veevers, 1976), along the Indian East Coast (Crawford, 1969) and along the Proto-Atlantic margin (Hurley et al., 1968; Hurley, 1972).

(D) The primary A-component

The mere fact that this primary magnetic component could be separated and determined despite three subsequent and dominant phases of magnetic overprinting may serve as an illustration of the resolution power of orthogonal projection figures (Zijderveld, 1967, 1975) in magnetically complex systems. In the Middle Cambrian Areyonga section (Tempe Formation, Illara Sandstone, Deception Formation) this A-component shows a very rapid alternation and intermixing of normal and reversed polarities (Fig. III.9B). Occasionally both polarities are present in various specimens from the same sample. This persistent dual polarity behaviour of the A-component in the Areyonga red-bed sequence contrasts sharply with the correlatable Middle Cambrian carbonate sequence from the Ross River sequence (Fig. III.9A). Throughout the Giles Creek Dolomite, reversed polarities seem to dominate completely. Sole exceptions are a section of mixed polarity in the middle of the massive upper member, and possibly a less well established mixed polarity sequence in the lower member (Fig. III.9A). Correlation of the whole dual polarity Areyonga section with either of these short mixed polarity intervals in the Giles Creek Dolomite seems a remote solution for

this discrepancy in polarity structure between both sections. It seems more likely that the dual polarity structure of the Areyonga red-bed sequence is due to a prolonged acquisition of magnetization upon diagenesis, lasting for at least a normal and a reversed polarity event. Similar observations (Fig. III.15) were noted also for red-bed sequences from the Flinders Ranges (Section I) and from Kangaroo Island (Section II). In the carbonate sequences studied here, however, such a dual polarity structure was notably absent or of very minor prominence. This may point to a shorter acquisition phase for the primary magnetization in case of such carbonate sediments. It is open to verification whether the difference observed here in polarity structure between the red beds and carbonate rocks, represents a more general occurrence. Clearly, application of the red bed results reported here for magnetostratigraphic purposes is subject to serious limitation.

The Middle Cambrian directions from the Amadeus Basin (Tempe Formation, Illara Sandstone, Deception Formation, Giles Creek Dolomite), agree well amongst themselves. This supports the suggested Middle Cambrian age of the Illara Sandstone and the Deception Formation (Wells et al., 1970), which was based on the superposition principle only. However, they clearly differ from the early-Late Cambrian results from the Shannon Formation. These Middle Cambrian results from the Amadeus Basin are also consistent with results from correlatable formations from the Adelaide "Geosyncline" (Figs. III.12B, 13), and imply a much reduced polar motion for the Middle Cambrian. Rapid polar motion began again in the late-Early Cambrian as is evidenced by the results from the Shannon Formation (Fig. III.14). Like the underlying Giles Creek Dolomite, this Shannon Formation (early-Late Cambrian) shows a well defined polarity pattern. The reversed polarity zone in the top part of the Giles Creek Dolomite continues into the basal part of the Shannon Formation, followed by a normal polarity sequence in its upper half (Figs. III.9A, 15). The declination record (Fig. III.10A) shows a swing from SW—NE pointing declinations in the Middle Cambrian to E—W pointing declinations in the early-Late Cambrian. Rapid polar movement probably continued during the Cambro-Ordovician as indicated by the sequence of pole positions from the B-component (Figs. III.12A, 14).

The latest Proterozoic—early Palaeozoic interval is thus characterized by periods of much reduced polar movement (quasi static intervals) alternating with periods of rapid polar movement (Figs. III.14, 15). Concentric directions obtained by Kirschvink (1978b) from formations spanning the Precambrian—Cambrian boundary (Pertatataka Formation, Arumbera Sandstone s.l., Eninta Formation and the Todd River Dolomite) led him to the suggestion of a quasi static interval for this period. The alternative explanation of magnetic resetting due to the Petermann Ranges Orogeny (600 m.y.) seems somewhat unlikely as its effects are mostly localized to the southwestern margin of the Amadeus Basin, whereas Kirschvink sampled in the northeastern part. Early Cambrian data from the Hawker Group of the Flinders Ranges (Section I) may suggest rapid polar movement, although an alter-

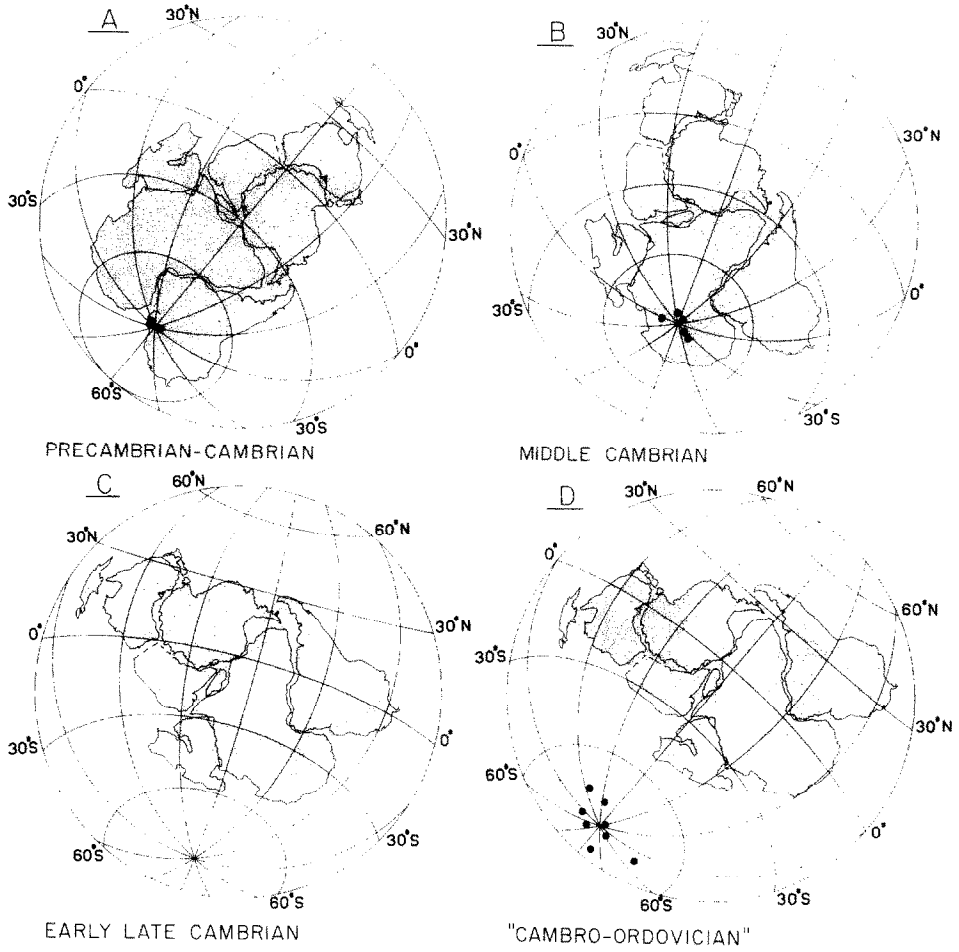


Fig. III.14. Palaeoposition of Gondwanaland for various periods, according to Australian data. A. Precambrian—Cambrian, according to Kirschvink's (1978b) results. B. Middle Cambrian, present studies. C. Early-Late Cambrian, present study (Shannon Formation). D. Cambro-Ordovician, present studies. Linear Polar projection.

native interpretation in terms of differential movement within the Australian plate (to be discussed) cannot be excluded. The Middle Cambrian was then another quasi static intervall followed by rapid polar movement during the Cambro-Ordovician.

IMPLICATIONS

Despite obvious problems in interpreting the polarity structure of arenitic red-bed sequences in both the Adelaide "Geosyncline" (Sections I and II)

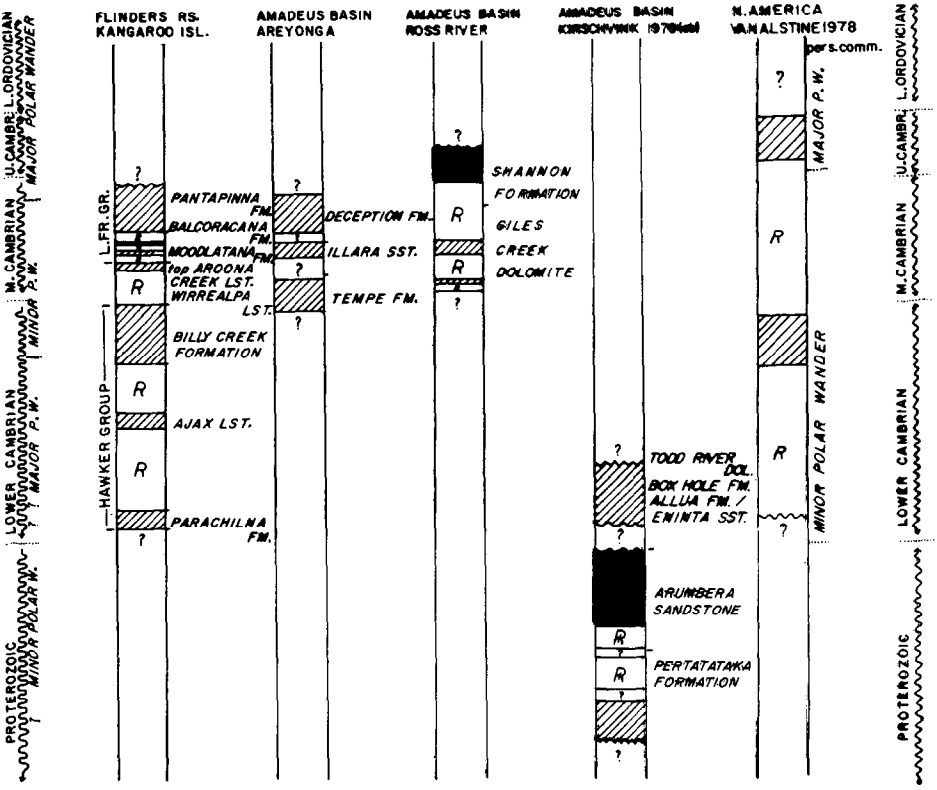


Fig. III.15. Magnetostratigraphic correlation of the late Proterozoic—Cambrian formations from the Adelaide "Geosyncline" and the Amadeus Basin, compared with a tentative polarity zonation for western North America.

and in the Amadeus Basin, a general polarity pattern for the Cambrian of Australia can be constructed (Fig. III.15). This shows a predominance of reversed polarities throughout the Early, Middle and earliest-Late Cambrian with some periods of mixed polarity intercalated, which are partly of doubtful nature as pointed out before (earliest Cambrian: Todd River Dolomite (Kirschvink, 1978a, b), Parachilna Formation; Ajax Limestone; late-Early Cambrian: Billy Creek Formation?; Kangaroo Island sequence?; early-Middle Cambrian: Aroona Creek Limestone, Wirrealpa Limestone; late-Middle Cambrian: Giles Creek Dolomite). This longlasting phase of reversed polarity is followed by a period of normal polarity starting in the early-Late Cambrian, shortly preceded by inception of a continuing rapid polar movement (Shannon Formation). Clearly, similar trends are also visible in the predominantly reversed polarity pattern of early Palaeozoic sequences from western North America (Fig. 15; D. van Alstine, personal communication, 1978). Although some discrepancies in timing may have to be updated, both

the Australian and the North American pattern may well serve as a basis for intercontinental correlation.

The good agreement between both the primary Middle Cambrian and the secondary Cambro-Ordovician results from the Amadeus Basin and from the Adelaide "Geosyncline" refute Harrington et al.'s (1973) hypothesis for substantial rifting in the Australian Plate during Middle and Late Cambrian times. An earlier, latest Proterozoic inition of the rifting as suggested by Harrington et al. (1973) cannot be ruled out, as a discrepancy in pole positions exists indeed between the Early Cambrian Todd River Dolomite results (Kirschvink, 1978b) from the Amadeus Basin and the cursory results from the Early Cambrian Hawker Group of the Flinders Ranges (Section I). It seems more likely, however, that the results represent various trajectories of the APW during a period of rapid polar wandering. This interpretation contrasts with McWilliams' (1977) analysis of newly obtained late Proterozoic palaeomagnetic results from Australia, which he interpreted in support of Harrington et al.'s (1973) model.

Clearly the early Palaeozoic part of the common apparent polar wander path for Gondwanaland as proposed by McElhinny et al. (1974b; McElhinny and Embleton, 1976) has to be updated considerably on the basis of new palaeomagnetic results from the early Palaeozoic of Australia as detailed in this series of studies (Sections I, II and III). The early Palaeozoic part of the originally proposed APW path was not completely satisfactorily defined as several pole positions, notably from the Adelaide "Geosyncline", appeared to be out of sequence, whilst some South American and African data did not conform to this path at all. This latter group of results is now supported, however, by the Cambro-Ordovician pole positions for Australia obtained here, and some of McWilliams (1977) results from late Proterozoic rocks from Southern Africa and from widely different areas in Australia can be interpreted in further support.

Palaeomagnetic studies on early Palaeozoic and older rocks face problems of (1) separation of individual components in a multicomponent magnetic system, (2) identification of the various components as primary and secondary magnetizations, (3) the ambiguity in polarity assignment. McWilliams' (1977) study is a fine example of separation of magnetic components through analytical and graphical (orthogonal projections) analysis of a multicomponent magnetization system. McWilliams found secondary magnetic components which adhere rather well to the presently found secondary B-components of Cambro-Ordovician age (Fig. III.13), in data from Southern Africa (Nosib Group-Blaubeker Formation — NBF, Nosib Group — NGR), and from Australia (Egan Tillite (Kimberley Basin) — ETS, Chambers Bluff Volcanics (Officer Basin) — CBVS1, the Wooltana Volcanics — WVS and the Merinjina Tillite — MTS, in the Northern Flinders Ranges). Moreover, after reversing of his polarity interpretation comparable results from the Chuas—Maieberg Formation (CMF) and the Lower Nama Group (LNG) from Southern Africa may be better interpreted in terms of a secondary

Cambro-Ordovician magnetization than in terms of a primary magnetization as originally suggested by McWilliams. McWilliams' (1977) results from the King Island Volcanics (KIV) and the King Island dykes (KID) also fit rather well with this pattern. These formations are of an unspecified but probably early Palaeozoic age, so such a Cambro-Ordovician magnetic component may equally well represent either a primary or a secondary magnetization.

An updated version of the common APW for Gondwanaland as originally proposed by McElhinny et al. (1974b; McElhinny and Embleton, 1976) is presented in Fig. III.13A, B. The most significant change has been made to the early Palaeozoic part of the APW (Fig. III.13A) with the addition of the Cambro-Ordovician loop. The suggested halfloops in the Cambrian part of the APW are open to refinement. In the absence of better data, some of the earlier Australian results had to be included, in which possibly an equally present overprinting has not yet been identified. The younger part of the common APW (late Palaeozoic—Middle Jurassic) remained largely unchanged. The pole positions for the detailed Devono-Carboniferous ASO component show good agreement with the path (Fig. 13B).

McElhinny et al. (1974b; McElhinny and Embleton, 1976) suggested backwards continuation of the common APW till at least 750 m.y. ago, which infers a long standing unity of Gondwanaland. An alternative solution was recently proposed by McWilliams (1977) who noted that pole positions from the various Gondwana continents appear to be concentrated mainly in different parts of the late Proterozoic loop. The African data are prominent in the "early" part, whereas the Australian data dominate the "later" half. This led McWilliams to an alternative solution of separate APW's for eastern (Australia, Antarctica and India) and western Gondwanaland (Africa, South America), which join each other in the Early Cambrian trajectory. Thus inferring a collision origin for at least part of the Pan-African orogenic belts. McWilliams' (1977) suggestion was based quite substantially on a reconfiguration of Australian pole positions according to the Cambrian fragmentation model (Harrington et al., 1973), which is not substantiated by the Middle Cambrian data presented here (Sections I—III). Moreover, the Cambro-Ordovician loop proposed here in the Gondwanaland APW facilitates alternative interpretations for the polarity and also for a secondary rather than a primary nature of the magnetization of various earlier palaeomagnetic results. Admittedly, a wide range of APW paths can be constructed through the palaeomagnetically ill defined late Proterozoic interval, and the here advanced combined Gondwanaland APW (Fig. III.13) is only proposed as one of the many possible trajectories. It is, however, a natural consequence of the model, based as it is on the present findings of thermo-chemically induced secondary magnetic components of Cambro-Ordovician age in various areas of Australia and possible beyond, notably observed in absence of contemporaneous deformation, and the widespread evidence for an enhanced heat flux in the Pan-African orogenic belts during Cambro-Ordovician times, that it suggests an ensialic rather than a collision origin as the more favourable model for the Pan-African belts in Gondwanaland.

CONCLUSIONS

Four magnetic components have been identified in a red-bed sequence from the Areyonga Gorge (central Amadeus Basin) and in a carbonate sequence from the Ross River Gorge (eastern Amadeus Basin).

(A) A recent component (RFC) of both normal and reversed polarity. In some of the formations studied, this RFC is clearly of latest Cainozoic origin, but an earlier acquisition, possibly related to a widespread middle Tertiary laterization event, cannot be excluded for other formations.

(B) A Late Devonian—Early Carboniferous component (ASO) induced prior to the main folding phase of the Alice Springs Orogeny (Early Carboniferous). This component is most dominantly present in the Ross River Gorge (nearly exclusively reversed polarities) and to a lesser extent in the Areyonga Gorge (reversed and normal polarities).

(C) A secondary component (B) of both normal and reversed polarity, which is very similar to a Late Cambrian—Early Ordovician secondary component which became induced in rocks from the Adelaide "Geosyncline" by enhanced thermo-chemical activity during a period of enhanced heat flux prior to the Delamarian Orogeny. In absence of indications for early Palaeozoic tectonic activity in the Amadeus Basin, a similar thermo-chemical acquisition is envisaged also for this B component.

(D) A primary magnetic component (A). Middle Cambrian directions from both the Areyonga section and the Ross River sequence group very well, whereas early-Late Cambrian directions from the Shannon Formation (Ross River) indicate onset of appreciably polar wandering.

The present data compared with Kirschvink's (1978b) Precambrian—Cambrian boundary period results from several Amadeus Basin sequences group into two quasi static intervals spanning the Precambrian—Cambrian boundary period and the Middle Cambrian, which alternate with periods of rather fast polar movement during the Early Cambrian and the Cambro-Ordovician.

Thermal demagnization study on red-bed sequences and carbonate sequences in the Adelaide "Geosyncline" and in the Amadeus Basin indicate that the former are generally less suited for magnetostratigraphic applications, as acquisition of their "primary" magnetization probably took place over a prolonged period. In contrast, the red-bed sequences generally showed a better concentration of component directions when compared with the carbonate sequences.

The combined data from the Amadeus Basin and the Adelaide "Geosyncline" indicate the Early- and Middle Cambrian as a dominantly reversed polarity interval with some periods of mixed polarity intercalated. A normal polarity interval seems to have started in the early-Late Cambrian. This pattern compares well with a preliminary early Palaeozoic polarity stratigraphy for North America.

Magnetic components similar to the thermo-chemically induced Cambro-

Ordovician components in the Amadeus Basin, and in the Adelaide "Geosyncline" were reported recently from widely different areas in Australia and in Southern Africa by McWilliams (1977), who also suggested as secondary early Palaeozoic origin for most of them. It is quite possible that this set of secondary magnetizations results from an enhanced thermo-chemical activity during Cambro-Ordovician times, associated with formation of Pan-African Orogenic belts of Gondwanaland.

The good agreement between Middle Cambrian poles from the Amadeus Basin and from the Adelaide "Geosyncline" refutes Harrington et al.'s (1973) hypothesis of fragmentation of the Australian plate during Middle and Late Cambrian times. A possible earlier fragmentation cannot be ruled out, however, on grounds of the present data.

An earlier proposed common APW for Gondwanaland (McElhinny et al., 1974b; McElhinny and Embleton, 1976) has been updated. Unity of Gondwanaland backwards till at least 750 m.y. ago seems well in accord with existing evidence, but alternative interpretations (McWilliams, 1977) remain equally possible.

ACKNOWLEDGEMENTS

Thanks are due to Drs. P.J. Burek, B.J.J. Embleton, M.E. Evans, J.W. Giddings, M. Idnurm, M.W. McElhinny, D. Vanalstine, A. Wells, an anonymous reviewer and in particular J. Kirschvink for discussion and criticism of the manuscript.

Drs. P.W. Schmidt and J. Kirschvink were of very valuable assistance during the fieldwork, and Dr. B. Daily guided the author during a short excursion in the northern Flinders Ranges.

The cryogenic magnetometer used in this series of studies was made available through courtesy of Dr. E. Shoemaker of the California Institute of Technology. Interfacing with a Digico minicomputer and programming was done respectively by Drs. J. Kirschvink and M.O. McWilliams.

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